

Learning Operators by Regularized Stochastic Gradient Descent with Operator-valued Kernels[†]

Jia-Qi Yang¹ and Lei Shi^{1,2}

¹School of Mathematical Sciences,
Fudan University, Shanghai, 200433, China

² Shanghai Key Laboratory for Contemporary Applied Mathematics,
Fudan University, Shanghai, 200433, China

Abstract

We consider a class of statistical inverse problems involving the estimation of a regression operator from a Polish space to a separable Hilbert space, where the target lies in a vector-valued reproducing kernel Hilbert space induced by an operator-valued kernel. To address the associated ill-posedness, we analyze regularized stochastic gradient descent (SGD) algorithms in both online and finite-horizon settings. The former uses polynomially decaying step sizes and regularization parameters, while the latter adopts fixed values. Under suitable structural and distributional assumptions, we establish dimension-independent bounds for prediction and estimation errors. The resulting convergence rates are near-optimal in expectation, and we also derive high-probability estimates that imply almost sure convergence. Our analysis introduces a general technique for obtaining high-probability guarantees in infinite-dimensional settings. We illustrate the practical scope of our framework with applications to structured prediction and parametric PDEs, providing examples that reflect how the approach can be applied in practice.

Keywords: Nonlinear operator learning, Operator-valued kernel, Regularized stochastic gradient descent, Convergence analysis

1 Introduction

In this work, we consider the nonlinear regression model

$$y = h^\dagger(x) + \epsilon, \quad (1.1)$$

where x belongs to a Polish space $\mathcal{X}$, and y lies in a separable Hilbert space $\mathcal{Y}$. The pair (x, y) is distributed according to an unknown probability measure ρ on $\mathcal{X} \times \mathcal{Y}$. The regression operator $h^\dagger : \mathcal{X} \rightarrow \mathcal{Y}$ is defined as the conditional expectation

$$h^\dagger(x) = \mathbb{E}[y|x],$$

and the noise ϵ is a centered $\mathcal{Y}$ -valued random variable with finite variance $\sigma^2 := \mathbb{E}[\|\epsilon\|_{\mathcal{Y}}^2]$, assumed independent of x . The objective is to recover $h^\dagger$ from independent, identically distributed samples of (x, y) .

[†] The work described in this paper is supported by the National Natural Science Foundation of China under Grant Nos. 12171093 and 12571099. Email addresses: jqyang24@m.fudan.edu.cn (J.-Q. Yang), leishi@fudan.edu.cn (L. Shi). The corresponding author is Lei Shi.

This formulation encompasses a broad class of statistical inverse problems in infinite-dimensional settings, where the forward map $h^\dagger$ must be recovered from noisy, indirect observations. Traditional approaches to inverse problems often employ explicit regularization techniques, such as Tikhonov regularization or spectral filtering, to stabilize the solution, typically under certain assumptions on the forward operator’s structure. In our work, we also use explicit regularization by penalizing the norm in a vector-valued reproducing kernel Hilbert space (RKHS). However, unlike classical deterministic frameworks, our approach studies stochastic gradient descent algorithms in infinite-dimensional function spaces with streaming data¹. This requires new analytical tools to establish convergence and error bounds under operator regularity conditions.

A key motivation for this model arises from structured output prediction, where the outputs exhibit functional or combinatorial structure. In many practical scenarios, the output space is not finite-dimensional but consists of complex objects such as functions, sequences, or graphs. To address these tasks, surrogate learning approaches embed the structured output into a Hilbert space and learn a continuous operator mapping inputs to embedded outputs. Applications include image completion (Weston et al., 2002), label ranking (Korba et al., 2018), graph prediction (Brouard et al., 2016a), and protein function prediction (Ciliberto et al., 2020). This framework also includes *functional output regression*, which has become increasingly relevant with the growing availability of functional data in science and engineering. Here, both the input and output may be infinite-dimensional, motivating the development of operator-theoretic regression approaches. These applications have spurred the growth of the operator learning paradigm, which aims to learn mappings between function spaces directly from data. A prominent example is learning solution operators for parameterized partial differential equations (PDEs), where the operator maps coefficient or boundary condition functions to PDE solutions (Li et al., 2020; Bhattacharya et al., 2021; Lu et al., 2021; Lei et al., 2024). In this paper, we focus on learning algorithms that operate directly on function spaces rather than on finite-dimensional discretizations. This functional viewpoint preserves the intrinsic geometry and infinite-dimensional structure of the problem, enabling discretization-invariant analysis and broad applicability. Our setting naturally leads to the study of regularized learning in vector-valued RKHSs, where the regression operator is modeled within a rich nonparametric hypothesis space induced by operator-valued kernels.

We introduce a supervised learning framework based on model (1.1). Consider a data set $\{(x_t, y_t)\}_{t=1}^T$ generated by model (1.1), or equivalently, drawn independently from the distribution ρ . To estimate $h^\dagger$, we minimize the regularized functional $\mathcal{E}(h) + \lambda \|h\|_{\mathcal{H}}^2$ over all $h \in \mathcal{H}$, where $\mathcal{H}$ is some Hilbert space, $\mathcal{E}(h) := \mathbb{E} [\|h(x) - y\|_{\mathcal{Y}}^2]$ denotes the mean squared error, and $\lambda > 0$ is a regularization parameter. In this paper, we adopt a non-parametric approach to solve the nonlinear model (1.1), assuming that $\mathcal{H}$ is a vector-valued reproducing kernel Hilbert space (RKHS) induced by an operator-valued kernel K (Kadri et al., 2012, 2016; Brouard et al., 2016b; Brogat-Motte et al., 2022).

To illustrate our algorithm, we introduce some notations along with basic concepts from operator theory (Conway, 2000). Consider a linear operator $A : \mathcal{H}_1 \rightarrow \mathcal{H}_2$, where both $(\mathcal{H}_1, \langle \cdot, \cdot \rangle_{\mathcal{H}_1}, \|\cdot\|_{\mathcal{H}_1})$ and $(\mathcal{H}_2, \langle \cdot, \cdot \rangle_{\mathcal{H}_2}, \|\cdot\|_{\mathcal{H}_2})$ are Hilbert spaces. The set of bounded linear operators from $\mathcal{H}_1$ to $\mathcal{H}_2$ forms a Banach space under the operator norm $\|A\| = \sup_{\|f\|_{\mathcal{H}_1}=1} \|Af\|_{\mathcal{H}_2}$, denoted by $\mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$, or simply $\mathcal{B}(\mathcal{H}_1)$ when $\mathcal{H}_1 = \mathcal{H}_2$. We call an operator $A \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ Hilbert-Schmidt if it holds $\sum_{k \geq 1} \|Ae_k\|_{\mathcal{H}_2}^2 < \infty$ for some (equivalently, any) orthonormal basis $\{e_k\}_{k \geq 1}$ of $\mathcal{H}_1$. The set of Hilbert-Schmidt operators from $\mathcal{H}_1$ to $\mathcal{H}_2$ forms a Hilbert space under the Hilbert-Schmidt inner product $\langle A, B \rangle_{\text{HS}} = \sum_{k \geq 1} \langle Ae_k, Be_k \rangle_{\mathcal{H}_2}$ and the induced norm $\|\cdot\|_{\text{HS}}$, denoted by $\mathcal{B}_{\text{HS}}(\mathcal{H}_1, \mathcal{H}_2)$. The adjoint of A , denoted by A^* , is the unique operator satisfying $\langle Af, f' \rangle_{\mathcal{H}_2} = \langle f, A^*f' \rangle_{\mathcal{H}_1}$ for all $f \in \mathcal{H}_1$ and $f' \in \mathcal{H}_2$. If $A \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$, then $A^* \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ and $\|A\| = \|A^*\|$. An operator $A \in \mathcal{B}(\mathcal{H}_1)$ is called self-adjoint if $A^* = A$, and positive if it is self-adjoint and satisfies $\langle Af, f \rangle_{\mathcal{H}_1} \geq 0$ for every $f \in \mathcal{H}_1$. Let $(\mathcal{H}_{\mathcal{K}}, \langle \cdot, \cdot \rangle_{\mathcal{H}_{\mathcal{K}}}, \|\cdot\|_{\mathcal{H}_{\mathcal{K}}})$ denote the RKHS generated by the scalar-valued kernel $\mathcal{K} : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$. Here, we say K is a scalar-valued kernel if it is a real, symmetric, and positive-definite bivariate function. A mapping $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{B}(\mathcal{Y})$ is called an operator-valued kernel (Schwartz, 1964; Micchelli and Pontil,

¹Here, “streaming data” refers to a setting in which samples arrive sequentially and are processed one at a time, so that the estimator can be updated online without needing repeated access to the full dataset.

2005) on $\mathcal{X}$ if:

- (1) For any $x, x' \in \mathcal{X}$, $K(x, x')$ is the adjoint operator of $K(x', x)$, i.e., $K(x, x')^* = K(x', x)$;
- (2) For any $n \in \mathbb{N}$, $\{x_i\}_{i=1}^n \subset \mathcal{X}$ and $\{y_i\}_{i=1}^n \subset \mathcal{Y}$, it holds that $\sum_{i=1}^n \langle K(x_i, x_j)y_i, y_j \rangle_{\mathcal{Y}} \geq 0$.

Note that the function $K(x, \cdot)y : \mathcal{X} \rightarrow \mathcal{Y}$ is well-defined for $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. The vector-valued RKHS $\mathcal{H}$ is the completion of the linear span of $\{K(x, \cdot)y : x \in \mathcal{X}, y \in \mathcal{Y}\}$ with inner product $\langle K(x, \cdot)y, K(x', \cdot)y' \rangle_{\mathcal{H}} = \langle K(x, x')y, y' \rangle_{\mathcal{Y}}$. Moreover, the reproducing property holds:

$$\langle K(x, \cdot)y, f \rangle_{\mathcal{H}} = \langle y, f(x) \rangle_{\mathcal{Y}}, \quad \forall (x, y) \in \mathcal{X} \times \mathcal{Y} \text{ and } f \in \mathcal{H}.$$

For more details on vector-valued RKHSs, see (Micchelli and Pontil, 2005; Carmeli et al., 2006, 2010). Furthermore, when $\mathcal{Y} = \mathbb{R}$, K reduces to a scalar-valued kernel.

The construction of operator-valued kernels plays an important role in our setting. A common choice is

$$K(x, x') = \mathcal{K}(x, x')W, \quad (1.2)$$

where $\mathcal{K}$ is a scalar-valued kernel and $W \in \mathcal{B}(\mathcal{Y})$ is a positive linear operator. In multi-task learning, W is typically a finite-dimensional matrix that facilitates information sharing among tasks (Evgeniou et al., 2005; Caponnetto et al., 2008). For some functional output learning problems, W is selected to be a multiplication or an integral operator (Kadri et al., 2010, 2011). Additionally, some works on functional regression (Lian, 2007) and structured output learning (Brouard et al., 2011; Ciliberto et al., 2016, 2020; Brogat-Motte et al., 2022) directly construct operator-valued kernels by setting W to be the identity operator. In (Kadri et al., 2012), the kernels are taken to be the finite combinations of operator-valued kernels. For other constructions, see (Kadri et al., 2016).

We briefly outline some algorithms for solving model (1.1). Kadri et al. (2016) address model (1.1) using spectral decomposition of block operator matrices, while Kadri et al. (2012) propose a block-coordinate descent method. To handle limited training data, Brogat-Motte et al. (2022) leverage the structure of the target output and propose a reduced-rank method to solve model (1.1). Note that all these existing algorithms follow the batch learning² paradigms. In this paper, we adopt a stochastic gradient descent (SGD) algorithm derived from the Tikhonov regularization scheme to solve the model (1.1), aiming to learn the nonlinear operators from streaming data. This algorithm is well-suited for real-time operator learning, enabling continuous adaptation without retaining historical data. This challenge has also been addressed in previous work on operator learning (Yeung et al., 2019; Hoi et al., 2021). The performance of the resulting estimator h can be evaluated using the prediction error $\mathcal{E}(h) - \mathcal{E}(h^\dagger) = \mathbb{E} \left[\|h(x) - h^\dagger(x)\|_{\mathcal{Y}}^2 \right]$ and the estimation error $\|h - h^\dagger\|_{\mathcal{H}}^2$, where $\mathcal{E}(h) := \mathbb{E}[\|h(x) - y\|_{\mathcal{Y}}^2]$. To illustrate our algorithm, we define the minimizer of the regularized least squares problem as

$$h_\lambda := \arg \min_{h \in \mathcal{H}} \{ \mathcal{E}(h) + \lambda \|h\|_{\mathcal{H}}^2 \}, \quad (1.3)$$

where $\lambda > 0$ is the regularization parameter. This paper focuses on two important settings of the SGD algorithm: one with constant step sizes and regularization parameters, and the other with decaying step sizes and regularization parameters. Hereinafter, we use $\mathbf{0}$ to denote the zero element in a Hilbert space.

The finite-horizon setting. In this setting, we assume access to finite i.i.d. samples $\{z_t = (x_t, y_t)\}_{t=1}^T$, where the sample size $T < \infty$ is known in advance. We aim to solve the regularized problem (1.3), where the parameter λ depends on T . The SGD algorithm proceeds by updating the current estimator h_t to h_{t+1} using a single sample at the t -th iterate with a constant step size and regularization parameter. Specifically, the iteration begins with $h_1 = \mathbf{0}$ and is recursively defined as

$$h_{t+1} = h_t - \eta_T (K(x_t, \cdot)(h_t(x_t) - y_t) + \lambda_T h_t), \quad t = 1, \dots, T, \quad (1.4)$$

²Here, “batch learning” refers to a setting in which the estimator is trained from a fixed dataset.

where the step size (learning rate) η_T and the regularization parameter λ_T are appropriately chosen based on the sample size T . The update in iteration (1.4) arises from a one-sample stochastic approximation of $2\mathbb{E}[K(x, \cdot)(h(x) - y)] + 2\lambda h$, which corresponds to the Fréchet derivative (Dunford and Schwartz, 1988) of $\mathcal{E}(h) + \lambda \|h\|_{\mathcal{H}}^2$. Implementing an efficient warm start can be non-trivial when new data points become available in the future.

The online setting. In this setting, the sample size T may be unknown in advance or even infinite, which is well-suited for scenarios that require real-time iterative updates. To accommodate this setting, we update the regularization parameter λ_t such that h_{t+1} follows the regularization path (Zhu and Liu, 2021) h_{λ_t} ,³ ensuring that $h_t - h_{\lambda_t} \rightarrow \mathbf{0}$ and $h_{\lambda_t} \rightarrow h^\dagger$ in the norm $\|\cdot\|_{\mathcal{H}}$ (or in the semi-norm associated with prediction error) as t increases. This leads to the following iterative scheme, initialized with $h_1 = \mathbf{0}$:

$$h_{t+1} = h_t - \eta_t (K(x_t, \cdot)(h_t(x_t) - y_t) + \lambda_t h_t), \quad t \geq 1. \quad (1.5)$$

Moreover, we let both η_t and λ_t decay polynomially with respect to t , enabling stability and convergence of the solution while adapting to streaming data and mitigating overfitting.

In this paper, we study both settings of the SGD algorithm. We express the iterative forms (1.4) and (1.5) in a unified manner as the form given in (1.5). For decaying step sizes adopted in the online setting, we set the step size as $\eta_t = \bar{\eta}(t + t_0)^{-\theta_1}$ for all $t \geq 1$, where $\theta_1 \in (0, 1)$, $\bar{\eta} > 0$, and $t_0 > 0$. The regularization parameter is defined as $\lambda_t = \bar{\lambda}(t + t_0)^{-\theta_2}$, where $\theta_2 \in (0, 1)$, $\bar{\lambda} > 0$. We emphasize that in the online setting, both $\bar{\eta}$ and $\bar{\lambda}$ are constants independent of t and the total number of iterations (e.g., the sample size) T . For the constant step sizes and regularization parameters adopted in the finite-horizon setting, we set $\eta_t = \eta_1 T^{-\theta_3}$ and $\lambda_t = \lambda_1 T^{-\theta_4}$ for $t = 1, 2, \dots, T$, where $\theta_3 \in (0, 1)$, $\theta_4 > 0$, and $\eta_1, \lambda_1 > 0$. In this finite-horizon setting, the step sizes and regularization parameters explicitly depend on the total number of iterations T .

Throughout the paper, we impose the following assumption on the operator-valued kernels.

Assumption 1. *The vector-valued RKHS $\mathcal{H}$ is generated by the operator-valued kernel $K(x, x') = \mathcal{K}(x, x')I$, where $\mathcal{K}$ is the scalar-valued kernel with $\|\mathcal{K}\|_\infty := \sup_{x \in \mathcal{X}} \mathcal{K}(x, x) \leq \kappa^2$ for some constant $\kappa > 0$, and I is the identity operator on $\mathcal{Y}$.*

This simple construction of operator-valued kernels has been adopted in previous works, e.g., (Brogat-Motte et al., 2022; Battle et al., 2024). Note that all elements h in the vector-valued RKHS $\mathcal{H}$ are measurable. Furthermore, when choosing kernels as in (1.2), it follows from (Carmeli et al., 2010, Example 5) that K is a Mercer [resp. $\mathcal{C}_0$]⁴ kernel if $\mathcal{K}$ is Mercer [resp. $\mathcal{C}_0$], implying that all operators in $\mathcal{H}$ are continuous. This choice of kernel, $K(x, x') = \mathcal{K}(x, x')I$, provides a mathematically tractable and commonly adopted setting in which the vector-valued RKHS structure allows for a complete convergence analysis of the regularized SGD algorithm in infinite-dimensional operator learning. In particular, each output component is associated with the same scalar kernel $\mathcal{K}$, which facilitates the analysis while preserving the infinite-dimensional nature of the operator-valued RKHS.

At the same time, this construction does not explicitly encode correlations between different output directions. More general operator-valued kernels, such as those incorporating a compact operator W , can, in principle, capture such dependencies. We discuss these generalizations in Appendix E, though a complete convergence analysis in these settings is beyond the scope of the present work. This discussion emphasizes both the practical relevance and the inherent limitations of the identity-operator construction used throughout the main text.

The two types of step sizes considered in this paper have been extensively studied in the previous literature on SGD in various settings. The seminal work (Smale and Yao, 2006) shows that the step size serves as an implicit form of regularization, thus improving the algorithm’s generalization and

³Regularization path refers to the trajectory of solutions h_λ as the regularization parameter λ varies, characterizing how the learned model evolves under different levels of regularization.

⁴That is, the Banach space of continuous functions vanishing at infinity with the uniform norm.

robustness. Our recent work (Shi and Yang, 2024) investigates operator learning via the SGD algorithm between Hilbert spaces, deriving bounds for both prediction and estimation errors in expectation. However, the SGD algorithm in (Shi and Yang, 2024) does not incorporate the regularization term. On the other hand, in the context of operator learning, research on the almost-sure convergence of SGD algorithms is still scarce. Only a few works, including (Tarres and Yao, 2014; Berthier et al., 2020; Varre et al., 2021), have considered the almost-sure convergence in finite-dimensional output settings but either assume a noise-free scenario or provide convergence results that do not directly extend to operator learning problems. This clearly identifies a significant gap in the existing literature. Consequently, while advancements have occurred in finite-dimensional settings, substantial challenges related to operator learning and the fundamental role of regularization remain largely unexplored.

This paper aims to fill this gap by rigorously analyzing the regularized SGD algorithm applied to the nonlinear operator regression problem described in (1.1). Our main contributions are summarized as follows: First, we introduce specific regularity assumptions on $h^\dagger$ (or its associated Hilbert-Schmidt operator, as defined in Proposition 2.1), which effectively capture the intrinsic features of infinite-dimensional regression problems. Under these assumptions, we derive bounds in expectation for the prediction and estimation errors of the regularized SGD algorithm, showing improvements compared to the unregularized SGD algorithm studied in (Shi and Yang, 2024). Second, we propose a novel technique for establishing high-probability bounds, ensuring almost-sure convergence via the Borel–Cantelli lemma. High-probability convergence provides a stronger guarantee than expectation-based bounds alone, moving beyond average-case performance. Crucially, we demonstrate that the introduction and careful tuning of regularization parameters are essential not only for achieving these high-probability bounds but also for significantly enhancing the convergence behaviors of the SGD algorithm. This underscores the superiority of our regularized approach over the unregularized framework considered in (Shi and Yang, 2024), highlighting the necessity of regularization for robust probabilistic guarantees. Lastly, the resulting convergence rates are demonstrated to be near-optimal, aligning closely with the minimax lower bounds established in (Shi and Yang, 2024), thus reinforcing the theoretical soundness and effectiveness of our proposed algorithm.

The rest of the paper is organized as follows. Section 2 introduces the main theoretical results and assumptions, and briefly discusses the key error decomposition and the relation to the unregularized analysis. Section 3 provides further discussion of the framework and its broader context. In Section 4, we illustrate the practical scope of our framework through applications to structured prediction and parametric PDEs. Section 5 presents a brief numerical experiment. Section A (located in the supplementary material) performs an error decomposition tailored to the regularized SGD algorithm. Building on this, Section B provides essential intermediate estimates used in the subsequent analysis. Sections C and D are devoted to establishing bounds on the prediction and estimation errors, first in expectation and then with high probability. For clarity and conciseness, some technical proofs are presented in the appendix. Appendix E discusses more general operator-valued kernels, while Appendix F addresses the connection with the PCA encoder-decoder framework.

2 Main Results

This section introduces regularity conditions on the structure and smoothness of the target operator and the input random variables. We then present our main theorems. We begin with some notations for further statements. Denote $\mathbb{N}_T$ as the set $\{1, 2, \dots, T\}$. The rank-one operator $f \otimes g \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$ is defined by $f \otimes g(g') := \langle g, g' \rangle_{\mathcal{H}_1} f$, where $g, g' \in \mathcal{H}_1$ and $f \in \mathcal{H}_2$. We denote $\text{Tr}(A)$ as the trace of a self-adjoint and compact operator $A \in \mathcal{B}(\mathcal{H}_1)$. Let $\mathbb{E}$ and $\mathbb{E}_{z_t}$ denote the expectation with respect to the distribution ρ and the sample $z_t := (x_t, y_t)$, respectively. For $k \in \mathbb{N}_T$, let $\mathbb{E}_{z_1, \dots, z_k}$ denote the expectation with respect to $\{z_i\}_{i=1}^k$, abbreviated as $\mathbb{E}_{z^k}$. Recall that $\mathcal{K}$ is the scalar-valued kernel. Since $\mathcal{X}$ is separable, the RKHS $\mathcal{H}_{\mathcal{K}}$ induced by $\mathcal{K}$ is also separable. The operator $C = \mathbb{E}[\phi(x) \otimes \phi(x)]$, defined by $\phi(x) := \mathcal{K}(x, \cdot) \in \mathcal{H}_{\mathcal{K}}$, is self-adjoint, compact, and satisfies $\|C\| \leq \|C\|_{\text{HS}} \leq \kappa^2$. Thus, for any $r > 0$, the operator C^r is also self-adjoint and compact. Moreover, it is straightforward to verify

that

$$\|C^{1/2}\|_{\text{HS}}^2 = \text{Tr}(C) = \mathbb{E}[\|\phi(x)\|_{\mathcal{H}_{\mathcal{K}}}^2] \leq \kappa^2.$$

With the aid of the following proposition, the iterative process in the vector-valued RKHS $\mathcal{H}$ can be equivalently reformulated as an iterative process in $\mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$.

Proposition 2.1. *The vector-valued RKHS $\mathcal{H}$, associated with the operator-valued kernel $K(x, x') = \mathcal{K}(x, x')W$, where W is a positive operator and $\mathcal{K}$ is a scalar-valued kernel, is isometrically isomorphic to $\mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \overline{W^{1/2}\mathcal{Y}}) \subset \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$. Specifically, for each $h \in \mathcal{H}$, there exists a unique $H \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \overline{W^{1/2}\mathcal{Y}})$ such that*

$$h(x) = W^{1/2}H\phi(x), \quad \forall x \in \mathcal{X},$$

and $\|h\|_{\mathcal{H}} = \|H\|_{\text{HS}}$.

The proof of Proposition 2.1 is deferred to Appendix A. When Proposition 2.1 is applied with $W = I$, the kernel coincides with that in Assumption 1, and the iteration (1.5) can be equivalently expressed as

$$\begin{cases} H_1 = \mathbf{0}, \\ H_{t+1} = H_t - \eta_t((H_t\phi(x_t) - y_t) \otimes \phi(x_t) + \lambda_t H_t), \\ h_t(\cdot) = H_t(\phi(\cdot)). \end{cases} \quad (2.1)$$

Hereinafter, we assume that $h^\dagger(x) = H^\dagger\phi(x)$, where $H^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$ is a Hilbert-Schmidt operator. Under this assumption, the nonlinear operator learning model (1.1) reduces to an infinite-dimensional linear model:

$$y = H^\dagger\phi(x) + \epsilon, \quad (2.2)$$

where the input and output are $\phi(x)$ and y , respectively. For $H \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$, let $\mathcal{E}(H) = \mathbb{E}[\|y - H\phi(x)\|_{\mathcal{Y}}^2]$, where $h(x) = H\phi(x)$, so that $\mathcal{E}(H) = \mathcal{E}(h)$. The prediction error is defined as $\mathcal{E}(H) - \mathcal{E}(H^\dagger) = \mathcal{E}(h) - \mathcal{E}(h^\dagger)$, and the estimation error is defined as $\mathbb{E}[\|H - H^\dagger\|_{\text{HS}}^2]$. According to Proposition 2.1, we have $\mathbb{E}[\|h - h^\dagger\|_{\mathcal{H}}^2] = \mathbb{E}[\|H - H^\dagger\|_{\text{HS}}^2]$, which implies that the prediction error and estimation error for the estimator $h(\cdot) = H\phi(\cdot)$ in the original model (1.1) coincide with those of H in the linearized model (1.1). Therefore, it suffices to analyze the convergence rates of the errors of H_t associated with the SGD iteration (2.1) in $\mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$. Although this is not directly required for our theoretical analysis, we emphasize for clarity that the iterative form of SGD derived from minimizing the regularized objective functional $\mathcal{E}(H) + \lambda\|H\|_{\text{HS}}^2$ corresponds exactly to the iteration given in (2.1), which is equivalent to (1.5).

2.1 Assumptions

To conduct the convergence analysis, we need the following assumptions.

Assumption 2 (Regularity condition of $H^\dagger$). *There exists a Hilbert-Schmidt operator $S^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$ and a positive parameter $r > 0$, such that:*

$$H^\dagger = S^\dagger C^r.$$

This assumption, introduced in (Shi and Yang, 2024), characterizes the regularity of the target operator $H^\dagger$ via its relation to the operator C . The parameter r serves as a smoothness index—larger values of r indicate higher regularity of $H^\dagger$. In the special case where $\mathcal{Y} = \mathbb{R}$, the Riesz representation theorem implies that $H^\dagger$ corresponds to an element $g^\dagger = C^r g$ in $\mathcal{H}_{\mathcal{K}}$ for some $g \in \mathcal{H}_{\mathcal{K}}$. This is exactly the regularity condition widely adopted in the convergence analysis of non-parametric regression in real-valued RKHSs (Ying and Pontil, 2008; Dieuleveut and Bach, 2016; Berthier et al., 2020; Guo et al., 2023).

Assumption 3 (Spectral decay condition of C). *There exists $s \in (0, 1]$ such that:*

$$\text{Tr}(C^s) < +\infty.$$

This condition is automatically satisfied for any $s \geq 1$ (as $\text{Tr}(C) < \kappa^2$), and it imposes constraints on the decay rate of the eigenvalues of the operator C . Let $\{a_k\}_{k \geq 1}$ denote the non-increasing sequence of eigenvalues of C . Under this condition, the eigenvalues exhibit polynomial decay, specifically satisfying

$$a_k \leq \text{Tr}(C^s)^{\frac{1}{s}} k^{-\frac{1}{s}}.$$

A sufficient (though not necessary) condition for this assumption is that $a_k = O(k^{-\frac{1}{s}-\epsilon})$ for some $\epsilon > 0$. For a detailed discussion on this condition, we refer the reader to (Guo et al., 2023). In this paper, Assumption 3 with some $0 < s < 1$ is introduced to derive sharper error bounds. Together with Assumption 2, Assumption 3 leads to improved convergence rates. This condition, commonly known as the capacity condition, was first introduced in (Dieuleveut et al., 2017) and has since been widely adopted in the literature, including (Pillaud-Vivien et al., 2018; Brogat-Motte et al., 2022; Guo et al., 2023; Shi and Yang, 2024), as a way to capture the intrinsic complexity of infinite-dimensional learning problems. Assumptions 2 and 3 are essential to establish dimension-free convergence analysis. As we will show, the resulting convergence rates depend explicitly on the parameters r and s , reflecting the regularity of the target operator and the capacity of the input random variables, respectively.

The following assumption is only required for establishing error bounds in expectation. Recall that $\phi(x) := \mathcal{K}(x, \cdot) \in \mathcal{H}_{\mathcal{K}}$ for some scalar-valued kernel $\mathcal{K}$.

Assumption 4 (Moment condition of $\phi(x)$). *There exists a constant $c > 0$ such that for any compact linear operator $A \in \mathcal{B}(\mathcal{H}_{\mathcal{K}})$,*

$$\mathbb{E} \left[\|A\phi(x)\|_{\mathcal{H}_{\mathcal{K}}}^4 \right] \leq c \left(\mathbb{E} \left[\|A\phi(x)\|_{\mathcal{H}_{\mathcal{K}}}^2 \right] \right)^2.$$

According to (Shi and Yang, 2024, Proposition 2.1), this assumption is equivalent to

$$\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_{\mathcal{K}}}^4 \right] \leq c \left(\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \right] \right)^2, \quad \forall f \in \mathcal{H}_{\mathcal{K}}. \quad (2.3)$$

Condition (2.3) holds, for example, when $\phi(x)$ is strictly sub-Gaussian, implying that all linear functionals of $\phi(x)$ have bounded kurtosis. Similar assumptions have been adopted in several papers (Yuan and Cai, 2010; Cai and Yuan, 2012; Guo et al., 2023; Shi and Yang, 2024). To further deepen our understanding, we now present a novel characterization of Assumption 4, which is analogous to the idea discussed in (Liu and Shi, 2024).

Proposition 2.2. *Consider the principal component decomposition of $\phi(x)$:*

$$\phi(x) = \bar{\phi} + \sum_{k \geq 1} \sqrt{\lambda_k} \xi_k \phi_k, \quad (2.4)$$

where $\bar{\phi} := \mathbb{E}[\phi(x)]$, and $\{(\lambda_k, \phi_k)\}_{k \geq 1}$ are the eigenvalue-eigenvector pairs of the covariance operator $\Sigma := \mathbb{E}[(\phi(x) - \bar{\phi}) \otimes (\phi(x) - \bar{\phi})]$. The sequence $\{\xi_k\}_{k \geq 1}$ consists of zero-mean, uncorrelated real-valued random variables with $\mathbb{E}[\xi_k^2] = 1$. If, in addition, $\{\xi_k\}_{k \geq 1}$ are independent, then Assumption 4 (or equivalently, (2.3)) holds provided that $\{\mathbb{E}[\xi_k^4]\}_{k \geq 1}$ are uniformly bounded. That is, there exists a constant $C > 0$ such that

$$\mathbb{E}[\xi_k^4] \leq C, \quad \forall k \geq 1.$$

The proof of the above proposition is presented in Appendix B. The next assumption is used to derive high-probability error bounds.

Assumption 5 (Boundedness condition of y). *There exists some constant $M_\rho > 0$ such that*

$$\|y\|_{\mathcal{Y}} \leq M_\rho$$

almost surely.

Assumption 5 applies, in particular, to the surrogate approach to structured prediction (Brouard et al., 2016b; Ciliberto et al., 2016; Korba et al., 2018; Ciliberto et al., 2020; Brogat-Motte et al., 2022), see Subsection 4.1. It is also relevant to PDE operator-learning settings, including benchmark examples such as Darcy-type flow and Navier–Stokes equations (Kovachki et al., 2021; Lanthaler et al., 2022; Lanthaler, 2023), provided that the corresponding solutions satisfy suitable uniform a priori bounds.

2.2 Error Bounds in Expectation

In this subsection, we assume that Assumption 1 holds, Assumption 2 holds with $S^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_K, \mathcal{Y})$ and $r > 0$, Assumption 3 holds with $0 < s \leq 1$, and Assumption 4 holds with $c > 0$. Theorem 2.3 and Theorem 2.4 provide the convergence rates of prediction error and estimation error in expectation for the online setting. In contrast, Theorem 2.5 and 2.6 focus on the finite-horizon setting.

For notational simplicity, we specify here the parameter dependence of the constants appearing in the following theorems. The constants $c_{1,1}$ and $c_{1,2}$ depend only on $\bar{\eta}$, $\bar{\lambda}$, θ_1 , t_0 , r , s , $\|S^\dagger\|_{\text{HS}}$, σ^2 , and c ; the constants $c_{1,3}$ and $c_{1,4}$ depend only on η_1 , λ_1 , θ_3 , θ_4 , r , s , $\|S^\dagger\|_{\text{HS}}$, σ^2 , and c .

Theorem 2.3. *Suppose that Assumption 1, Assumption 2, Assumption 3 and Assumption 4 are satisfied. Define $\{h_t\}_{t \geq 1}$ through (2.1) with step sizes $\{\eta_t = \bar{\eta}(t + t_0)^{-\theta_1}\}_{t \geq 1}$ and regularization parameters $\{\lambda_t = \bar{\lambda}(t + t_0)^{-\theta_2}\}_{t \geq 1}$, where $0 < \theta_1 < 1$, $0 < \theta_2 < 1$ and $\bar{\eta}\bar{\lambda} > \theta_2 \min\{r, 1\}$. Additionally, let t_0 satisfy $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$, $t_0 \geq \exp\{\frac{1}{\theta_1}\}$, and*

$$c_4 \sqrt{c} t_0^{-\theta_1} \log t_0 < 1,$$

where $c_4 = c_4(\bar{\eta}, \bar{\lambda}, \theta_1, s, c)$ is a constant independent of t_0 , as specified in Proposition B.9. Choose $\theta_1 = \min\left\{\frac{2r+1}{2r+2}, \frac{2}{3}\right\}$ and $\theta_2 = 1 - \theta_1$. Then for any $T \geq 1$,

$$\mathbb{E}_{z^T} [\mathcal{E}(h_{T+1}) - \mathcal{E}(h^\dagger)] \leq c_{1,1} \begin{cases} (T + t_0)^{-\theta_1}, & \text{when } s < 1, \\ (T + t_0)^{-\theta_1} \log(T + t_0), & \text{when } s = 1. \end{cases}$$

Here the constant $c_{1,1}$ is independent of T .

Remark 1. *In the theorem above, we set $\theta_1 + \theta_2 = 1$, as this choice leads to the most favorable convergence rates achievable within our framework. The condition on t_0 is necessary for the proof, while the constraint on $\bar{\eta}\bar{\lambda}$ serves to accelerate convergence. When $\theta_1 + \theta_2 \neq 1$, the resulting rates are slower. In such cases, one can set $t_0 = 0$ and choose a small $\bar{\eta}\bar{\lambda}$; the corresponding analysis is similar and more straightforward, so we omit it for brevity.*

In Theorem 2.3, since the constant c_4 is independent of t_0 , one can choose t_0 sufficiently large to satisfy the required conditions. Compared to Assumption 3 with $s = 1$, the stronger assumption with $0 < s < 1$ only removes a logarithmic factor in the convergence rate. It is also clear that the convergence rate saturates at $r = 1/2$, i.e., increasing r beyond $1/2$ does not yield further improvement. This saturation phenomenon under increasing source regularity is not specific to the present setting and has also been observed in related kernel-learning literature; see, for example, (Li et al., 2024a). According to (Shi and Yang, 2024, Theorem 2.9), the result is minimax optimal (up to a logarithmic term) when $s = 1$ and $r < 1/2$. Compared to the unregularized SGD algorithm analyzed in (Shi and Yang, 2024, Theorem 2.4), adding a regularization term here leads to faster convergence. Specifically, while the prediction error rate of unregularized SGD in (Shi and Yang, 2024) saturates at $r = (1 - s)/2$, the regularized SGD in our work improves the saturation level to $r = 1/2$.

The following theorem provides the convergence rate for the estimation error.

Theorem 2.4. *Under the conditions of Theorem 2.3, choose $\theta_1 = \min \left\{ \frac{s+2r}{1+s+2r}, \frac{2+s}{3+s} \right\}$ and $\theta_2 = 1 - \theta_1$. Then for any $T \geq 1$,*

$$\mathbb{E}_{z^T} \left[\|h_{T+1} - h^\dagger\|_{\mathcal{H}}^2 \right] \leq c_{1,2} (T + t_0)^{-\min \left\{ \frac{2r}{1+s+2r}, \frac{2}{3+s} \right\}}.$$

Here the constant $c_{1,2}$ is independent of T .

The convergence rate of the estimation error saturates at $r = 1$, which improves the convergence of unregularized SGD in (Shi and Yang, 2024, Theorem 2.6), where the rate saturates at $r = \frac{1-s}{2}$. Moreover, in (Guo et al., 2023, Theorem 3) and (Shi and Yang, 2024, Theorem 2.6), we cannot guarantee the convergence of the estimation error with decaying step sizes for $s = 1$, whereas adding a regularization term addresses this issue. According to (Shi and Yang, 2024, Theorem 2.9), the convergence rate with decaying step sizes is minimax optimal when $r < 1$.

Next, we present the convergence rates for prediction and estimation errors with constant step sizes and regularization parameters, where both depend on the total number of iterations T (i.e., the total sample size).

Theorem 2.5. *Suppose that Assumption 1, Assumption 2, Assumption 3 and Assumption 4 are satisfied. Define $\{h_t\}_{t \in \mathbb{N}_T}$ through (2.1) with step sizes $\{\eta_t = \eta_1 T^{-\theta_3}\}_{t \in \mathbb{N}_T}$ and regularization parameters $\{\lambda_t = \lambda_1 T^{-\theta_4}\}_{t \in \mathbb{N}_T}$, where $T \geq 2$, $\eta_1(\kappa^2 + \lambda_1) \leq 1$, and*

$$\eta_1 < \frac{1}{6c\kappa^2 \left(1 + \frac{1}{2e\theta_3}\right)}.$$

Choose $\theta_3 = \frac{2r+1}{2r+2}$ and $\theta_4 \geq \frac{2r+1}{(2r+2) \min\{2r+1, 2\}}$. Then

$$\mathbb{E}_{z^T} [\mathcal{E}(h_{T+1}) - \mathcal{E}(h^\dagger)] \leq c_{1,3} \begin{cases} T^{-\frac{2r+1}{2r+2}}, & \text{when } s < 1, \\ T^{-\frac{2r+1}{2r+2}} \log T, & \text{when } s = 1. \end{cases}$$

Here the constant $c_{1,3}$ is independent of T .

Theorem 2.6. *Under the conditions of Theorem 2.5, choose*

$$\theta_3 = \frac{2r+s}{1+2r+s} \quad \text{and} \quad \theta_4 \geq \frac{2r}{(1+2r+s) \min\{2r, 2\}}.$$

Then

$$\mathbb{E}_{z^T} \left[\|h_{T+1} - h^\dagger\|_{\mathcal{H}}^2 \right] \leq c_{1,4} T^{-\frac{2r}{1+2r+s}}.$$

Here the constant $c_{1,4}$ is independent of T .

In the case of constant step sizes and regularization parameters, the prediction error achieves the minimax optimal rate when $s = 1$, and the estimation error performs so for any $s > 0$, as established by the minimax lower bounds in (Shi and Yang, 2024). Unlike the scenario with decaying step sizes and regularization parameters, no saturation occurs when these parameters are held constant. It is also noteworthy that the convergence rates and the choice of θ_3 in the above two theorems align with those in Theorems 2.5 and 2.7 of (Shi and Yang, 2024), which analyze the unregularized SGD algorithm with constant step sizes. This contrasts with the case of decaying step size, where adding a regularization term leads to improved rates. When employing constant step sizes, introducing regularization does not improve the convergence rates; in fact, an improperly chosen θ_4 (not sufficiently large) may degrade performance. The unregularized SGD can be viewed as the limiting case corresponding to $\theta_4 = \infty$.

2.3 High-probability Error Bounds

In this subsection, we assume Assumption 1 holds, Assumption 2 holds with $S^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$ and $r > 0$, Assumption 3 holds with $0 < s \leq 1$, and Assumption 5 holds with $M_\rho > 0$. We derive high-probability error bounds for both the prediction and estimation errors in both the online and finite-horizon settings. These error bounds guarantee almost-sure convergence of the regularized SGD algorithm, providing a stronger guarantee than bounds in expectation, as convergence is ensured with high probability across all realizations. The notation $a \lesssim b$ denotes $a \leq Cb$ for some constant C independent of t , T , and the confidence level δ .

For notational simplicity, we specify here the parameter dependence of the constants appearing in the following theorems and corollaries. The constants $c_{2,1}$, $\tilde{c}_{2,1}$, $c_{2,2}$, and $\tilde{c}_{2,2}$ depend only on $\bar{\eta}$, $\bar{\lambda}$, θ_1 , t_0 , r , s , $\|S^\dagger\|_{\text{HS}}$, and M_ρ ; the constants $c_{2,3}$ and $c_{2,4}$ depend only on η_1 , λ_1 , θ_3 , θ_4 , r , s , $\|S^\dagger\|_{\text{HS}}$, and M_ρ .

The following theorem establishes the prediction error bounds in the online setting.

Theorem 2.7. *Suppose that Assumption 1, Assumption 2, Assumption 3 and Assumption 5 are satisfied. Define $\{h_t\}_{t \geq 1}$ through (2.1) with step sizes $\{\eta_t = \bar{\eta}(t+t_0)^{-\theta_1}\}_{t \geq 1}$ and regularization parameters $\{\lambda_t = \bar{\lambda}(t+t_0)^{-\theta_2}\}_{t \geq 1}$, where $\bar{\eta}\bar{\lambda} > \max\{\theta_2 \min\{r, 1\}, \theta_1, 2\theta_1 - \frac{1}{2}\}$ and $(t_0+1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$. Choose*

$$\theta_1 = \begin{cases} \frac{2r+1}{2r+2}, & \text{when } r < \frac{1}{2}, \\ \frac{2}{3}, & \text{when } r \geq \frac{1}{2}, \end{cases}$$

and $\theta_2 = 1 - \theta_1$. Then for any $T \geq 1$ and $\delta \in (0, 2/e)$, with probability at least $1 - 2\delta$, the following holds:

(1) If $s < 1$,

$$\begin{aligned} \mathcal{E}(h_{T+1}) - \mathcal{E}(h^\dagger) &\leq c_{2,1} \left((T+t_0)^{-\theta_1} + (T+t_0)^{1-3\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \\ &\lesssim (T+t_0)^{-\theta_1} \log^4 \frac{2}{\delta}. \end{aligned}$$

(2) If $s = 1$,

$$\begin{aligned} \mathcal{E}(h_{T+1}) - \mathcal{E}(h^\dagger) &\leq c_{2,1} \left((T+t_0)^{-\theta_1} + (T+t_0)^{1-3\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta} \right) \log(T+t_0) \log^2 \frac{2}{\delta} \\ &\lesssim (T+t_0)^{-\theta_1} \log(T+t_0) \log^4 \frac{2}{\delta}. \end{aligned}$$

Here the constant $c_{2,1}$ is independent of T and δ .

The following corollary, as a natural extension of Theorem 2.7, establishes a uniform high-probability bound that holds simultaneously for all $t \geq 1$.

Corollary 2.8. *Under the conditions of Theorem 2.7, choose*

$$\theta_1 = \begin{cases} \frac{2r+1}{2r+2}, & \text{when } r < \frac{1}{2}, \\ \frac{2}{3}, & \text{when } r \geq \frac{1}{2}, \end{cases}$$

and $\theta_2 = 1 - \theta_1$. Then, with probability at least $1 - 2\delta$, for all $1 \leq t < \infty$, the following holds:

$$\mathcal{E}(h_{t+1}) - \mathcal{E}(h^\dagger) \leq \tilde{c}_{2,1} \begin{cases} (t+t_0)^{-\theta_1} \log^4(t+t_0) \log^4 \frac{2}{\delta}, & \text{when } s < 1, \\ (t+t_0)^{-\theta_1} \log^5(t+t_0) \log^4 \frac{2}{\delta}, & \text{when } s = 1. \end{cases}$$

Here the constant $\tilde{c}_{2,1}$ is independent of t and δ .

In Theorem 2.9 and Corollary 2.10, we focus on the estimation error in the online setting.

Theorem 2.9. *Under the conditions of Theorem 2.7, choose*

$$\theta_1 = \begin{cases} \frac{1+2r+s}{3+2r+s}, & \text{when } r < \frac{1-s}{2}, \\ \frac{2\min\{r,1\}+s}{1+2\min\{r,1\}+s}, & \text{when } r \geq \frac{1-s}{2}, \end{cases}$$

and $\theta_2 = 1 - \theta_1$. Then for any $T \geq 1$ and $\delta \in (0, 2/e)$, with at least $1 - 2\delta$ probability, the following holds:

(1) If $r < \frac{1-s}{2}$,

$$\|h_{T+1} - h^\dagger\|_{\mathcal{H}}^2 \leq c_{2,2}(T+t_0)^{-\frac{4r}{3+2r+s}} \log^2(T+t_0) \log^4 \frac{2}{\delta}.$$

(2) If $r \geq \frac{1-s}{2}$,

$$\begin{aligned} \|h_{T+1} - h^\dagger\|_{\mathcal{H}}^2 &\leq c_{2,2} \left((T+t_0)^{-\frac{2\min\{r,1\}}{1+2\min\{r,1\}+s}} + (T+t_0)^{-\frac{4\min\{r,1\}+s-1}{1+2\min\{r,1\}+s}} \log^2(T+t_0) \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \\ &\lesssim (T+t_0)^{-\frac{2\min\{r,1\}}{1+2\min\{r,1\}+s}} \log^4 \frac{2}{\delta}. \end{aligned}$$

Here the constant $c_{2,2}$ is independent of T and δ .

Corollary 2.10. *Under the conditions of Theorem 2.7, choose θ_1 and θ_2 as in Theorem 2.9. Then, for any $\delta \in (0, 2/e)$, with probability at least $1 - 2\delta$, for all $1 \leq t < \infty$, the following holds:*

$$\|h_{t+1} - h^\dagger\|_{\mathcal{H}}^2 \leq \begin{cases} \tilde{c}_{2,2}(t+t_0)^{-\frac{4r}{3+2r+s}} \log^6(t+t_0) \log^4 \frac{2}{\delta}, & \text{when } r < \frac{1-s}{2}, \\ \tilde{c}_{2,2}(t+t_0)^{-\frac{2\min\{r,1\}}{1+2\min\{r,1\}+s}} \log^4(t+t_0) \log^4 \frac{2}{\delta}, & \text{when } r \geq \frac{1-s}{2}. \end{cases}$$

Here the constant $\tilde{c}_{2,2}$ is independent of t and δ .

The following two theorems provide high-probability convergence rates for the prediction and estimation errors, respectively, in the finite-horizon setting.

Theorem 2.11. *Suppose that Assumption 1, Assumption 2, Assumption 3 and Assumption 5 are satisfied. Define $\{h_t\}_{t \in \mathbb{N}_T}$ through (2.1) with step sizes $\{\eta_t = \eta_1 T^{-\theta_3}\}_{t \in \mathbb{N}_T}$ and regularization parameters $\{\lambda_t = \lambda_1 T^{-\theta_4}\}_{t \in \mathbb{N}_T}$, where $T \geq 2$ and $\eta_1(\kappa^2 + \lambda_1) \leq 1$. Choose $\theta_3 = \frac{2r+1}{2r+2}$ and $\theta_4 \geq \frac{2r+1}{(2r+2)\min\{2r+1,2\}}$. Then, for any $\delta \in (0, 2/e)$, with probability at least $1 - 2\delta$,*

$$\begin{aligned} \mathcal{E}(h_{T+1}) - \mathcal{E}(h^\dagger) &\leq c_{2,3} \begin{cases} T^{-\theta_3} \log^2 \frac{2}{\delta} + T^{1-3\theta_3} \log^2 T \log^4 \frac{2}{\delta}, & \text{when } s < 1, \\ T^{-\theta_3} \log T \log^2 \frac{2}{\delta} + T^{1-3\theta_3} \log^3 T \log^4 \frac{2}{\delta}, & \text{when } s = 1, \end{cases} \\ &\lesssim \log^4 \frac{2}{\delta} \begin{cases} T^{-\theta_3}, & \text{when } s < 1, \\ T^{-\theta_3} \log T, & \text{when } s = 1. \end{cases} \end{aligned}$$

Here the constant $c_{2,3}$ is independent of T and δ .

Theorem 2.12. *Under the conditions of Theorem 2.11, choose*

$$\theta_3 = \begin{cases} \frac{1+2r+s}{3+2r+s}, & \text{when } r < \frac{1-s}{2}, \\ \frac{2r+s}{1+2r+s}, & \text{when } r \geq \frac{1-s}{2}, \end{cases}$$

and

$$\theta_4 \geq \begin{cases} \frac{2r}{(3+2r+s)r}, & \text{when } r < \frac{1-s}{2}, \\ \frac{r}{(1+2r+s)\min\{r,1\}}, & \text{when } r \geq \frac{1-s}{2}. \end{cases}$$

Then, for any $\delta \in (0, 2/e)$, with probability at least $1 - 2\delta$, the following holds:

(1) If $r < \frac{1-s}{2}$,

$$\begin{aligned}\|h_{T+1} - h^\dagger\|_{\mathcal{H}}^2 &\leq c_{2,4} \left(T^{-\frac{1+2r-s}{3+2r+s}} + T^{-\frac{4r}{3+2r+s}} \log^2 T \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \\ &\lesssim T^{-\frac{4r}{3+2r+s}} \log^2 T \log^4 \frac{2}{\delta}.\end{aligned}$$

(2) If $r \geq \frac{1-s}{2}$,

$$\begin{aligned}\|h_{T+1} - h^\dagger\|_{\mathcal{H}}^2 &\leq c_{2,4} \left(T^{-\frac{2r}{1+2r+s}} + T^{-\frac{4r+s-1}{1+2r+s}} \log^2 T \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \\ &\lesssim T^{-\frac{2r}{1+2r+s}} \log^4 \frac{2}{\delta}.\end{aligned}$$

Here the constant $c_{2,4}$ is independent of T and δ .

Building upon the results from the previous theorems, we note that Theorem 2.11 for $s = 1$ and Theorem 2.12 for $r \geq \frac{1-s}{2}$ achieve the minimax lower bound derived in Theorem 2.9 of (Shi and Yang, 2024), up to a logarithmic factor.

A key decomposition. To help the reader better understand the proof strategy, we briefly indicate the common quantity underlying our error analysis, namely

$$\|(H_{T+1} - H^\dagger) C^\alpha\|_{\text{HS}}^2, \quad 0 \leq \alpha \leq \frac{1}{2}.$$

By the identity

$$\mathcal{E}(H) - \mathcal{E}(H^\dagger) = \|(H - H^\dagger) C^{1/2}\|_{\text{HS}}^2,$$

the choice $\alpha = \frac{1}{2}$ corresponds to the prediction error, while $\alpha = 0$ corresponds to the estimation error. The full details are given in Supplementary Material, Section A; here we only record the corresponding decompositions and refer the reader there for the precise definitions of the auxiliary quantities.

In the expectation analysis, we derive

$$\mathbb{E}_{z^T} \left[\|(H_{T+1} - H^\dagger) C^\alpha\|_{\text{HS}}^2 \right] \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4,$$

where

$$\begin{aligned}\mathcal{T}_1 &:= 2 \|(H_{\lambda_T} - H^\dagger) C^\alpha\|_{\text{HS}}^2, \\ \mathcal{T}_2 &:= 6 \left\| H_{\lambda_0} C^\alpha \prod_{t=1}^T (I - \eta_t (C + \lambda_t I)) \right\|_{\text{HS}}^2, \\ \mathcal{T}_3 &:= 6 \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) C^\alpha \prod_{j=t}^T (I - \eta_j (C + \lambda_j I)) \right\|_{\text{HS}}^2, \\ \mathcal{T}_4 &:= 6 \sqrt{c} \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{z^{t-1}} \|(H_t - H^\dagger) \phi(x_t)\|_{\mathcal{Y}}^2 + \sigma^2 \right) \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j (C + \lambda_j I))^2 \right).\end{aligned}$$

Here $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{T}_3$, and $\mathcal{T}_4$ are referred to as the approximation term, the initial term, the drift term, and the sample term, respectively. In the finite-horizon setting, the drift term $\mathcal{T}_3$ vanishes since the regularization parameter is constant.

For the high-probability analysis, the corresponding decomposition takes the form

$$\|(H_{T+1} - H^\dagger) C^\alpha\|_{\text{HS}}^2 \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + 6 \left\| \sum_{t=1}^T \chi_t \right\|_{\text{HS}}^2,$$

where $\{\chi_{it}\}$ is the stochastic fluctuation sequence appearing in the high-probability analysis; see Supplementary Material, Section D. The deterministic terms $\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3$ are estimated in the same way as in the expectation analysis, whereas the last stochastic term is handled differently. In expectation, it is controlled through $\mathcal{T}_4$ using Assumption 4. In the high-probability setting, it is treated by a truncation argument together with martingale techniques.

Relation to the unregularized analysis. The decomposition above also clarifies the relation between the present paper and our earlier unregularized SGD analysis (Shi and Yang, 2024). While the basic operator-learning setting and the overall expectation-based framework are similar, the explicit regularization term changes the recursion in an essential way. In particular, it introduces the regularization path $\{H_{\lambda_t}\}$ and the drift term $\mathcal{T}_3$, and it changes the treatment of $\mathcal{T}_2$ and $\mathcal{T}_4$, especially in the critical regime $\theta_1 + \theta_2 = 1$.

These changes are reflected both in the analysis and in the resulting guarantees. At the level of results, explicit regularization yields sharper conclusions in the online setting, especially for the estimation error. At the methodological level, it leads to a different decomposition and a different treatment of the stochastic recursion. At the probabilistic level, it also enables high-probability bounds and hence almost sure convergence in the infinite-dimensional setting.

3 Discussion

This section places the abstract framework studied in this paper in a broader operator-learning context and further interprets its functional-analytic meaning. We first show that, for a suitable class of scalar kernels, the induced vector-valued RKHS is norm-equivalent to a Bochner–Sobolev space, thereby giving a concrete interpretation of both the hypothesis space and the estimation error in our analysis. We then discuss the relation of our results to recent operator learning literature and conclude with several remarks on possible extensions and on the scope and limitations of the present theory.

Bochner–Sobolev spaces. In the discussion below, we further assume that the input space $\mathcal{X} \subset \mathbb{R}^d$ is a bounded Lipschitz domain, equipped with the Lebesgue measure. For a Banach space E and $1 \leq p \leq \infty$, let $W^{k,p}(\mathcal{X}; E)$ denote the Bochner–Sobolev space of E -valued functions on $\mathcal{X}$ whose weak derivatives up to order k belong to $L^p(\mathcal{X}; E)$; see (Amann et al., 1995; Aubin, 2000; Cardanobile and Mugnolo, 2009). When E is a Hilbert space, we write

$$H^k(\mathcal{X}; E) = W^{k,2}(\mathcal{X}; E).$$

Since our output space $\mathcal{Y}$ is a separable Hilbert space, the space $H^k(\mathcal{X}; \mathcal{Y})$ is well defined.

The next proposition shows that, for a suitable class of scalar kernels, the induced vector-valued RKHS is norm-equivalent to a Bochner–Sobolev space.

Proposition 3.1. *Suppose that $\mathcal{K}$ is a translation-invariant scalar-valued kernel on $\mathbb{R}^d$ whose Fourier transform satisfies*

$$\widehat{\mathcal{K}}(\xi) \asymp (1 + \|\xi\|_2^2)^{-k}, \quad \xi \in \mathbb{R}^d,$$

for some $k > d/2$. Then the RKHS induced by the restriction of $\mathcal{K}$ to $\mathcal{X} \times \mathcal{X}$ is norm-equivalent to $H^k(\mathcal{X})$.

Define the operator-valued kernel

$$K : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{B}(\mathcal{Y}), \quad K(x, x') = \mathcal{K}(x, x') I,$$

where I denotes the identity operator on $\mathcal{Y}$. Then the corresponding vector-valued RKHS $\mathcal{H}$ is norm-equivalent to $H^k(\mathcal{X}; \mathcal{Y})$.

This proposition is essentially a specialization of the vector-valued Sobolev RKHS characterization developed in (Li et al., 2024b, Section 6). The underlying argument combines three ingredients: the norm equivalence between scalar Matérn-type RKHSs and Sobolev spaces on $\mathcal{X}$ (Wendland, 2004, Corollary 10.13 and Theorem 10.46); the Hilbert-valued Sobolev representation $H^k(\mathcal{X}; E) \cong E \widehat{\otimes} H^k(\mathcal{X})$, where $\widehat{\otimes}$ denotes the Hilbert tensor product, from (Aubin, 2000, Theorem 12.7.1); and the fact that an operator-valued kernel of the form $K(x, x') = \mathcal{K}(x, x')I$ induces a vector-valued RKHS canonically identified with $\mathcal{Y} \widehat{\otimes} \mathcal{H}_{\mathcal{K}}$; see (Carmeli et al., 2010, Example 5(i)). A concrete example covered by this proposition is provided by Matérn kernels with Sobolev smoothness index k . Consequently, in such settings, the estimation error in our analysis can be interpreted in terms of a Sobolev norm.

Relation to operator learning. The present work studies operator learning from the perspective of nonparametric kernel methods. In the setting $K(x, x') = \mathcal{K}(x, x')I$, the operator-valued kernel K is L^2 -universal if and only if the scalar kernel $\mathcal{K}$ is L^2 -universal⁵. This provides a qualitative parallel to the universal approximation results established for neural operator architectures such as FNO (Kovachki et al., 2021), DeepONet (Lanthaler et al., 2022), and PCA-Net (Lanthaler, 2023).

Much of the existing quantitative theory for operator learning concerns approximation, parameter complexity, and sample complexity for achieving accuracy ϵ . A substantial part of this theory either treats highly structured operator classes, such as infinite-dimensional holomorphic maps (Adcock et al., 2024a), Fréchet differentiable operators learned via prespecified neural operators (Cheng et al., 2026), or derives rates for concrete PDEs by showing that neural architectures can emulate classical numerical schemes such as Fourier–Galerkin discretizations and fixed-point iterations (Kovachki et al., 2021; Lanthaler et al., 2022; Lanthaler, 2023). Related statistical learning theory has also been developed for neural operators from the viewpoint of empirical risk minimization and statistical learning theory (Reinhardt et al., 2024), while recent theory-to-practice gap results reveal substantial sampling barriers for neural operators in relevant approximation spaces (Grohs et al., 2025). For broader classes, algebraic rates are generally unavailable: for Lipschitz operator classes and Gaussian–Sobolev settings, both parameter and sample complexity may exhibit curse-of-complexity behavior (Adcock et al., 2024b; Lanthaler, 2023; Kovachki et al., 2024), while near-sharp minimax risk bounds for Lipschitz operators were recently established in (Adcock et al., 2025).

In contrast to works that focus primarily on approximation, architectural complexity, or sample complexity for specific operator-learning models, the present paper studies a structured nonparametric kernel framework and analyzes regularized SGD as a concrete learning algorithm in infinite-dimensional spaces. Under RKHS-based regularity assumptions, we establish algebraic convergence rates for both prediction and estimation errors, in expectation and with high probability, in both online and finite-horizon settings. Our results therefore complement the existing operator-learning literature by highlighting the roles of optimization and regularization, and by providing almost sure convergence guarantees.

Remarks on extensions. The regularized SGD analysis also extends beyond Assumption 1 to more general operator-valued kernels; see Appendix E. Since this extension does not include the noncompact case $K(x, x') = \mathcal{K}(x, x')I$, the two settings are complementary. Together they cover a broader class of operator learning problems.

We next describe a viewpoint that links the prediction error, the estimation error, and Assumption 2. As noted in (Yang and Shi, 2025, Remark 3), Assumption 2 is equivalent to $h^\dagger \in \text{ran}(L_K^{r+1/2})$, where $\rho_{\mathcal{X}}$ denotes the marginal distribution of ρ on $\mathcal{X}$, and $L_K : L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y}) \rightarrow L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y})$ is the integral operator induced by K . Restricting L_K to $\ker(L_K)^\perp$, we see that for $h \in \mathcal{H}$, the prediction error is $\|h - h^\dagger\|_{L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y})}^2$, while the estimation error can be written as $\|h - h^\dagger\|_{\mathcal{H}}^2 = \|L_K^{-1/2}(h - h^\dagger)\|_{L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y})}^2$. More generally, for $\beta \in (0, 1)$, the quantity $\|L_K^{-\beta/2}(h - h^\dagger)\|_{L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y})}^2$ measures the error in the interpolation space $[L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y}), \mathcal{H}]_{\beta, 2}$; see (Yang and Shi, 2025, Theorem

⁵Though elementary, this equivalence does not seem to be stated explicitly in the literature. Indeed, by (Carmeli et al., 2010, Example 5(i)), $\mathcal{H} \cong \mathcal{Y} \widehat{\otimes} \mathcal{H}_{\mathcal{K}}$, and by (Aubin, 2000, Theorem 12.6.1), $L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y}) \cong \mathcal{Y} \widehat{\otimes} L^2(\mathcal{X}, \rho_{\mathcal{X}})$, where $\rho_{\mathcal{X}}$ is any probability measure on $\mathcal{X}$.

2.3). By the interpolation inequality,

$$\|L_K^{-\beta/2}(h - h^\dagger)\|_{L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y})}^2 \leq \|h - h^\dagger\|_{L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y})}^{2(1-\beta)} \|h - h^\dagger\|_{\mathcal{H}}^{2\beta}.$$

Applying this to the estimator h_{T+1} and taking expectation with respect to the sample, Hölder's inequality yields

$$\text{interpolation-space error} \leq (\text{prediction error})^{1-\beta} \times (\text{estimation error})^\beta.$$

Under the setting of Proposition 3.1, if $\rho_{\mathcal{X}}$ is equivalent to the uniform measure on $\mathcal{X}$, so that $L^2(\mathcal{X}, \rho_{\mathcal{X}})$ and $L^2(\mathcal{X})$ coincide with equivalent norms, then the interpolation space $[L^2(\mathcal{X}, \rho_{\mathcal{X}}; \mathcal{Y}), \mathcal{H}]_{\beta, 2}$ corresponds to the Bochner-Sobolev space $H^{\beta k}(\mathcal{X}; \mathcal{Y})$; see (Li et al., 2024b, Section 6).

4 Applications and Extensions

This section discusses several applications and related connections of our results. We begin with structured prediction problems, then turn to a class of parametric PDEs. A brief discussion of the connection with the PCA encoder-decoder framework is deferred to Appendix F.

4.1 Applications to Structured Prediction

Now, we formulate the surrogate approach for structured prediction as an application example of our model (1.1). In structured prediction, the input takes values in $\mathcal{X}$ and the output takes values in $\mathcal{Z}$, where $\mathcal{X}$ is a Polish space and $\mathcal{Z}$ represents the structured output space. A structured loss function $\mathcal{D} : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$ is defined on $\mathcal{Z}$ to measure the discrepancy between the true and the predicted outputs. Let x denote the input random variable and z the output random variable. Given a set of independent and identically distributed input-output samples, our goal is to learn a mapping from the inputs to the structured outputs. To this end, we minimize the prediction error defined by

$$\mathcal{R}(f) := \mathbb{E}[\mathcal{D}(f(x), z)],$$

where f is an estimator of $f^\dagger$. The function $f^\dagger : \mathcal{X} \rightarrow \mathcal{Z}$ is the minimizer of $\mathcal{R}$, i.e., $f^\dagger = \arg \min_f \mathcal{R}(f)$.

We focus on the case where the loss function $\mathcal{D}$ is induced by a scalar-valued kernel $k_{\mathcal{Z}} : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$. Specifically, denote the RKHS induced by $k_{\mathcal{Z}}$ by $(\mathcal{Y}, \|\cdot\|_{\mathcal{Y}})$, and embed $\mathcal{Z}$ into $\mathcal{Y}$ via the canonical feature map $\phi(z) := k_{\mathcal{Z}}(z, \cdot)$. We then define the structured loss as $\mathcal{D}(z, z') = \|\phi(z) - \phi(z')\|_{\mathcal{Y}}^2$. Building on extensive research on kernels for structured objects (Gärtner, 2003), this class of loss functions addresses various structured prediction problems. Instead of directly learning $f^\dagger$, we adopt a surrogate model $h^\dagger : \mathcal{X} \rightarrow \mathcal{Y}$, where $h^\dagger(x) := \mathbb{E}[y|x]$ and $y := \phi(z)$ is a random variable taking values in $\mathcal{Y}$. This reduces the original structured prediction task to the model (1.1). We then reformulate the original structured prediction problem as the following surrogate nonlinear operator learning problem:

$$\min_{h: \mathcal{X} \rightarrow \mathcal{Y}} \mathbb{E}[\|h(x) - y\|_{\mathcal{Y}}^2].$$

We solve this problem using the SGD algorithm presented in this paper, which yields an approximation of $h^\dagger$, denoted by $\hat{h}$. During prediction, we use a decoding operator D defined as

$$D(h)(\cdot) := \arg \min_{z \in \mathcal{Z}} \{\|h(\cdot) - \phi(z)\|_{\mathcal{Y}}\}$$

for any estimator h , as detailed in (Ciliberto et al., 2016; Brogat-Motte et al., 2022). Let $\hat{f} = D(\hat{h})$, denote the estimator for $f^\dagger$ obtained via the algorithm. The surrogate approach for structured prediction is illustrated in Figure 1.

According to (Ciliberto et al., 2016), the following properties hold:

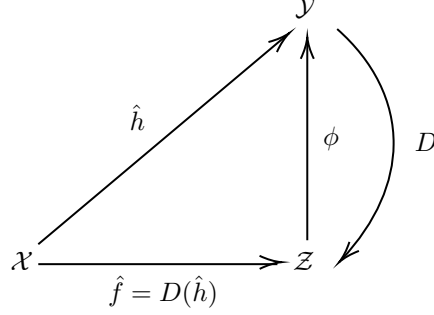


Figure 1: Surrogate approach for structured prediction

- (1) Fisher Consistency: $D(h^\dagger) = f^\dagger$ almost surely.
- (2) Comparison Inequality:

$$\mathcal{R}(\hat{f}) - \mathcal{R}(f^\dagger) \lesssim \left(\mathbb{E} \left[\|\hat{h}(x) - h^\dagger(x)\|_{\mathcal{Y}}^2 \right] \right)^{\frac{1}{2}}.$$

Thus, to bound $\mathcal{R}(\hat{f}) - \mathcal{R}(f^\dagger)$, it suffices to bound $\mathbb{E}[\|\hat{h}(x) - h^\dagger(x)\|_{\mathcal{Y}}^2] = \mathcal{E}(\hat{h}) - \mathcal{E}(h^\dagger)$, as conducted in this paper. This guarantees decay rates of the prediction error under mild assumptions.

In many structured prediction tasks, such as those in natural language processing (e.g., sequence labeling and machine translation) or time series forecasting, data often arrive sequentially or in streams. In such cases, SGD is particularly well suited, as it allows for incremental model updates with each new data point. This makes it an effective tool for structured prediction in streaming or time-dependent environments.

4.2 Applications to Parametric PDEs

In this subsection, we apply regularized SGD to a class of parameter-dependent elliptic PDEs, following the setting of (Burman et al., 2026). This example shows that the abstract RKHS framework considered in our analysis naturally covers a concrete class of parametric PDEs, and that the corresponding estimation norm admits a Sobolev interpretation.

Let $\Omega_x \subset \mathbb{R}^d$, with $d \in \{2, 3\}$, be either a convex polygonal/polyhedral domain or a bounded $C^{1,1}$ domain, and let $\Omega_\zeta = [0, 1]^{N_\zeta}$, where $N_\zeta \geq 1$ may be large. We denote by $x \in \Omega_x$ the spatial variable and by $\zeta \in \Omega_\zeta$ the parameter vector. Consider the following Schrödinger-type elliptic parameter-dependent PDE:

$$-\Delta u(x, \zeta) + q(x, \zeta) u(x, \zeta) = f(x, \zeta), \quad (x, \zeta) \in \Omega_x \times \Omega_\zeta, \quad (4.1)$$

subject to homogeneous Dirichlet boundary conditions on $\partial\Omega_x$. Here, $f = f(x, \zeta)$ and $q = q(x, \zeta)$ denote the source term and the potential, respectively. Our interest is in the dependence of the solution on the parameter ζ .

To ensure existence, uniqueness, and regularity of the solution, we impose the following assumptions on q and f :

- (1) **Coefficients (boundedness and coercivity).** There exist constants $0 < q_{\min} \leq q_{\max} < \infty$ such that

$$q_{\min} \leq q(x, \zeta) \leq q_{\max} \quad \text{for a.e. } (x, \zeta) \in \Omega_x \times \Omega_\zeta,$$

where

$$q(x, \zeta) = q_0(x) + \sum_{i=1}^{N_\zeta} \zeta_i q_i(x), \quad \zeta \in \Omega_\zeta,$$

with $q_i \in L^\infty(\Omega_x)$ for $i = 0, 1, \dots, N_\zeta$.

(2) **Source term.** The source term f depends affinely on ζ , namely,

$$f(x, \zeta) = f_0(x) + \sum_{i=1}^{N_\zeta} \zeta_i f_i(x), \quad \zeta \in \Omega_\zeta,$$

with $f_i \in H^{-1}(\Omega_x)$ for $i = 0, 1, \dots, N_\zeta$.

As a consequence, $f \in W^{k,\infty}(\Omega_\zeta; H^{-1}(\Omega_x))$ for every $k \geq 0$.

Here, $H_0^1(\Omega_x)$, $H^{-1}(\Omega_x)$, and $L^\infty(\Omega_x)$ respectively denote the Sobolev, dual Sobolev, and essentially bounded function spaces on Ω_x . We also use the Bochner–Sobolev spaces $W^{k,p}(\Omega_\zeta; X)$, where X is a Banach space and $1 \leq p \leq \infty$, as defined in Section 3. In particular, when X is a Hilbert space, we write

$$H^k(\Omega_\zeta; X) = W^{k,2}(\Omega_\zeta; X).$$

Under the above assumptions, (4.1) admits a unique weak solution

$$u \in L^2(\Omega_\zeta; H_0^1(\Omega_x)),$$

and, moreover, by (Burman et al., 2026, Theorem 2.1), for any $k \geq 0$,

$$\|u\|_{W^{k,\infty}(\Omega_\zeta; H_0^1(\Omega_x))} \lesssim \|f\|_{W^{k,\infty}(\Omega_\zeta; H^{-1}(\Omega_x))},$$

with a constant independent of ζ . We thus define the parameter-to-solution map

$$h^\dagger : \Omega_\zeta \rightarrow H_0^1(\Omega_x), \quad h^\dagger(\zeta) = u(\cdot, \zeta).$$

In particular,

$$h^\dagger \in W^{k,\infty}(\Omega_\zeta; H_0^1(\Omega_x)).$$

Since Ω_ζ has finite measure, it follows that

$$W^{k,\infty}(\Omega_\zeta; H_0^1(\Omega_x)) \hookrightarrow H^k(\Omega_\zeta; H_0^1(\Omega_x)),$$

and hence $h^\dagger \in H^k(\Omega_\zeta; H_0^1(\Omega_x))$. Furthermore, $h^\dagger \in W^{k,\infty}(\Omega_\zeta; H^m(\Omega_x))$ under a uniform elliptic shift property (Burman et al., 2026, Remark 2.1) of order m .

This regularity fits naturally into the RKHS framework studied in this paper. Indeed, if $\mathcal{K}$ is a scalar kernel on Ω_ζ whose RKHS is norm-equivalent to $H^k(\Omega_\zeta)$, for example a Matérn-type kernel of Sobolev smoothness k , then the associated vector-valued RKHS generated by $K(\zeta, \zeta') = \mathcal{K}(\zeta, \zeta')I$ is norm-equivalent to $H^k(\Omega_\zeta; H_0^1(\Omega_x))$. Hence $h^\dagger$ belongs to the corresponding RKHS, and the estimation error in our analysis can be interpreted through the Bochner–Sobolev norm with respect to ζ .

As a further example, consider the finite-dimensional parametric elliptic diffusion problem

$$-\nabla \cdot (a(x, \zeta) \nabla u(x, \zeta)) = f(x), \quad \zeta \in \Omega_\zeta \subset \mathbb{R}^{N_\zeta},$$

subject to homogeneous Dirichlet boundary conditions, where

$$a(x, \zeta) = a_0(x) + \sum_{j=1}^{N_\zeta} \zeta_j a_j(x)$$

depends affinely on the parameter and satisfies a uniform ellipticity condition. In this setting, classical results show that the parameter-to-solution map $\zeta \mapsto u(\cdot, \zeta)$ is analytic with values in $H_0^1(\Omega_x)$; see Cohen et al. (2011) (see also Cohen and DeVore (2015)). Since this map admits an analytic extension to a complex neighborhood of the bounded finite-dimensional parameter domain Ω_ζ , one obtains

$$\zeta \mapsto u(\cdot, \zeta) \in W^{m,\infty}(\Omega_\zeta; H_0^1(\Omega_x)) \subset H^m(\Omega_\zeta; H_0^1(\Omega_x))$$

for every finite m . This also provides a natural class of examples for the present framework, in which the estimation error can again be interpreted in terms of a Sobolev norm.

5 Numerical Experiments

In the experiments, we represent inputs through the first m basis functions of $\mathcal{X} = L^2[0, 1]$,

$$e_k(u) = \sqrt{2} \cos(k\pi u), \quad u \in [0, 1], \quad 1 \leq k \leq m,$$

and generate

$$x = \sum_{k=1}^m \sqrt{\mu_k} \xi_k e_k, \quad \mu_k = k^{-\alpha},$$

where $\alpha > 0$, $m \gg 1$, and $\xi_k \sim \mathcal{N}(0, 1)$ are independent. We consider the scalar kernel

$$\mathcal{K}(x, x') = \sum_{k \geq 1} \nu_k \langle x, e_k \rangle_{L^2} \langle x', e_k \rangle_{L^2}, \quad \nu_k = k^{-\beta},$$

and define the operator-valued kernel $K(x, x') = \mathcal{K}(x, x') I_{\mathcal{Y}}$ with $\mathcal{Y} = L^2[0, 1]$. We choose the target operator in the form $H^\dagger = S^\dagger C^r$, where C is the covariance operator associated with $\mathcal{K}$. The observations are generated by

$$y_t = h^\dagger(x_t) + \epsilon_t, \quad \epsilon_t = \frac{\sigma}{\sqrt{m}} \sum_{k=1}^m \zeta_k^t e_k,$$

where $\zeta_k^t \sim \mathcal{N}(0, 1)$ are independent. The experimental results are displayed in Figure 2; see Appendix G for additional details.

In practice, both the step sizes and the regularization parameters can be tuned according to validation performance, and the output iterate can be determined by validation-based early stopping. This stopping strategy can be regarded as a form of implicit regularization. Our theoretical results also indicate that the benefit of explicit regularization is especially pronounced under a decaying step-size schedule.

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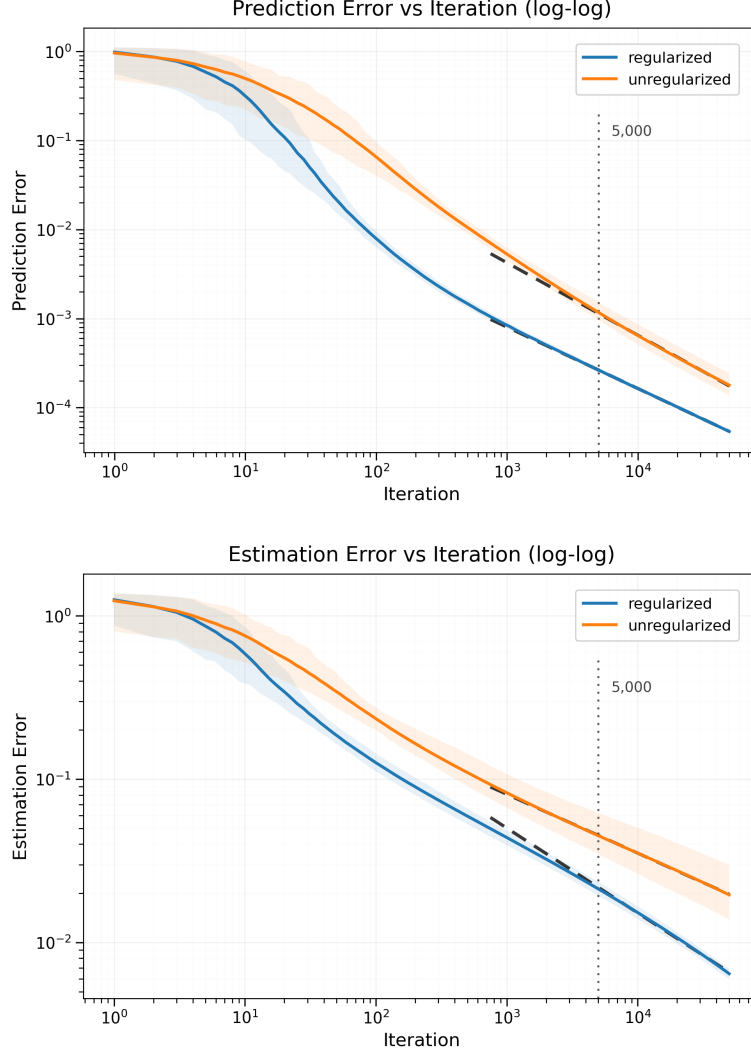


Figure 2: Log-log plots of the prediction and estimation errors versus the iteration number in the online setting, comparing regularized SGD with unregularized SGD. We set $m = 200$, $\alpha = 0.3$, $\beta = 0.9$, $r = 1.0$, and $\sigma = 0.1$. Each experiment is run for 50,000 iterations and repeated 100 times. The shaded regions represent the empirical 90% confidence bands. Both axes are shown on a logarithmic scale to highlight the polynomial decay behavior, and linear interpolation is used for $t > 5000$. The results suggest that regularization yields smaller errors and improved stability. In particular, regularized SGD consistently outperforms unregularized SGD in prediction error and shows an even more pronounced advantage in estimation error.

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Supplementary Material

for “Learning Operators by Regularized Stochastic Gradient Descent
with Operator-valued Kernels”

This supplementary material provides all technical proofs and auxiliary results that support the main paper. It includes the error decomposition, intermediate estimates, and convergence analyses for both expectation and high probability, as well as additional technical details in the appendix.

- Section A presents the error decomposition tailored to the regularized SGD algorithm.
- In Section B, we provide essential intermediate estimates used in the subsequent analysis.
- Section C and Section D establish bounds on the prediction and estimation errors—first in expectation, then with high probability.
- For clarity and conciseness, some of the technical proofs are presented in the appendix.

For readability, we use labels and cross-references throughout the supplement to avoid repeating definitions and intermediate estimates.

A Error Decomposition

In this section, we present the error decomposition employed in the convergence analysis of upper bounds. We begin with several useful observations.

For any $H \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$, by the definition of $\mathcal{E}(H)$ in the main text,

$$\begin{aligned}\mathcal{E}(H) - \mathcal{E}(H^\dagger) &= \mathbb{E} [\|y - H\phi(x)\|_{\mathcal{Y}}^2] - \mathbb{E} [\|y - H^\dagger\phi(x)\|_{\mathcal{Y}}^2] \\ &= \mathbb{E} [\|(H^\dagger - H)\phi(x) + \epsilon\|_{\mathcal{Y}}^2] - \sigma^2 \\ &= \mathbb{E} [\|(H - H^\dagger)\phi(x)\|_{\mathcal{Y}}^2] + 2\mathbb{E} [\langle \epsilon, (H^\dagger - H)\phi(x) \rangle_{\mathcal{Y}}].\end{aligned}$$

Since ϵ is a centered noise independent of x , we have $\mathbb{E} [\langle \epsilon, (H^\dagger - H)\phi(x) \rangle_{\mathcal{Y}}] = 0$. Therefore,

$$\mathcal{E}(H) - \mathcal{E}(H^\dagger) = \mathbb{E} [\|(H - H^\dagger)\phi(x)\|_{\mathcal{Y}}^2].$$

Furthermore, suppose that $\{f_j\}_{j \geq 1}$ is an orthonormal basis of the separable Hilbert space $\mathcal{Y}$. We express $(H - H^\dagger)\phi(x)$ using a Fourier expansion:

$$\begin{aligned}\mathcal{E}(H) - \mathcal{E}(H^\dagger) &= \mathbb{E} \left[\sum_{j \geq 1} \langle (H - H^\dagger)\phi(x), f_j \rangle_{\mathcal{Y}}^2 \right] \\ &= \sum_{j \geq 1} \mathbb{E} \left[\langle (H - H^\dagger)\phi(x) \otimes \phi(x)(H - H^\dagger)^* f_j, f_j \rangle_{\mathcal{Y}} \right] \\ &= \left\| (H - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2.\end{aligned}\tag{A.1}$$

Our goal in this paper is to estimate $\|(H - H^\dagger)C^\alpha\|_{\text{HS}}^2$ for $\alpha = 0$ or $1/2$, corresponding respectively to the estimation error and the prediction error, for a given estimator H .

We define the regularizing operator as

$$H_\lambda := \arg \min_{H \in \mathcal{B}_{\text{HS}}(\mathcal{H}_K, \mathcal{Y})} \mathcal{E}(H) + \lambda \|H\|_{\text{HS}}^2 \quad (\text{A.2})$$

for some $\lambda > 0$. By computing the Fréchet derivative on H , we obtain

$$H_\lambda = H^\dagger C(C + \lambda I)^{-1} = S^\dagger C^{1+r}(C + \lambda I)^{-1}, \quad (\text{A.3})$$

where the final equality follows from Assumption 2.

As introduced in Section 1, we consider two types of step sizes and regularization parameters. Both can be uniformly expressed in the following form:

$$\begin{cases} \eta_t = \bar{\eta}(t + t_0)^{-\theta_1}, \\ \lambda_t = \bar{\lambda}(t + t_0)^{-\theta_2}, \end{cases} \quad (\text{A.4})$$

where $t_0 \geq 0$, η_t is the step size with $\theta_1 \in [0, 1)$ and λ_t denotes the regularization parameter with $\theta_2 \in [0, 1)$. To avoid confusion, we clarify the parameter settings below:

1. The online setting. In this setting, in (A.4) we require that $\theta_1, \theta_2 \in (0, 1)$, $t_0 > 0$, and $\bar{\eta}, \bar{\lambda} > 0$ be constants independent of the current iteration t .
2. The finite-horizon setting. In this setting, we set

$$\eta_t \equiv \bar{\eta}, \quad \lambda_t \equiv \bar{\lambda}, \quad t = 1, 2, \dots, T,$$

where $\bar{\eta} = \eta_1 T^{-\theta_3}$ and $\bar{\lambda} = \lambda_1 T^{-\theta_4}$, with $t_0 = \theta_1 = \theta_2 = 0$, $\theta_3 \in (0, 1)$, and $\theta_4 > 0$. Unlike the decaying case, here $\bar{\eta} = \bar{\eta}(T)$ and $\bar{\lambda} = \bar{\lambda}(T)$ depend on T , while η_1 and λ_1 are constants independent of T .

Lemma A.1. *Let $\{H_t\}_{t \geq 1}$ be defined as (2.1). Then, we have*

$$\begin{aligned} H_{t+1} - H_{\lambda_t} &= (H_t - H_{\lambda_{t-1}})(I - \eta_t(C + \lambda_t I)) \\ &\quad + (H_{\lambda_{t-1}} - H_{\lambda_t})(I - \eta_t(C + \lambda_t I)) + \eta_t \mathcal{B}_t, \end{aligned} \quad (\text{A.5})$$

where I denotes the identity operator, and $\mathcal{B}_t$ is defined by

$$\mathcal{B}_t = (H_t - H^\dagger)C + (y_t - H_t \phi(x_t)) \otimes \phi(x_t).$$

Moreover, for any $t \in \mathbb{N}_T$, it holds that $\mathbb{E}_{z_t}[\mathcal{B}_t] = 0$.

Proof. From (A.3), we have $H^\dagger C = H_{\lambda_t}(C + \lambda_t I)$. Combining this with the update rule in algorithm (2.1), we obtain the equality in (A.5), which can be directly verified.

Note that H_t depends on z^{t-1} and is independent of z_t . Therefore, we have

$$\begin{aligned} \mathbb{E}_{z_t}[\mathcal{B}_t] &= (H_t - H^\dagger)C + \mathbb{E}_{z_t}[(y_t - H_t \phi(x_t)) \otimes \phi(x_t)] \\ &= (H_t - H^\dagger)C + \mathbb{E}_{z_t}[(H^\dagger - H_t)\phi(x_t) + \epsilon_t] \otimes \phi(x_t). \end{aligned}$$

Since ϵ_t is a centered noise independent of x_t , it follows that $\mathbb{E}_{z_t}[\epsilon_t \otimes \phi(x_t)] = 0$. Hence,

$$\mathbb{E}_{z_t}[\mathcal{B}_t] = (H_t - H^\dagger)C + (H^\dagger - H_t)C = 0.$$

The proof is then completed. $\square$

We set $\lambda_0 = t_0^{-\theta_2}$ in the online setting, and $\lambda_0 = \bar{\lambda}$ in the finite-horizon setting. Let $\prod_{j=T+1}^T (I - \eta_j(C + \lambda_j I)) = I$. By applying induction to the equality (A.5), we derive the following key identity used in the error decomposition:

$$\begin{aligned}
H_{T+1} - H_{\lambda_T} &= (H_T - H_{\lambda_{T-1}})(I - \eta_T(C + \lambda_T I)) \\
&\quad + (H_{\lambda_{T-1}} - H_{\lambda_T})(I - \eta_T(C + \lambda_T I)) + \eta_T \mathcal{B}_T \\
&= \dots \\
&= -H_{\lambda_0} \prod_{t=1}^T (I - \eta_t(C + \lambda_t I)) + \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \\
&\quad + \sum_{t=1}^T \eta_t \mathcal{B}_t \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)).
\end{aligned} \tag{A.6}$$

In the next proposition, we decompose the expectation of the prediction error (when $\alpha = 1/2$) and the estimation error (when $\alpha = 0$), $\mathbb{E}_{z^T} \left[\|(H_{T+1} - H^\dagger) C^\alpha\|_{\text{HS}}^2 \right]$, into four terms that can be estimated separately.

Its derivation, as well as the subsequent expectation estimates, builds on ideas from earlier work. In the scalar case, Ying and Pontil (2008) studied kernel SGD for functional regression. For Hilbert-space-valued outputs, Ciliberto et al. (2020); Brogat-Motte et al. (2022) used the isometric identification between RKHSs of the form $K = \mathcal{K}I$ and Hilbert–Schmidt operator spaces (an identification already available in (Carmeli et al., 2010)) to analyze least-squares-type methods through an equivalent Hilbert–Schmidt operator formulation in that setting. Later, Guo et al. (2023); Shi and Yang (2024) extended the analysis of (Ying and Pontil, 2008) to the linear-operator setting and studied SGD in that framework.

The present proposition builds on this line of work, but the introduction of regularization under Assumption 1 requires additional terms and estimates that are specific to the regularized SGD analysis developed in this paper. In this sense, while the overall decomposition strategy is connected to the existing literature, the parts of the argument involving regularization are new in the present paper.

Proposition A.2. *Let $\{H_t\}_{t \in \mathbb{N}_T}$ be defined as (2.1). Suppose that Assumption 4 holds with some $c > 0$. Then, for any $T \geq 1$ and $0 \leq \alpha \leq \frac{1}{2}$, the following inequality holds:*

$$\mathbb{E}_{z^T} \left[\|(H_{T+1} - H^\dagger) C^\alpha\|_{\text{HS}}^2 \right] \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4, \tag{A.7}$$

where

$$\begin{aligned}
\mathcal{T}_1 &:= 2 \|(H_{\lambda_T} - H^\dagger) C^\alpha\|_{\text{HS}}^2, \\
\mathcal{T}_2 &:= 6 \left\| H_{\lambda_0} C^\alpha \prod_{t=1}^T (I - \eta_t(C + \lambda_t I)) \right\|_{\text{HS}}^2, \\
\mathcal{T}_3 &:= 6 \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) C^\alpha \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2, \\
\mathcal{T}_4 &:= 6\sqrt{c} \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{z^{t-1}} \|(H_t - H^\dagger) \phi(x_t)\|_{\mathcal{Y}}^2 + \sigma^2 \right) \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right).
\end{aligned} \tag{A.8}$$

Proof. Since $(H_{T+1} - H^\dagger) C^\alpha = (H_{T+1} - H_{\lambda_T}) C^\alpha + (H_{\lambda_T} - H^\dagger) C^\alpha$, we have

$$\begin{aligned}
\mathbb{E}_{z^T} \left[\|(H_{T+1} - H^\dagger) C^\alpha\|_{\text{HS}}^2 \right] &= \mathbb{E}_{z^T} \left[\|(H_{T+1} - H_{\lambda_T}) C^\alpha + (H_{\lambda_T} - H^\dagger) C^\alpha\|_{\text{HS}}^2 \right] \\
&\leq 2 \mathbb{E}_{z^T} \left[\|(H_{T+1} - H_{\lambda_T}) C^\alpha\|_{\text{HS}}^2 \right] + 2 \|(H_{\lambda_T} - H^\dagger) C^\alpha\|_{\text{HS}}^2.
\end{aligned}$$

We aim to bound $\mathbb{E}_{z^T} [\|(H_{T+1} - H_{\lambda_T})C^\alpha\|_{\text{HS}}^2]$. From the equality (A.6), it follows that

$$\begin{aligned} (H_{T+1} - H_{\lambda_T})C^\alpha &= -H_{\lambda_0}C^\alpha \prod_{t=1}^T (I - \eta_t(C + \lambda_t I)) + \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t})C^\alpha \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \\ &\quad + \sum_{t=1}^T \eta_t \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) =: J_1 + J_2 + J_3. \end{aligned} \quad (\text{A.9})$$

Then,

$$\mathbb{E}_{z^T} [\|(H_{T+1} - H_{\lambda_T})C^\alpha\|_{\text{HS}}^2] \leq 3\|J_1\|_{\text{HS}}^2 + 3\|J_2\|_{\text{HS}}^2 + 3\mathbb{E}_{z^T} [\|J_3\|_{\text{HS}}^2].$$

We express $\mathbb{E}_{z^T} [\|J_3\|_{\text{HS}}^2] = \mathbb{E}_{z^T} \left[\left\| \sum_{t=1}^T \eta_t \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \right]$ as

$$\sum_{t=1}^T \sum_{t'=1}^T \eta_t \eta_{t'} \mathbb{E}_{z^T} \left[\left\langle \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)), \mathcal{B}_{t'} C^\alpha \prod_{j=t'+1}^T (I - \eta_j(C + \lambda_j I)) \right\rangle_{\text{HS}} \right].$$

Using the property $\mathbb{E}_{z_t}[\mathcal{B}_t] = 0$, for $t > t'$, we obtain

$$\begin{aligned} &\mathbb{E}_{z^T} \left[\left\langle \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)), \mathcal{B}_{t'} C^\alpha \prod_{j=t'+1}^T (I - \eta_j(C + \lambda_j I)) \right\rangle_{\text{HS}} \right] \\ &= \mathbb{E}_{z^{t-1}} \mathbb{E}_{z_t} \left[\left\langle \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)), \mathcal{B}_{t'} C^\alpha \prod_{j=t'+1}^T (I - \eta_j(C + \lambda_j I)) \right\rangle_{\text{HS}} \right] \\ &= \mathbb{E}_{z^{t-1}} \left[\left\langle \mathbb{E}_{z_t} \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)), \mathcal{B}_{t'} C^\alpha \prod_{j=t'+1}^T (I - \eta_j(C + \lambda_j I)) \right\rangle_{\text{HS}} \right] = 0. \end{aligned}$$

Similarly, the above equality also holds for $t < t'$. Consequently, there holds

$$\mathbb{E}_{z^T} [\|J_3\|_{\text{HS}}^2] = \sum_{t=1}^T \mathbb{E}_{z^T} \left[\left\| \eta_t \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \right]. \quad (\text{A.10})$$

Using the property $\mathbb{E}_{z_t}[\mathcal{B}_t] = 0$ again, we have

$$\mathcal{B}_t = -\mathbb{E}_{z_t}[(y_t - H_t \phi(x_t)) \otimes \phi(x_t)] + (y_t - H_t \phi(x_t)) \otimes \phi(x_t).$$

Denote $\eta_t [(y_t - H_t \phi(x_t)) \otimes \phi(x_t)] C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))$ by $\mathcal{A}$, then substituting $\mathcal{A}$ into (A.10) yields that

$$\begin{aligned} &\mathbb{E}_{z^T} \left[\left\| \eta_t \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \right] = \mathbb{E}_{z^{t-1}} \mathbb{E}_{z_t} [\|-\mathbb{E}_{z_t}[\mathcal{A}] + \mathcal{A}\|_{\text{HS}}^2] \\ &\leq \mathbb{E}_{z^T} [\|\mathcal{A}\|_{\text{HS}}^2] = \mathbb{E}_{z^T} \left[\left\| \eta_t [(y_t - H_t \phi(x_t)) \otimes \phi(x_t)] C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \right]. \end{aligned} \quad (\text{A.11})$$

Take $\{e_i\}_{i \geq 1}$ to be an orthonormal basis of Hilbert space $\mathcal{H}_K$. Since C is self-adjoint, by (A.10), (A.11) and the definition of the Hilbert-Schmidt norm, there holds

$$\begin{aligned}
& \mathbb{E}_{z^T} [\|J_3\|_{\text{HS}}^2] \\
& \leq \sum_{t=1}^T \mathbb{E}_{z^t} \left[\sum_{i \geq 1} \left\| \eta_t [(y_t - H_t \phi(x_t)) \otimes \phi(x_t)] C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) e_i \right\|_{\mathcal{H}_K}^2 \right] \\
& = \sum_{t=1}^T \mathbb{E}_{z^t} \left[\sum_{i \geq 1} \|\eta_t (y_t - H_t \phi(x_t))\|_{\mathcal{Y}}^2 \left\langle \phi(x_t), C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) e_i \right\rangle_{\mathcal{H}_K}^2 \right] \\
& = \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z^t} \left[\|y_t - H_t \phi(x_t)\|_{\mathcal{Y}}^2 \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \right] \\
& = \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z^{t-1}} \mathbb{E}_{x_t} \left[\mathbb{E}_{\epsilon_t} \|(H^\dagger - H_t) \phi(x_t) + \epsilon_t\|_{\mathcal{Y}}^2 \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \right], \tag{A.12}
\end{aligned}$$

where we use $y_t = H^\dagger \phi(x_t) + \epsilon_t$ in the last equality. It is obvious that

$$\mathbb{E}_{\epsilon_t} [\|(H^\dagger - H_t) \phi(x_t) + \epsilon_t\|_{\mathcal{Y}}^2] = \|(H_t - H^\dagger) \phi(x_t)\|_{\mathcal{Y}}^2 + \sigma^2,$$

where $\sigma^2 = \mathbb{E}[\|\epsilon\|_{\mathcal{Y}}^2]$ is the variance of ϵ . Substitute it back into (A.12) and use the Cauchy-Schwarz inequality. Then we obtain

$$\begin{aligned}
\mathbb{E}_{z^T} [\|J_3\|_{\text{HS}}^2] & \leq \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z^{t-1}} \mathbb{E}_{x_t} \left[\left(\|(H^\dagger - H_t) \phi(x_t)\|_{\mathcal{Y}}^2 + \sigma^2 \right) \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \right] \\
& \leq \sum_{t=1}^T \eta_t^2 \left(\mathbb{E}_{z^{t-1}} \sqrt{\mathbb{E}_{x_t} \|(H_t - H^\dagger) \phi(x_t)\|_{\mathcal{Y}}^4} + \sigma^2 \right) \\
& \quad \times \left(\mathbb{E}_{x_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^4 \right)^{1/2} \\
& \leq \sqrt{c} \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{z^{t-1}} \|(H_t - H^\dagger) \phi(x_t)\|_{\mathcal{Y}}^2 + \sigma^2 \right) \\
& \quad \times \mathbb{E}_{x_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2,
\end{aligned}$$

where the last inequality is due to Assumption 4 and Proposition C.1 in Appendix C. Since

$$\begin{aligned}
\mathbb{E}_{x_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 & = \sum_{i \geq 1} \mathbb{E}_{x_t} \left\langle C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t), e_i \right\rangle_{\mathcal{H}_K}^2 \\
& = \sum_{i \geq 1} \left\langle C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 e_i, e_i \right\rangle_{\mathcal{H}_K} \\
& = \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right),
\end{aligned}$$

there holds

$$\mathbb{E}_{z^T} [\|J_3\|_{\text{HS}}^2] \leq \sqrt{c} \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{z^{t-1}} \|(H_t - H^\dagger)\phi(x_t)\|_{\mathcal{Y}}^2 + \sigma^2 \right) \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right),$$

which finishes our proof. $\square$

Hereafter, we refer to $\mathcal{T}_1$ as the approximation error, $\mathcal{T}_2$ as the initial error, $\mathcal{T}_3$ as the drift error, and $\mathcal{T}_4$ as the sample error, respectively.

We now present the error decomposition of $\|(H_{T+1} - H^\dagger)C^\alpha\|_{\text{HS}}^2$, which serves to establish a high-probability upper bound. For any random variable μ taking values in $\mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$, we denote the L^∞ norm of $\|\mu\|_{\text{HS}}$ by $\|\mu\|_{L_{\text{HS}}^\infty}$.

Proposition A.3. *Let $\{H_t\}_{t \in \mathbb{N}_T}$ be defined as (2.1). Suppose that Assumption 5 holds for some $M_\rho > 0$. Then, for any $T \geq 1$ and $0 \leq \alpha \leq \frac{1}{2}$, the quantity $\|(H_{T+1} - H^\dagger)C^\alpha\|_{\text{HS}}^2$ admits the decomposition*

$$\|(H_{T+1} - H^\dagger)C^\alpha\|_{\text{HS}}^2 \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + 6 \left\| \sum_{t=1}^T \chi_t \right\|_{\text{HS}}^2, \quad (\text{A.13})$$

where $\mathcal{T}_1$, $\mathcal{T}_2$, and $\mathcal{T}_3$ are defined in (A.8), and $\chi_t = \eta_t \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))$ satisfies

$$\|\chi_t\|_{\text{HS}} \leq 2\eta_t \kappa \left(M_\rho + \kappa \|H_t\|_{L_{\text{HS}}^\infty} \right) \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|, \quad \forall t \in \mathbb{N}_T. \quad (\text{A.14})$$

Proof. The proof follows a similar strategy to the previous proposition. As in the proof of Proposition A.2, we readily obtain

$$\|(H_{T+1} - H^\dagger)C^\alpha\|_{\text{HS}}^2 \leq 2\|(H_{T+1} - H_{\lambda_T})C^\alpha\|_{\text{HS}}^2 + 2\|(H_{\lambda_T} - H^\dagger)C^\alpha\|_{\text{HS}}^2$$

and

$$\|(H_{T+1} - H_{\lambda_T})C^\alpha\|_{\text{HS}}^2 \leq 3\|J_1\|_{\text{HS}}^2 + 3\|J_2\|_{\text{HS}}^2 + 3\|J_3\|_{\text{HS}}^2,$$

where J_1 , J_2 , and J_3 are defined in (A.9). Defining $\chi_t = \eta_t \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))$, we then have $J_3 = \sum_{t=1}^T \chi_t$. Since $\mathcal{B}_t$ can be expressed as

$$\mathcal{B}_t = (y_t - H_t \phi(x_t)) \otimes \phi(x_t) - \mathbb{E}_{z_t} [(y_t - H_t \phi(x_t)) \otimes \phi(x_t)],$$

it follows from Assumption 5 that

$$\|\mathcal{B}_t\|_{\text{HS}} \leq 2\|(y_t - H_t \phi(x_t)) \otimes \phi(x_t)\|_{L_{\text{HS}}^\infty} \leq 2\kappa \left(M_\rho + \kappa \|H_t\|_{L_{\text{HS}}^\infty} \right).$$

Thus,

$$\|\chi_t\|_{\text{HS}} \leq 2\eta_t \kappa \left(M_\rho + \kappa \|H_t\|_{L_{\text{HS}}^\infty} \right) \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|.$$

The proof is then finished. $\square$

B Intermediate Estimates for Error Analysis

In this section, we derive bounds for $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{T}_3$, and $\mathcal{T}_4$. The bounds for $\mathcal{T}_1$ and $\mathcal{T}_2$ are presented in a unified form encompassing the finite-horizon setting. The term $\mathcal{T}_3$ arises exclusively in the online setting and is therefore analyzed only within that context. For $\mathcal{T}_4$, we provide separate bounds corresponding to these two settings. These intermediate results play an important role in the subsequent analysis of prediction and estimation errors, both in expectation and with high probability.

B.1 Bounding Approximation Error

We bound $\mathcal{T}_1$ in the following proposition.

Proposition B.1. *Under the Assumption 2 with $r > 0$, there exists a constant $c_1 = c_1(\|S^\dagger\|_{\text{HS}}, r, \alpha)$ independent of $t_0, T, \bar{\eta}$, and $\bar{\lambda}$, such that*

$$\mathcal{T}_1 = 2 \|(H_{\lambda_T} - H^\dagger)C^\alpha\|_{\text{HS}}^2 \leq c_1 \lambda_T^{\min\{2(r+\alpha), 2\}}.$$

Proof. We know from (A.3) that

$$\begin{aligned} H_{\lambda_T} - H^\dagger &= H^\dagger C(C + \lambda_T I)^{-1} - H^\dagger \\ &= -\lambda_T H^\dagger (C + \lambda_T I)^{-1} = -\lambda_T S^\dagger C^r (C + \lambda_T I)^{-1}, \end{aligned}$$

where the last identity uses Assumption 2. It then follows that

$$\begin{aligned} \|(H_{\lambda_T} - H^\dagger)C^\alpha\|_{\text{HS}} &= \|\lambda_T S^\dagger C^{r+\alpha} (C + \lambda_T I)^{-1}\|_{\text{HS}} \\ &\leq \lambda_T \|S^\dagger\|_{\text{HS}} \|C^{r+\alpha} (C + \lambda_T I)^{-1}\|. \end{aligned} \tag{B.1}$$

Since

$$\begin{aligned} \|C^{r+\alpha} (C + \lambda_T I)^{-1}\| &\leq \sup_{0 \leq x \leq \kappa^2} \frac{x^{r+\alpha}}{x + \lambda_T} \\ &\leq \begin{cases} \kappa^{2(r+\alpha-1)}, & \text{when } r + \alpha \geq 1, \\ (r + \alpha)^{r+\alpha} (1 - r - \alpha)^{1-r-\alpha} \lambda_T^{r+\alpha-1}, & \text{when } r + \alpha < 1, \end{cases} \end{aligned} \tag{B.2}$$

combining (B.1) with (B.2), there exists a constant c_1 such that

$$\|(H_{\lambda_T} - H^\dagger)C^\alpha\|_{\text{HS}}^2 \leq c_1 \lambda_T^{\min\{2(r+\alpha), 2\}}.$$

It is clear that c_1 is independent of $t_0, T, \bar{\eta}$, and $\bar{\lambda}$, which completes this proof. $\square$

B.2 Bounding Initial Error

The following two lemmas will be used repeatedly throughout our analysis.

Lemma B.2. *Let $\beta > 0$ and let η_t, λ_t be defined as (A.4). Let l, m be integers satisfying $1 \leq l \leq m$. Suppose that $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$. Then, the following estimates hold:*

$$\begin{aligned} (1) \quad &\|C^\beta \prod_{t=l}^m (I - \eta_t (C + \lambda_t I))\| \leq \exp \left\{ -\sum_{t=l}^m \eta_t \lambda_t \right\} \frac{2(\kappa^{2\beta} + (\beta/e)^\beta)}{1 + (\sum_{t=l}^m \eta_t)^\beta}. \\ (2) \quad &\left\| C^\beta \prod_{t=l}^m (I - \eta_t (C + \lambda_t I))^2 \right\| \leq \left(\frac{\beta}{2e} \right)^\beta (\sum_{t=l}^m \eta_t)^{-\beta} \exp \left\{ -2 \sum_{t=l}^m \eta_t \lambda_t \right\}. \\ (3) \quad &\left\| C^\beta \prod_{t=l}^m (I - \eta_t (C + \lambda_t I))^2 \right\| \leq \exp \left\{ -2 \sum_{t=l}^m \eta_t \lambda_t \right\} \frac{2(\kappa^{2\beta} + (\beta/(2e))^\beta)}{1 + (\sum_{t=l}^m \eta_t)^\beta}. \end{aligned}$$

Proof. Recall that C (defined in Section 2) is self-adjoint and compact. By the definition of the operator norm, we have

$$\begin{aligned} \left\| C^\beta \prod_{t=l}^m (I - \eta_t (C + \lambda_t I)) \right\| &\leq \sup_{0 \leq x \leq \kappa^2} x^\beta \prod_{t=l}^m (1 - \eta_t (x + \lambda_t)) \\ &\leq \sup_{0 \leq x \leq \kappa^2} x^\beta \exp \left\{ -\sum_{t=l}^m \eta_t (x + \lambda_t) \right\} \\ &= \left(\frac{\beta}{e} \right)^\beta \left(\sum_{t=l}^m \eta_t \right)^{-\beta} \exp \left\{ -\sum_{t=l}^m \eta_t \lambda_t \right\}, \end{aligned} \tag{B.3}$$

where the first inequality follows from the fact $1 - \eta_t(x + \lambda_t) \geq 0$ for all $t \geq 1$ and $0 \leq x \leq \kappa^2$, which is ensured by the condition $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$. On the other hand, we also have

$$\left\| C^\beta \prod_{t=l}^m (I - \eta_t(C + \lambda_t I)) \right\| \leq \kappa^{2\beta} \prod_{t=l}^m (1 - \eta_t \lambda_t) \leq \kappa^{2\beta} \exp \left\{ - \sum_{t=l}^m \eta_t \lambda_t \right\}. \quad (\text{B.4})$$

Applying the inequality $\min\{a, b\} \leq \frac{2}{1/a+1/b}$, $\forall a, b > 0$ and combining (B.3) with (B.4), we obtain

$$\left\| C^\beta \prod_{t=l}^m (I - \eta_t(C + \lambda_t I)) \right\| \leq \exp \left\{ - \sum_{t=l}^m \eta_t \lambda_t \right\} \frac{2(\kappa^{2\beta} + (\beta/e)^\beta)}{1 + (\sum_{t=l}^m \eta_t)^\beta}.$$

Now, using (B.4) once more, there holds

$$\begin{aligned} \left\| C^\beta \prod_{t=l}^m (I - \eta_t(C + \lambda_t I))^2 \right\| &= \left\| C^{\beta/2} \prod_{t=l}^m (I - \eta_t(C + \lambda_t I)) \right\|^2 \\ &\leq \left(\frac{\beta}{2e} \right)^\beta \left(\sum_{t=l}^m \eta_t \right)^{-\beta} \exp \left\{ -2 \sum_{t=l}^m \eta_t \lambda_t \right\}. \end{aligned}$$

Moreover, since $\left\| C^\beta \prod_{t=l}^m (I - \eta_t(C + \lambda_t I))^2 \right\| \leq \kappa^{2\beta} \exp \{ - \sum_{t=l}^m \eta_t \lambda_t \}$, applying $\min\{a, b\} \leq \frac{2}{1/a+1/b}$ again yields

$$\left\| C^\beta \prod_{t=l}^m (I - \eta_t(C + \lambda_t I))^2 \right\| \leq \exp \left\{ -2 \sum_{t=l}^m \eta_t \lambda_t \right\} \frac{2(\kappa^{2\beta} + (\beta/(2e))^\beta)}{1 + (\sum_{t=l}^m \eta_t)^\beta}.$$

This completes the proof. $\square$

The next lemma establishes lower bounds for $\sum_{t=l}^m \eta_t$ and $\sum_{t=1}^T \eta_t \lambda_t$.

Lemma B.3. *Let $0 \leq \theta_1 < 1$, $0 \leq \theta_2 < 1$, and η_t, λ_t be defined as (A.4). Then the following bounds hold for $1 \leq l \leq m$ with $l \in \mathbb{N}$:*

$$(1) \quad \sum_{t=l}^m \eta_t \geq \frac{\bar{\eta}}{1-\theta_1} [(m+t_0+1)^{1-\theta_1} - (l+t_0)^{1-\theta_1}].$$

(2)

$$\sum_{t=l}^m \eta_t \lambda_t \geq \begin{cases} \frac{\bar{\eta}\bar{\lambda}}{1-\theta_1-\theta_2} [(m+t_0+1)^{1-\theta_1-\theta_2} - (l+t_0)^{1-\theta_1-\theta_2}], & \text{when } \theta_1 + \theta_2 \neq 1, \\ \bar{\eta}\bar{\lambda} \log \left(\frac{m+t_0+1}{l+t_0} \right), & \text{when } \theta_1 + \theta_2 = 1. \end{cases}$$

In particular, when $l = 1$ and $m = T$ with $T \geq t_0 + 1$, we have:

$$(3) \quad \sum_{t=1}^T \eta_t \geq \frac{1-2^{\theta_1-1}}{1-\theta_1} \bar{\eta}(T+t_0)^{1-\theta_1}.$$

(4)

$$\sum_{t=1}^T \eta_t \lambda_t \geq \begin{cases} \frac{\bar{\eta}\bar{\lambda}}{1-\theta_1-\theta_2} (1-2^{\theta_1+\theta_2-1})(T+t_0)^{1-\theta_1-\theta_2}, & \text{when } 0 \leq \theta_1 + \theta_2 < 1, \\ \bar{\eta}\bar{\lambda} \log \left(\frac{T+t_0}{t_0+1} \right), & \text{when } \theta_1 + \theta_2 = 1, \\ \frac{\bar{\eta}\bar{\lambda}}{\theta_1+\theta_2-1} (1-2^{1-\theta_1-\theta_2})(t_0+1)^{1-\theta_1-\theta_2}, & \text{when } \theta_1 + \theta_2 > 1. \end{cases}$$

Proof. We bound the summation $\sum_{t=l}^m \eta_t$ using

$$\begin{aligned} \sum_{t=l}^m \eta_t &= \bar{\eta} \sum_{t=l}^m (t+t_0)^{-\theta_1} \geq \bar{\eta} \int_l^{m+1} (x+t_0)^{-\theta_1} dx \\ &= \frac{\bar{\eta}}{1-\theta_1} [(m+t_0+1)^{1-\theta_1} - (l+t_0)^{1-\theta_1}]. \end{aligned}$$

For the specific case where $l = 1$, $m = T$, and $T \geq t_0 + 1$, we obtain

$$\begin{aligned} \sum_{t=1}^T \eta_t &\geq \frac{\bar{\eta}}{1-\theta_1} [(T+t_0+1)^{1-\theta_1} - (t_0+1)^{1-\theta_1}] \\ &\geq \frac{1-2^{\theta_1-1}}{1-\theta_1} \bar{\eta} (T+t_0)^{1-\theta_1}. \end{aligned}$$

Next, we analyze the summation involving λ_t using the same estimate as before:

$$\begin{aligned} \sum_{t=l}^m \eta_t \lambda_t &= \bar{\eta} \bar{\lambda} \sum_{t=l}^m (t+t_0)^{-\theta_1-\theta_2} \geq \bar{\eta} \bar{\lambda} \int_l^{m+1} (x+t_0)^{-\theta_1-\theta_2} dx \\ &= \begin{cases} \frac{\bar{\eta} \bar{\lambda}}{1-\theta_1-\theta_2} [(m+t_0+1)^{1-\theta_1-\theta_2} - (l+t_0)^{1-\theta_1-\theta_2}], & \text{when } \theta_1 + \theta_2 \neq 1, \\ \bar{\eta} \bar{\lambda} \log \left(\frac{m+t_0+1}{l+t_0} \right), & \text{when } \theta_1 + \theta_2 = 1. \end{cases} \end{aligned}$$

For the case $l = 1$, $m = T$, and $T \geq t_0 + 1$, we obtain:

$$\sum_{t=1}^T \eta_t \lambda_t \geq \begin{cases} \frac{\bar{\eta} \bar{\lambda}}{1-\theta_1-\theta_2} (1-2^{\theta_1+\theta_2-1}) (T+t_0)^{1-\theta_1-\theta_2}, & \text{when } \theta_1 + \theta_2 < 1, \\ \bar{\eta} \bar{\lambda} \log \left(\frac{T+t_0}{t_0+1} \right), & \text{when } \theta_1 + \theta_2 = 1, \\ \frac{\bar{\eta} \bar{\lambda}}{\theta_1+\theta_2-1} (1-2^{1-\theta_1-\theta_2}) (t_0+1)^{1-\theta_1-\theta_2}, & \text{when } \theta_1 + \theta_2 > 1. \end{cases}$$

The proof is then finished. $\square$

Next, based on the two lemmas above, we provide a unified upper bound for $\mathcal{T}_2$ under the following two settings:

- (1) $0 < \theta_1 < 1$, $0 < \theta_2 < 1$, and $t_0 > 0$, corresponding to the online setting;
- (2) $\theta_1 = \theta_2 = t_0 = 0$, corresponding to the finite-horizon setting.

The bound in both cases is established by the following proposition.

Proposition B.4. *Suppose that Assumption 2 holds with $S^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$ and $r > 0$. Then, for any $T \geq t_0 + 1$, $t_0 \geq 0$, $0 \leq \theta_1 < 1$, and $0 \leq \theta_2 < 1$, the quantity*

$$\mathcal{T}_2 = 6 \left\| H_{\lambda_0} C^\alpha \prod_{t=1}^T (I - \eta_t (C + \lambda_t I)) \right\|_{\text{HS}}^2$$

admits the following bound:

$$\mathcal{T}_2 \leq c_2 \bar{\eta}^{-2(r+\alpha)} \begin{cases} (T+t_0)^{-2(r+\alpha)(1-\theta_1)} \exp\{-\tau \bar{\eta} \bar{\lambda} (T+t_0)^{1-\theta_1-\theta_2}\}, & \text{when } 0 \leq \theta_1 + \theta_2 < 1, \\ (t_0+1)^{2\bar{\eta} \bar{\lambda}} (T+t_0)^{-2(r+\alpha)(1-\theta_1)-2\bar{\eta} \bar{\lambda}}, & \text{when } \theta_1 + \theta_2 = 1, \\ (T+t_0)^{-2(r+\alpha)(1-\theta_1)}, & \text{when } \theta_1 + \theta_2 > 1, \end{cases}$$

where $c_2 = c_2(\|S^\dagger\|_{\text{HS}}, r, \alpha, \theta_1)$ and $\tau = \tau(\theta_1, \theta_2)$ are positive constants independent of t_0 , T , $\bar{\eta}$, and $\bar{\lambda}$.

Proof. According to equality (A.3), it follows that

$$\begin{aligned} \left\| H_{\lambda_0} C^\alpha \prod_{t=1}^T (I - \eta_t (C + \lambda_t I)) \right\|_{\text{HS}}^2 &= \left\| S^\dagger C^{r+\alpha+1} (C + \lambda_0 I)^{-1} \prod_{t=1}^T (I - \eta_t (C + \lambda_t I)) \right\|_{\text{HS}}^2 \\ &\leq \|S^\dagger\|_{\text{HS}}^2 \left\| C^{2(r+\alpha)} \prod_{t=1}^T (I - \eta_t (C + \lambda_t I))^2 \right\|. \end{aligned} \quad (\text{B.5})$$

By applying (2) in Lemma B.2 with $\beta = 2(r + \alpha)$ and Lemma B.3 (3), the following inequality holds:

$$\begin{aligned}
& \left\| H_{\lambda_0} C^\alpha \prod_{t=1}^T (I - \eta_t (C + \lambda_t I)) \right\|_{\text{HS}}^2 \\
& \leq \|S^\dagger\|_{\text{HS}}^2 \left(\frac{r + \alpha}{e} \right)^{2(r+\alpha)} \left(\sum_{t=1}^T \eta_t \right)^{-2(r+\alpha)} \exp \left\{ -2 \sum_{t=1}^T \eta_t \lambda_t \right\} \\
& \leq \|S^\dagger\|_{\text{HS}}^2 \left(\frac{(r + \alpha)(1 - \theta_1)}{e(1 - 2^{\theta_1 - 1})} \right)^{2(r+\alpha)} \bar{\eta}^{-2(r+\alpha)} (T + t_0)^{-2(r+\alpha)(1 - \theta_1)} \exp \left\{ -2 \sum_{t=1}^T \eta_t \lambda_t \right\}.
\end{aligned} \tag{B.6}$$

Next, using (4) in Lemma B.3, the exponential term can be bounded as:

$$\exp \left\{ -2 \sum_{t=1}^T \eta_t \lambda_t \right\} \leq \begin{cases} \exp \{ -\tau \bar{\eta} \bar{\lambda} (T + t_0)^{1 - \theta_1 - \theta_2} \}, & \text{when } 0 \leq \theta_1 + \theta_2 < 1, \\ (t_0 + 1)^{2\bar{\eta} \bar{\lambda}} (T + t_0)^{-2\bar{\eta} \bar{\lambda}}, & \text{when } \theta_1 + \theta_2 = 1, \\ 1, & \text{when } \theta_1 + \theta_2 > 1, \end{cases}$$

where $\tau = \frac{2}{1 - \theta_1 - \theta_2} (1 - 2^{\theta_1 + \theta_2 - 1})$. Therefore, the bound for $\mathcal{T}_2$ becomes:

$$\begin{aligned}
\mathcal{T}_2 &= 6 \left\| H_{\lambda_0} C^\alpha \prod_{t=1}^T (I - \eta_t (C + \lambda_t I)) \right\|_{\text{HS}}^2 \\
&\leq c_2 \bar{\eta}^{-2(r+\alpha)} \begin{cases} (T + t_0)^{-2(r+\alpha)(1 - \theta_1)} \exp \{ -\tau \bar{\eta} \bar{\lambda} (T + t_0)^{1 - \theta_1 - \theta_2} \}, & \text{when } 0 \leq \theta_1 + \theta_2 < 1, \\ (t_0 + 1)^{2\bar{\eta} \bar{\lambda}} (T + t_0)^{-2(r+\alpha)(1 - \theta_1) - 2\bar{\eta} \bar{\lambda}}, & \text{when } \theta_1 + \theta_2 = 1, \\ (T + t_0)^{-2(r+\alpha)(1 - \theta_1)}, & \text{when } \theta_1 + \theta_2 > 1, \end{cases} \tag{B.7}
\end{aligned}$$

where $c_2 = 6 \|S^\dagger\|_{\text{HS}}^2 \left(\frac{(r+\alpha)(1-\theta_1)}{e(1-2^{\theta_1-1})} \right)^{2(r+\alpha)}$ is independent of t_0 , T , $\bar{\eta}$, and $\bar{\lambda}$.

The desired result is established and the proof is complete. $\square$

Proposition B.4 is written in a unified form in order to cover both the online setting and the finite-horizon setting. In particular, in the finite-horizon case, the bound is obtained by applying the proposition with $\theta_1 = \theta_2 = 0$, while θ_3 and θ_4 enter through the definitions of the constant step size $\bar{\eta}$ and regularization parameter $\bar{\lambda}$.

B.3 Bounding Drift Error

In the finite-horizon setting, where $\lambda_t = \bar{\lambda}$ is fixed depending on T , we have $\mathcal{T}_3 = 0$. Therefore, it is sufficient to bound $\mathcal{T}_3$ under the regime of decaying step sizes and regularization parameters. In what follows, we focus on the setting where $0 < \theta_1 < 1$ and $0 < \theta_2 < 1$.

Lemma B.5. *Suppose that Assumption 2 holds with $S^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y})$ and $r > 0$ and let $t_0 \geq 1$. Then, for any $t \geq 1$, the following bound holds:*

$$\|H_{\lambda_{t-1}} - H_{\lambda_t}\|_{\text{HS}} \leq \tilde{c}_3 \bar{\lambda}^{\min\{r, 1\}} (t + t_0)^{-\theta_2 \min\{r, 1\} - 1},$$

where $\tilde{c}_3 = \tilde{c}_3(\|S^\dagger\|_{\text{HS}}, r, \theta_2)$ is a constant independent of t_0 , t , $\bar{\eta}$, and $\bar{\lambda}$.

Proof. Based on the expression for H_λ in (A.3) and under Assumption 2, we deduce that

$$\begin{aligned}
\|H_{\lambda_{t-1}} - H_{\lambda_t}\|_{\text{HS}} &= \|H^\dagger C(C + \lambda_{t-1}I)^{-1} - H^\dagger C(C + \lambda_t I)^{-1}\|_{\text{HS}} \\
&= |\lambda_t - \lambda_{t-1}| \|S^\dagger C^{r+1}(C + \lambda_t I)^{-1}(C + \lambda_{t-1}I)^{-1}\|_{\text{HS}} \\
&\leq \|S^\dagger\|_{\text{HS}} |\lambda_t - \lambda_{t-1}| \|C^r(C + \lambda_t I)^{-1}\| \\
&\leq \|S^\dagger\|_{\text{HS}} \bar{\lambda} |(t + t_0 - 1)^{-\theta_2} - (t + t_0)^{-\theta_2}| \begin{cases} \kappa^{2r-2}, & \text{when } r \geq 1, \\ r^r(1-r)^{1-r}\lambda_t^{r-1}, & \text{when } r < 1, \end{cases}
\end{aligned} \tag{B.8}$$

where the last inequality uses the fact that

$$\|C^r(C + \lambda_t I)^{-1}\| \leq \sup_{0 \leq x \leq \kappa^2} \{x^r(x + \lambda_t)^{-1}\} \leq \begin{cases} \kappa^{2r-2}, & \text{when } r \geq 1, \\ r^r(1-r)^{1-r}\lambda_t^{r-1}, & \text{when } r < 1. \end{cases}$$

Applying the mean value theorem, there exists $\xi \in (0, 1)$ such that

$$\begin{aligned}
|(t + t_0 - 1)^{-\theta_2} - (t + t_0)^{-\theta_2}| &= \theta_2(t + t_0 - \xi)^{-(\theta_2+1)} \leq \theta_2(t + t_0 - 1)^{-(\theta_2+1)} \\
&\leq 2^{\theta_2+1}\theta_2(t + t_0)^{-(\theta_2+1)},
\end{aligned} \tag{B.9}$$

where the last inequality uses $t + t_0 - 1 \geq (t + t_0)/2$. Substituting (B.9) into (B.8), we arrive at

$$\begin{aligned}
\|H_{\lambda_{t-1}} - H_{\lambda_t}\|_{\text{HS}} &\leq 2^{\theta_2+1}\theta_2 \|S^\dagger\|_{\text{HS}} \bar{\lambda} (t + t_0)^{-(\theta_2+1)} \begin{cases} \kappa^{2r-2}, & \text{when } r \geq 1, \\ r^r(1-r)^{1-r}\lambda_t^{r-1}, & \text{when } r < 1, \end{cases} \\
&\leq \tilde{c}_3 \bar{\lambda}^{\min\{r,1\}} (t + t_0)^{-\theta_2 \min\{r,1\}-1},
\end{aligned}$$

where $\tilde{c}_3$ is a constant independent of t_0 , t , $\bar{\eta}$, and $\bar{\lambda}$.

The proof is complete. $\square$

Now, we derive an error bound for $\mathcal{T}_3$ in the case $\alpha = 0$, applied to the analysis of estimation error.

Proposition B.6. *Suppose that Assumption 2 holds with $S^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_K, \mathcal{Y})$ and $r > 0$. Set $\alpha = 0$ in $\mathcal{T}_3$ and assume $t_0 \geq 1$. Let $T \geq t_0 + 1$ when $\theta_1 + \theta_2 < 1$, and $T \geq 1$ otherwise. Additionally, assume $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$ and $\bar{\eta}\bar{\lambda} > \theta_2 \min\{r, 1\}$. Then, the following bound holds for $\mathcal{T}_3$:*

$$\mathcal{T}_3 = 6 \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \leq c_3 \begin{cases} 1, & \text{when } \theta_1 + \theta_2 > 1, \\ (T + t_0)^{-2\theta_2 \min\{r,1\}}, & \text{when } \theta_1 + \theta_2 \leq 1, \end{cases}$$

where $c_3 = c_3(t_0, \bar{\lambda}, \bar{\eta}, \|S^\dagger\|_{\text{HS}}, r, \theta_1, \theta_2)$ is a constant independent of T .

Proof. By Lemma B.5, we deduce that

$$\begin{aligned}
&\left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}} \\
&\leq \tilde{c}_3 \bar{\lambda}^{\min\{r,1\}} \sum_{t=1}^T (t + t_0)^{-\theta_2 \min\{r,1\}-1} \left\| \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|.
\end{aligned} \tag{B.10}$$

Since C is self-adjoint and compact, with the operator norm $\|C\| \leq \kappa^2$ (see Section 2), it follows that

$$\begin{aligned}
&\left\| \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\| \leq \sup_{0 \leq x \leq \kappa^2} \prod_{j=t}^T (1 - \eta_j(x + \lambda_j)) \\
&\leq \sup_{0 \leq x \leq \kappa^2} \left\{ \exp \left\{ - \sum_{j=t}^T \eta_j(x + \lambda_j) \right\} \right\} \leq \exp \left\{ - \sum_{j=t}^T \eta_j \lambda_j \right\}.
\end{aligned} \tag{B.11}$$

Hence, we have the following bound for $\left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j (C + \lambda_j I)) \right\|_{\text{HS}}$:

$$\tilde{c}_3 \bar{\lambda}^{\min\{r,1\}} \sum_{t=1}^T (t + t_0)^{-\theta_2 \min\{r,1\} - 1} \exp \left\{ - \sum_{j=t}^T \eta_j \lambda_j \right\}.$$

From (2) in Lemma B.3, we know that $\exp \left\{ - \sum_{j=t}^T \eta_j \lambda_j \right\}$ is bounded by

$$\begin{cases} \exp \left\{ - \frac{\bar{\eta} \bar{\lambda}}{1 - \theta_1 - \theta_2} [(T + t_0 + 1)^{1 - \theta_1 - \theta_2} - (t + t_0)^{1 - \theta_1 - \theta_2}] \right\}, & \text{when } \theta_1 + \theta_2 \neq 1, \\ \exp \left\{ - \bar{\eta} \bar{\lambda} \log \left(\frac{T + t_0 + 1}{t + t_0} \right) \right\}, & \text{when } \theta_1 + \theta_2 = 1. \end{cases}$$

Thus, we obtain the following bound for the exponential term:

$$\begin{aligned} & \exp \left\{ - \sum_{j=t}^T \eta_j \lambda_j \right\} \\ & \leq \begin{cases} \exp \left\{ - \frac{\bar{\eta} \bar{\lambda}}{1 - \theta_1 - \theta_2} [(T + t_0 + 1)^{1 - \theta_1 - \theta_2} - (t + t_0)^{1 - \theta_1 - \theta_2}] \right\}, & \text{when } \theta_1 + \theta_2 < 1, \\ (t + t_0)^{\bar{\eta} \bar{\lambda}} (T + t_0)^{-\bar{\eta} \bar{\lambda}}, & \text{when } \theta_1 + \theta_2 = 1, \\ 1, & \text{when } \theta_1 + \theta_2 > 1. \end{cases} \end{aligned}$$

We next consider the three cases corresponding to $\theta_1 + \theta_2 > 1$, $\theta_1 + \theta_2 = 1$, and $\theta_1 + \theta_2 < 1$.

Case 1: When $\theta_1 + \theta_2 > 1$, we obtain

$$\begin{aligned} & \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j (C + \lambda_j I)) \right\|_{\text{HS}} \\ & \leq \tilde{c}_3 \bar{\lambda}^{\min\{r,1\}} \sum_{t=1}^T (t + t_0)^{-\theta_2 \min\{r,1\} - 1} \leq \tilde{c}_3 \frac{t_0^{-\theta_2 \min\{r,1\}}}{\theta_2 \min\{r,1\}} \bar{\lambda}^{\min\{r,1\}}. \end{aligned}$$

Case 2: When $\theta_1 + \theta_2 = 1$, we derive the following bound:

$$\begin{aligned} & \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j (C + \lambda_j I)) \right\|_{\text{HS}} \\ & \leq \tilde{c}_3 \bar{\lambda}^{\min\{r,1\}} (T + t_0)^{-\bar{\eta} \bar{\lambda}} \sum_{t=1}^T (t + t_0)^{-\theta_2 \min\{r,1\} - 1 + \bar{\eta} \bar{\lambda}} \\ & \leq \frac{\tilde{c}_3}{\bar{\eta} \bar{\lambda} - \theta_2 \min\{r,1\}} \bar{\lambda}^{\min\{r,1\}} (T + t_0)^{-\theta_2 \min\{r,1\}}, \end{aligned}$$

where the last inequality uses the condition that $\bar{\eta} \bar{\lambda} > \theta_2 \min\{r,1\}$.

Case 3: When $\theta_1 + \theta_2 < 1$, there holds that

$$\begin{aligned} & \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j (C + \lambda_j I)) \right\|_{\text{HS}} \leq \tilde{c}_3 \bar{\lambda}^{\min\{r,1\}} \sum_{t=1}^T (t + t_0)^{-\theta_2 \min\{r,1\} - 1} \\ & \quad \times \exp \left\{ - \frac{\bar{\eta} \bar{\lambda}}{1 - \theta_1 - \theta_2} [(T + t_0 + 1)^{1 - \theta_1 - \theta_2} - (t + t_0)^{1 - \theta_1 - \theta_2}] \right\}. \end{aligned}$$

Now, we estimate the summation in the last inequality. Since $T \geq t_0 + 1$, we have $t + t_0 \leq \frac{3}{4}(T + t_0 + 1)$ when $t \leq \frac{T}{2}$. By splitting the summation into two parts, from 1 to $T/2$ and from $T/2$ to T , we deduce that

$$\sum_{t=1}^T (t + t_0)^{-\theta_2 \min\{r,1\} - 1} \exp \left\{ - \frac{\bar{\eta} \bar{\lambda}}{1 - \theta_1 - \theta_2} [(T + t_0 + 1)^{1 - \theta_1 - \theta_2} - (t + t_0)^{1 - \theta_1 - \theta_2}] \right\}$$

$$\begin{aligned}
&\leq \sum_{t=1}^{T/2} (t+t_0)^{-\theta_2 \min\{r,1\}-1} \exp \left\{ -\frac{\bar{\eta}\bar{\lambda}}{1-\theta_1-\theta_2} \left(1 - (3/4)^{1-\theta_1-\theta_2}\right) (T+t_0)^{1-\theta_1-\theta_2} \right\} \\
&\quad + \sum_{t=T/2}^T (t+t_0)^{-\theta_2 \min\{r,1\}-1} \\
&\leq \frac{t_0^{-\theta_2 \min\{r,1\}}}{\theta_2 \min\{r,1\}} \exp \left\{ -\frac{\bar{\eta}\bar{\lambda}}{1-\theta_1-\theta_2} \left(1 - (3/4)^{1-\theta_1-\theta_2}\right) (T+t_0)^{1-\theta_1-\theta_2} \right\} \\
&\quad + \frac{4^{\theta_2 \min\{r,1\}} - 1}{\theta_2 \min\{r,1\}} (T+t_0)^{-\theta_2 \min\{r,1\}}.
\end{aligned}$$

Using the fact that for any constants $k, \gamma > 0$, there exists a constant m such that

$$\exp \{ -k(T+t_0)^{1-\theta_1-\theta_2} \} \leq m (T+t_0)^{-\gamma},$$

we conclude from this that

$$\begin{aligned}
\mathcal{T}_3 &= 6 \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) \prod_{j=t}^T (I - \eta_j (C + \lambda_j I)) \right\|_{\text{HS}}^2 \\
&\leq c_3 \begin{cases} 1, & \text{when } \theta_1 + \theta_2 > 1, \\ (T+t_0)^{-2\theta_2 \min\{r,1\}}, & \text{when } \theta_1 + \theta_2 \leq 1, \end{cases}
\end{aligned}$$

where c_3 is a constant independent of T .

We then finish the proof. $\square$

Remark 2. We will only use the bound for the case $\theta_1 + \theta_2 = 1$ in the above proposition, as it provides better convergence rates than the other cases. Note that we cannot guarantee convergence of the estimation error when $\theta_1 + \theta_2 > 1$. However, convergence of the prediction error is assured when $\theta_1 + \theta_2 > 1$ with $0 < \theta_1, \theta_2 < 1$, and both the prediction and estimation errors converge when $\theta_1 + \theta_2 \leq 1$. The proofs for the remaining cases are similar and are omitted here to avoid repetition.

The following proposition plays a key role in deriving upper bounds for the drift and sample errors. Its technical proof is provided in Appendix D.

Proposition B.7. Let $v > 0$, $\theta \in \mathbb{R}$, $t_0 \geq 1$, and $T \geq t_0 + 1$. The step size η_t is defined as (A.4). Suppose $\bar{\eta}\bar{\lambda} > \theta - 1$ and $\theta_1 + \theta_2 = 1$. Then,

$$\sum_{t=1}^T \exp \left\{ -\sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \leq \delta_1 \begin{cases} (T+t_0)^{-\theta+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{-\theta+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{-\theta+1-v(1-\theta_1)}, & \text{when } v < 1, \end{cases}$$

where $\delta_1 = \delta_1(\bar{\lambda}, \bar{\eta}, \theta_1, \theta)$ is a constant independent of T and t_0 .

Remark 3. The inequality in Proposition B.7 remains valid for all $T \geq 1$, not only when $T \geq t_0 + 1$, provided that the constant δ_1 is allowed to depend on t_0 and is chosen sufficiently large. More specifically, there exists a constant $\delta_2 = \delta_2(t_0, \bar{\lambda}, \bar{\eta})$ such that, for any $T \geq 1$,

$$\sum_{t=1}^T \exp \left\{ -\sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \leq \delta_2 \begin{cases} (T+t_0)^{-\theta+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{-\theta+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{-\theta+1-v(1-\theta_1)}, & \text{when } v < 1. \end{cases}$$

Note that $\delta_2 = \delta_2(t_0, \bar{\lambda}, \bar{\eta}, \theta_1, \theta)$ depends on t_0 , whereas δ_1 does not.

The following proposition establishes a bound on $\mathcal{T}_3$ for the case $\alpha = \frac{1}{2}$, which is instrumental in analyzing the prediction error.

Proposition B.8. *Suppose that Assumption 2 holds with $S^\dagger \in \mathcal{B}_{\text{HS}}(\mathcal{H}_K, \mathcal{Y})$ and $r > 0$. Set $\alpha = 1/2$ in $\mathcal{T}_3$. Let $\theta_1 + \theta_2 = 1$, $t_0 \geq 1$ and $T \geq t_0 + 1$. Suppose that $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$ and $\bar{\eta}\bar{\lambda} > \theta_2 \min\{r, 1\}$. Then, there holds*

$$\mathcal{T}_3 = 6 \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) C^{1/2} \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \leq c_3 (T + t_0)^{-2\theta_2 \min\{r, 1\} + \theta_1 - 1}, \quad (\text{B.12})$$

where $c_3 = c_3(\bar{\lambda}, \bar{\eta}, \|S^\dagger\|_{\text{HS}}, r, \theta_1)$ is a constant independent of T and t_0 .

Proof. Applying Lemma B.5, we have

$$\begin{aligned} & \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) C^{1/2} \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}} \\ & \leq \tilde{c}_3 \bar{\lambda}^{\min\{r, 1\}} \sum_{t=1}^T (t + t_0)^{-\theta_2 \min\{r, 1\} - 1} \left\| C^{1/2} \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|. \end{aligned}$$

Using Lemma B.2 (1) with $\beta = \frac{1}{2}$, the above inequality is further bounded as

$$\begin{aligned} & \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) C^{1/2} \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}} \\ & \leq 2(\kappa + (1/(2e))^{1/2}) \tilde{c}_3 \bar{\lambda}^{\min\{r, 1\}} \sum_{t=1}^T \exp \left\{ - \sum_{j=t}^T \eta_j \lambda_j \right\} \frac{(t + t_0)^{-\theta_2 \min\{r, 1\} - 1}}{1 + \left(\sum_{j=t}^T \eta_j \right)^{1/2}} \\ & \leq 2(\kappa + (1/(2e))^{1/2}) \tilde{c}_3 \bar{\lambda}^{\min\{r, 1\}} \delta_1 (T + t_0)^{-\theta_2 \min\{r, 1\} - (1 - \theta_1)/2}, \end{aligned}$$

where in the last inequality we use Proposition B.7 with $\theta = \theta_2 \min\{r, 1\} + 1$ and $v = 1/2 < 1$. As a consequence,

$$\mathcal{T}_3 = 6 \left\| \sum_{t=1}^T (H_{\lambda_{t-1}} - H_{\lambda_t}) C^{1/2} \prod_{j=t}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \leq c_3 (T + t_0)^{-2\theta_2 \min\{r, 1\} + \theta_1 - 1},$$

where c_3 is a constant independent of T and t_0 .

The proof is then finished. $\square$

B.4 Bounding Sample Error

Let $\mathbb{E}_{z^0}[\xi] = \xi$ for any random variable ξ . The next proposition applies to the online setting.

Proposition B.9. *Suppose that Assumption 3 holds with $0 < s \leq 1$. If $t_0 \geq 1$, $T \geq t_0 + 1$, $0 < \theta_1 < 1$, $0 < \theta_2 < 1$, $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$, and the following condition holds for any $t \in \mathbb{N}_T$:*

$$\mathbb{E}_{z^{t-1}} \left\| (H_t - H^\dagger) \phi(x_t) \right\|_{\mathcal{Y}}^2 \left(= \mathbb{E}_{z^{t-1}} \left[\left\| (H_t - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \right] \right) \leq M, \quad (\text{B.13})$$

where M is independent of T . Then, the following bound holds for $\mathcal{T}_4$:

$$\mathcal{T}_4 = 6\sqrt{c} \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{z^{t-1}} \left\| (H_t - H^\dagger) \phi(x_t) \right\|_{\mathcal{Y}}^2 + \sigma^2 \right) \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right)$$

$$\leq c_4 (\sqrt{c}M + \sigma^2) \begin{cases} (T + t_0)^{-\theta_1}, & \text{when } 2\alpha > s, \\ (T + t_0)^{-\theta_1} \log(T + t_0), & \text{when } 2\alpha = s, \\ (T + t_0)^{-(1+s-2\alpha)\theta_1+s-2\alpha}, & \text{when } 2\alpha < s, \end{cases}$$

where $c_4 = c_4(\bar{\lambda}, \bar{\eta}, \theta_1, \alpha, s, c)$ is a constant independent of T , t_0 , and M .

Proof. Assumption 3 on C guarantees that

$$\text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right) \leq \text{Tr}(C^s) \left\| C^{1+2\alpha-s} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right\|.$$

Then we use (3) in Lemma B.2 with $\beta = 1 + 2\alpha - s$ and (B.13) to bound the operator norm as

$$\begin{aligned} & \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{Z^{t-1}} \|(H_t - H^\dagger)\phi(x_t)\|_y^2 + \sigma^2 \right) \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right) \\ & \leq \sum_{t=1}^T \eta_t^2 (\sqrt{c}M + \sigma^2) \text{Tr}(C^s) \left\| C^{1+2\alpha-s} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right\| \\ & \leq 2 \left(\kappa^{2+4\alpha-2s} + ((1+2\alpha-s)/(2e))^{1+2\alpha-s} \right) (\sqrt{c}M + \sigma^2) \text{Tr}(C^s) \\ & \quad \times \sum_{t=1}^T \exp \left\{ -2 \sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{\eta_t^2}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^{1+2\alpha-s}}. \end{aligned}$$

Now, applying Proposition B.7 with $\theta = 2\theta_1$ and $v = 1 + 2\alpha - s$, we get

$$\begin{aligned} \mathcal{T}_4 &= 6\sqrt{c} \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{Z^{t-1}} \|(H_t - H^\dagger)\phi(x_t)\|_y^2 + \sigma^2 \right) \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right) \\ &\leq c_4 (\sqrt{c}M + \sigma^2) \begin{cases} (T + t_0)^{-\theta_1}, & \text{when } 2\alpha > s, \\ (T + t_0)^{-\theta_1} \log(T + t_0), & \text{when } 2\alpha = s, \\ (T + t_0)^{-(1+s-2\alpha)\theta_1+s-2\alpha}, & \text{when } 2\alpha < s, \end{cases} \end{aligned}$$

where $c_4 = 12\sqrt{c} \left(\kappa^{2+4\alpha-2s} + ((1+2\alpha-s)/(2e))^{1+2\alpha-s} \right) \text{Tr}(C^s) \delta_1 \bar{\eta}^2$.

The proof is thus complete. $\square$

The next proposition is used to bound $\mathcal{T}_4$ in the finite-horizon setting. We define $0^0 := 0$ for convenience.

Proposition B.10. *Let $v \geq 0$, $\bar{\eta} = \eta_1 T^{-\theta_3}$, $0 < \theta_3 < 1$, $\theta_4 > 0$, and η_1, λ_1 be constants independent of T . Then, there exists a constant $\delta_3 = \delta_3(\lambda_1, \eta_1, \theta_3, \theta_4, v)$, independent of T , such that for any $T \geq 2$,*

$$\sum_{t=0}^{T-1} \frac{\exp \{-2\lambda_1 \eta_1 t T^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} \leq \delta_3 \begin{cases} T^{v\theta_3+(1-v)\min\{1, \theta_3+\theta_4\}}, & \text{when } 0 \leq v < 1, \\ T^{\theta_3} \log T, & \text{when } v = 1, \\ T^{\theta_3}, & \text{when } v > 1. \end{cases}$$

Proof. We divide the proof into three cases: $v = 0$, $v > 0$, and $0 < v < 1$ with $\theta_3 + \theta_4 < 1$. The third case is an improvement upon the analysis in the second case.

Case 1: $v = 0$

We first apply the inequality $1 - \exp\{-x\} \geq \exp\{-2\lambda_1\eta_1\}x$ for $0 \leq x \leq 2\lambda_1\eta_1$, yielding:

$$\begin{aligned} \sum_{t=0}^{T-1} \frac{\exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} &= \frac{1 - \exp\{-2\lambda_1\eta_1 T^{1-\theta_4-\theta_3}\}}{1 - \exp\{-2\lambda_1\eta_1 T^{-\theta_4-\theta_3}\}} \\ &\leq \frac{\exp\{2\lambda_1\eta_1\}}{2\lambda_1\eta_1} T^{\theta_3+\theta_4} (1 - \exp\{-2\lambda_1\eta_1 T^{1-\theta_4-\theta_3}\}). \end{aligned}$$

When $\theta_3 + \theta_4 \leq 1$, this term simplifies as

$$\sum_{t=0}^{T-1} \frac{\exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} \leq \frac{\exp\{2\lambda_1\eta_1\}}{2\lambda_1\eta_1} T^{\theta_3+\theta_4}.$$

When $\theta_3 + \theta_4 > 1$, we use the inequality $1 - \exp\{-x\} \leq x$ to obtain

$$\sum_{t=0}^{T-1} \frac{\exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} \leq \exp\{2\lambda_1\eta_1\}T.$$

Case 2: $v > 0$

We bound the summation as

$$\begin{aligned} \sum_{t=0}^{T-1} \frac{\exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} &\leq 1 + \int_0^{T-1} \frac{1}{1 + (t\bar{\eta})^v} dt \leq 1 + \frac{1}{\bar{\eta}} \int_0^{\bar{\eta}(T-1)} \frac{1}{1 + t^v} dt \\ &\leq 1 + \frac{1}{\bar{\eta}} \left(1 + \int_1^{\bar{\eta}(T-1)} t^{-v} dt \right) \\ &\leq 1 + \frac{1}{\bar{\eta}} + \frac{1}{\bar{\eta}} \begin{cases} \frac{(\bar{\eta}T)^{1-v}}{1-v}, & \text{when } 0 < v < 1, \\ \log(\bar{\eta}T), & \text{when } v = 1, \\ \frac{1}{v-1}, & \text{when } v > 1, \end{cases} \quad (\text{B.14}) \\ &\leq \delta_3 \begin{cases} T^{1-v+\theta_3v}, & \text{when } 0 < v < 1, \\ T^{\theta_3} \log T, & \text{when } v = 1, \\ T^{\theta_3}, & \text{when } v > 1, \end{cases} \end{aligned}$$

where δ_3 is a constant independent of T .

Case 3: $0 < v < 1$ with $\theta_3 + \theta_4 < 1$

In this case, a more refined estimation can be achieved compared to Case 2. We split the summation into three parts,

$$\begin{aligned} &\sum_{t=0}^{T-1} \frac{\exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} \\ &\leq 1 + \sum_{t=1}^{T^{\theta_3+\theta_4}} \frac{1}{1 + (t\bar{\eta})^v} + \sum_{t=T^{\theta_3+\theta_4}}^T \frac{\exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} \quad (\text{B.15}) \\ &=: 1 + \mathcal{A}_1 + \mathcal{A}_2. \end{aligned}$$

We estimate $\mathcal{A}_1$ in the same manner as in (B.14). Noting that $\bar{\eta}T^{\theta_3+\theta_4} = \eta_1 T^{\theta_4}$, we obtain

$$\begin{aligned} \mathcal{A}_1 &\leq \int_0^{T^{\theta_3+\theta_4}} \frac{1}{1 + (t\bar{\eta})^v} dt \leq \frac{1}{\bar{\eta}} + \frac{1}{\bar{\eta}} \int_1^{\bar{\eta}T^{\theta_3+\theta_4}} \frac{1}{t^v} dt \\ &\leq \frac{1}{\eta_1} T^{\theta_3} \left(1 + \frac{(\eta_1 T^{\theta_4})^{1-v}}{1-v} \right) \leq \frac{1}{\eta_1} \left(1 + \frac{\eta_1^{1-v}}{1-v} \right) T^{\theta_4(1-v)+\theta_3}. \end{aligned}$$

Now, we estimate $\mathcal{A}_2$. Since $T^{\theta_3+\theta_4} - 1 \geq kT^{\theta_3+\theta_4}$ for $T \geq 2$, where $k = 1 - 2^{-\theta_4-\theta_3}$, it follows that

$$\mathcal{A}_2 \leq \int_{kT^{\theta_3+\theta_4}}^T \frac{1}{1 + (t\bar{\eta})^v} \exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\} dt.$$

Letting $x = tT^{-\theta_4-\theta_3}$, we rewrite the above as

$$\mathcal{A}_2 \leq \eta_1^{-v} T^{(1-v)\theta_4+\theta_3} \int_k^{+\infty} x^{-v} \exp\{-2\lambda_1\eta_1 x\} dx.$$

Since the integral is finite and satisfies

$$\int_k^{+\infty} x^{-v} \exp\{-2\lambda_1\eta_1 x\} dx \leq k^{-v} \frac{\exp\{-2\lambda_1\eta_1 k\}}{2\lambda_1\eta_1} < \infty,$$

we combine the bounds for $\mathcal{A}_1$ and $\mathcal{A}_2$, and use (B.15), to conclude that

$$\sum_{t=0}^{T-1} \frac{\exp\{-2\lambda_1\eta_1 tT^{-\theta_4-\theta_3}\}}{1 + (t\bar{\eta})^v} \leq \left(1 + \frac{1}{\eta_1} \left(1 + \frac{\eta_1^{1-v}}{1-v}\right) + \eta_1^{-v} k^{-v} \frac{\exp\{-2\lambda_1\eta_1 k\}}{2\lambda_1\eta_1}\right) T^{(1-v)\theta_4+\theta_3}.$$

We then finish the proof. $\square$

The following proposition concerns the finite-horizon setting. Before presenting the result, we highlight the following distinctions: in this setting, we have $t_0 = \theta_1 = \theta_2 = 0$, the step size is fixed as $\eta_t \equiv \bar{\eta} = \eta_1 T^{-\theta_3}$, the regularization parameter is set to be $\lambda_t \equiv \bar{\lambda} = \lambda_1 T^{-\theta_4}$, where η_1 and λ_1 are constants independent of T . By contrast, under the online setting in the unified form (A.4), $\bar{\eta}$ and $\bar{\lambda}$ are fixed constants, while η_t and λ_t depend directly on the current iteration t , rather than on a prescribed total number of iterations.

Proposition B.11. *Suppose that Assumption 3 holds with $0 < s \leq 1$. Let $\alpha \in [0, \frac{1}{2}]$, and set the parameters $t_0 = \theta_1 = \theta_2 = 0$, $\eta_t \equiv \bar{\eta} = \eta_1 T^{-\theta_3}$ with $0 < \theta_3 < 1$ and $\lambda_t \equiv \bar{\lambda} = \lambda_1 T^{-\theta_4}$ with $\theta_4 > 0$. Additionally, assume that $T \geq 2$, $\eta_1(\kappa^2 + \lambda_1) \leq 1$ and there exists a constant $\widetilde{M}$ independent of T , such that for all $t \in \mathbb{N}_T$,*

$$\mathbb{E}_{z^{t-1}} \|(H_t - H^\dagger)\phi(x_t)\|_{\mathcal{Y}}^2 \left(= \mathbb{E}_{z^{t-1}} \left[\left\| (H_t - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \right] \right) \leq \widetilde{M}. \quad (\text{B.16})$$

Recall that

$$\mathcal{T}_4 = 6\sqrt{c} \sum_{t=1}^T \eta_t^2 \left(\sqrt{c} \mathbb{E}_{z^{t-1}} \|(H_t - H^\dagger)\phi(x_t)\|_{\mathcal{Y}}^2 + \sigma^2 \right) \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right).$$

Then, the following bound holds for $\mathcal{T}_4$:

$$\mathcal{T}_4 \leq \widetilde{c}_4 \begin{cases} T^{-(1-2\alpha+s)\theta_3+(s-2\alpha)\min\{1,\theta_3+\theta_4\}}, & \text{when } 2\alpha < s \leq 1+2\alpha, \\ T^{-\theta_3} \log T, & \text{when } 2\alpha = s, \\ T^{-\theta_3}, & \text{when } 2\alpha > s, \end{cases}$$

where $\widetilde{c}_4 = \widetilde{c}_4(\lambda_1, \eta_1, \theta_3, \theta_4, \alpha, s, \widetilde{M}, c)$ is a constant independent of T .

Proof. Applying the assumed condition (B.16) and Assumption 3, we get

$$\begin{aligned} \mathcal{T}_4 &\leq 6\sqrt{c} \left(\sqrt{c} \widetilde{M} + \sigma^2 \right) \bar{\eta}^2 \sum_{t=1}^T \text{Tr} \left(C^{1+2\alpha} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right) \\ &\leq 6\sqrt{c} \left(\sqrt{c} \widetilde{M} + \sigma^2 \right) \text{Tr}(C^s) \bar{\eta}^2 \sum_{t=1}^T \left\| C^{1+2\alpha-s} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right\|. \end{aligned}$$

If $1 + 2\alpha - s > 0$, applying Lemma B.2 (3) for $1 \leq t \leq T - 1$, we obtain the following estimate, which also holds for $t = T$,

$$\left\| C^{1+2\alpha-s} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right\| \leq \exp \left\{ -2(T-t)\bar{\lambda}\bar{\eta} \right\} \frac{2(\kappa^{2(1+2\alpha-s)} + (\frac{1+2\alpha-s}{2e})^{1+2\alpha-s})}{1 + ((T-t)\bar{\eta})^{1+2\alpha-s}}.$$

If $1 + 2\alpha - s = 0$, we derive the bound

$$\left\| C^{1+2\alpha-s} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right\| \leq \kappa^{2(1+2\alpha-s)} \exp \left\{ -2(T-t)\bar{\lambda}\bar{\eta} \right\}$$

for any $1 \leq t \leq T$. Thus, based on the above estimates, we obtain the bound for $\mathcal{T}_4$:

$$\mathcal{T}_4 \lesssim \bar{\eta}^2 \sum_{t=0}^{T-1} \frac{\exp \left\{ -2t\bar{\lambda}\bar{\eta} \right\}}{1 + (t\bar{\eta})^{1+2\alpha-s}},$$

where we use the notation $\lesssim$ to omit constants independent of T and t for simplicity, indicating an inequality up to a multiplicative constant.

Since $\bar{\eta} = \eta_1 T^{-\theta_3}$ and $\bar{\lambda} = \lambda_1 T^{-\theta_4}$, and applying Proposition B.10 with $v = 1 + 2\alpha - s$, we obtain

$$\mathcal{T}_4 \leq \tilde{c}_4 \begin{cases} T^{-(1-2\alpha+s)\theta_3 + (s-2\alpha)\min\{1, \theta_3 + \theta_4\}}, & \text{when } 2\alpha < s \leq 1 + 2\alpha, \\ T^{-\theta_3} \log T, & \text{when } 2\alpha = s, \\ T^{-\theta_3}, & \text{when } 2\alpha > s, \end{cases}$$

where $\tilde{c}_4 := 12(\kappa^{2(1+2\alpha-s)} + (\frac{1+2\alpha-s}{2e})^{1+2\alpha-s})\sqrt{c} \left(\sqrt{c}\widetilde{M} + \sigma^2 \right) \text{Tr}(C^s) \eta_1^2 \delta_3$ is a constant independent of T .

The proof is then complete. $\square$

B.5 Key Bounds for Estimating Prediction Error

In this subsection, we establish the key bounds for estimating the prediction error, specifically (B.13) in Proposition B.9 and (B.16) in Proposition B.11. The following proposition pertains to the online setting.

Proposition B.12. *Under Assumption 2, Assumption 3 and Assumption 4, if $\theta_1 + \theta_2 = 1$, $t_0 \geq 1$, $\bar{\eta}\bar{\lambda} > \theta_2 \min\{r, 1\}$, $t_0 \geq \exp\{\frac{1}{\theta_1}\}$, and $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$, then there exists a constant $M = M(\bar{\eta}, \bar{\lambda}, t_0, \theta_1, r, \sigma^2, \|S^\dagger\|_{\text{HS}}, s, c)$ independent of t , such that*

$$\mathbb{E}_{z^{t-1}} \left[\left\| (H_t - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right] \leq M, \quad \forall t \geq 1. \quad (\text{B.17})$$

Proof. The proposition is proved by induction. We have already bounded $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{T}_3$, and $\mathcal{T}_4$ through four propositions, where M in Proposition B.9 will share the same value during the induction process. An important fact we need to be aware of is that the bounds of $\mathcal{T}_2$, $\mathcal{T}_3$ and $\mathcal{T}_4$ require that $t \geq t_0 + 1$ when we bound $\mathbb{E}_{z^{t-1}} \left[\left\| (H_t - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right]$. Hence, we first bound $\mathbb{E}_{z^{t-1}} \left[\left\| (H_t - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right]$ when $t \leq \lfloor t_0 \rfloor + 1$. When $t = 1$,

$$\mathbb{E}_{z^0} \left[\left\| (H_1 - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \right] \leq \left\| H^\dagger C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \leq \kappa^2 \|H^\dagger\|_{\text{HS}}^2.$$

Note that $\mathcal{T}_1$, $\mathcal{T}_2$, and $\mathcal{T}_3$ are deterministic and can be regarded as functions of t . Define a function $f : \{1, 2, \dots\} \rightarrow \mathbb{R}$ iteratively, as $f(1) := \kappa^2 \|H^\dagger\|_{\text{HS}}^2$ and,

$$f(t+1) := \mathcal{T}_1(t+1) + \mathcal{T}_2(t+1) + \mathcal{T}_3(t+1) + 6\sqrt{c} \sum_{k=1}^t \eta_k^2 (\sqrt{c}f(k) + \sigma^2) \text{Tr} \left(C^2 \prod_{j=k+1}^t (I - \eta_j(C + \lambda_j I))^2 \right)$$

when $t > 1$. Then, by the error decomposition in Proposition A.2,

$$\mathbb{E}_{z^{t-1}} \left[\left\| (H_t - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right] \leq f(t) \quad (\text{B.18})$$

for any $t \geq 1$. Choose

$$M = f(\lfloor t_0 \rfloor + 1) + \frac{c_1 \bar{\lambda}^{\min\{2r+1, 2\}} + c_2 \bar{\eta}^{-2(r+\alpha)} + c_3 + c_4 \sigma^2 t_0^{-\theta_1} \log t_0}{1 - c_4 \sqrt{c} t_0^{-\theta_1} \log t_0}.$$

Then, (B.17) holds for any $t \leq \lfloor t_0 \rfloor + 1$. Suppose (B.17) holds until some $t \geq \lfloor t_0 \rfloor + 1$. For $t+1$, set $\alpha = 1/2$, corresponding to the prediction error $\mathbb{E}_{z^t} \left[\left\| (H_{t+1} - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right]$. Since Assumption 3 is satisfied with $s = 1$, we set $s = 1$ accordingly. Using Proposition A.2, Proposition B.1, Proposition B.4, Proposition B.8 and Proposition B.9, we obtain that

$$\begin{aligned} & \mathbb{E}_{z^t} \left[\left\| (H_{t+1} - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right] \\ & \leq c_1 (\bar{\lambda}(t+t_0)^{-\theta_2})^{\min\{2r+1, 2\}} + c_2 \bar{\eta}^{-2(r+\alpha)} (t_0+1)^{2\bar{\eta}\bar{\lambda}} (t+t_0)^{-2(r+\alpha)(1-\theta_1)-2\bar{\eta}\bar{\lambda}} \\ & \quad + c_3 (t+t_0)^{-2\theta_2 \min\{r, 1\} + \theta_1 - 1} + c_4 (\sqrt{c}M + \sigma^2) (t+t_0)^{-\theta_1} \log(t+t_0) \\ & \leq c_1 \bar{\lambda}^{\min\{2r+1, 2\}} + c_2 \bar{\eta}^{-2(r+\alpha)} + c_3 + c_4 (\sqrt{c}M + \sigma^2) (t+t_0)^{-\theta_1} \log(t+t_0) \\ & \leq c_1 \bar{\lambda}^{\min\{2r+1, 2\}} + c_2 \bar{\eta}^{-2(r+\alpha)} + c_3 + c_4 (\sqrt{c}M + \sigma^2) t_0^{-\theta_1} \log t_0. \end{aligned}$$

where the last inequality holds when $t_0 \geq \exp\{\frac{1}{\theta_1}\}$. Since c_4 is independent of t_0 , for sufficiently large t_0 , we have

$$c_4 \sqrt{c} t_0^{-\theta_1} \log t_0 < 1.$$

By (B.18), it follows that

$$\mathbb{E}_{z^t} \left[\left\| (H_{t+1} - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right] \leq M,$$

which advances the induction.

The proof is then complete. $\square$

Next, we establish a similar bound for the finite-horizon setting.

Proposition B.13. *Under Assumption 2 and Assumption 4, if $t_0 = \theta_2 = \theta_1 = 0$, suppose $\eta_1(\kappa^2 + \lambda_1) \leq 1$ and*

$$\eta_1 < \frac{1}{6c\kappa^2 \left(1 + \frac{1}{2e\theta_3}\right)}.$$

Then, for any $T \geq 2$, there exists a constant $\widetilde{M} = \widetilde{M}(\lambda_1, \eta_1, \theta_3, \|S^\dagger\|_{\text{HS}}, r, \sigma^2, c)$ independent of T , such that

$$\mathbb{E}_{z^{t-1}} \left[\left\| (H_t - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right] \leq \widetilde{M}, \quad (\text{B.19})$$

for any $t \in \mathbb{N}_T$.

Proof. We prove this proposition by induction. Set

$$\widetilde{M} = \kappa^2 \|H^\dagger\|_{\text{HS}}^2 + \frac{c_1 \lambda_1^{\min\{2r+1, 2\}} + c_2 \eta_1^{-(2r+1)} + 6\sqrt{c}\sigma^2 \kappa^2 \left(1 + \frac{1}{2e\theta_3}\right) \eta_1}{1 - 6c\kappa^2 \left(1 + \frac{1}{2e\theta_3}\right) \eta_1}.$$

For $t = 1$, it is clear that

$$\mathbb{E}_{z^0} \left[\left\| (H_1 - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \right] \leq \kappa^2 \|H^\dagger\|_{\text{HS}}^2 \leq \widetilde{M}.$$

Assume that (B.19) holds from 1 to t . We now prove that it also holds for $t + 1$. Using Proposition A.2, Proposition B.1, Proposition B.4 with $t_0 = 0$ and $\mathcal{T}_3 = 0$, we have

$$\mathbb{E}_{z^t} \left[\left\| (H_{t+1} - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right] \leq c_1 \lambda_1^{\min\{2r+1, 2\}} + c_2 \eta_1^{-(2r+1)} + \mathcal{T}_4. \quad (\text{B.20})$$

Note that Proposition B.11 cannot be used in the induction process, because the current step size η_t relies on the total number of iterations T . Therefore, we re-estimate $\mathcal{T}_4$. By the definition of $\mathcal{T}_4$ in Proposition A.2 and the induction hypothesis,

$$\begin{aligned} \mathcal{T}_4 &\leq 6\sqrt{c}(\sqrt{c}\widetilde{M} + \sigma^2) \bar{\eta}^2 \sum_{i=1}^t \text{Tr} \left(C^2 (I - \bar{\eta}(C + \bar{\lambda}I)^{2(t-i)}) \right) \\ &\leq 6\sqrt{c}(\sqrt{c}\widetilde{M} + \sigma^2) \text{Tr}(C) \bar{\eta}^2 \left(\kappa^2 + \sum_{i=1}^{t-1} \|C(I - \bar{\eta}(C + \bar{\lambda}I)^{2i})\| \right) \\ &\leq 6\sqrt{c}(\sqrt{c}\widetilde{M} + \sigma^2) \text{Tr}(C) \bar{\eta}^2 \left(\kappa^2 + \frac{1}{2e} \sum_{i=1}^{t-1} (\bar{\eta}i)^{-1} \exp\{-2\bar{\eta}\bar{\lambda}i\} \right), \end{aligned}$$

where we have used Lemma B.2 (2) with $\beta = 1$. Using $\eta_1 \kappa^2 \leq 1$, $\bar{\eta} = \eta_1 T^{-\theta_3}$ and $\sum_{i=1}^{t-1} i^{-1} \leq 1 + \log T$, it follows that

$$\begin{aligned} \mathcal{T}_4 &\leq 6\sqrt{c}(\sqrt{c}\widetilde{M} + \sigma^2) \text{Tr}(C) \eta_1 \left(1 + \frac{1}{2e} (1 + \log T) \right) T^{-\theta_3} \\ &\leq 6\sqrt{c}(\sqrt{c}\widetilde{M} + \sigma^2) \text{Tr}(C) \left(1 + \frac{1}{2e\theta_3} \right) \eta_1, \end{aligned}$$

where we have used the fact that $\sup_{x>0} (1 + \log x) x^{-\theta_3} = \frac{1}{\theta_3} \exp\{\theta_3 - 1\}$. Substituting this into (B.20) yields that

$$\begin{aligned} \mathbb{E}_{z^t} \left[\left\| (H_{t+1} - H^\dagger) C^{1/2} \right\|_{\text{HS}}^2 \right] &\leq c_1 \lambda_1^{\min\{2r+1, 2\}} + c_2 \eta_1^{-(2r+1)} + 6\sqrt{c}(\sqrt{c}\widetilde{M} + \sigma^2) \kappa^2 \left(1 + \frac{1}{2e\theta_3} \right) \eta_1 \\ &\leq \widetilde{M}, \end{aligned}$$

which advances the induction.

The proof is thus complete. $\square$

C Convergence Analysis in Expectation

In this section, we prove the error bounds in expectation provided by Subsection 2.2.

Proof of Theorem 2.3. Let $\theta_1 + \theta_2 = 1$ and $\alpha = \frac{1}{2}$. If $T \geq t_0 + 1$, from Proposition A.2, Proposition B.1, Proposition B.4, Proposition B.8, Proposition B.9 and Proposition B.12 with $\alpha = \frac{1}{2}$ and $0 < s \leq 1$,

there holds

$$\begin{aligned}\mathbb{E}_{z^T}[\mathcal{E}(H_{T+1}) - \mathcal{E}(H^\dagger)] &\leq c_1 (\bar{\lambda}(T+t_0)^{-\theta_2})^{\min\{2r+1,2\}} \\ &\quad + c_2 \bar{\eta}^{-(2r+1)} (t_0+1)^{2\bar{\eta}\bar{\lambda}} (T+t_0)^{-(2r+1)(1-\theta_1)-2\bar{\eta}\bar{\lambda}} \\ &\quad + c_3 (T+t_0)^{-2\theta_2 \min\{r,1\} + \theta_1 - 1} \\ &\quad + c_4 (\sqrt{c}M + \sigma^2) \begin{cases} (T+t_0)^{-\theta_1}, & \text{when } s < 1, \\ (T+t_0)^{-\theta_1} \log(T+t_0), & \text{when } s = 1. \end{cases}\end{aligned}$$

We choose $\theta_1 = \frac{2 \min\{r+1/2,1\}}{1+2 \min\{r+1/2,1\}}$ and $\theta_2 = \frac{1}{1+2 \min\{r+1/2,1\}}$, then

$$\mathbb{E}_{z^T}[\mathcal{E}(H_{T+1}) - \mathcal{E}(H^\dagger)] \leq c_{1,1} \begin{cases} (T+t_0)^{-\frac{2 \min\{r+1/2,1\}}{1+2 \min\{r+1/2,1\}}} \log(T+t_0), & \text{when } s = 1, \\ (T+t_0)^{-\frac{2 \min\{r+1/2,1\}}{1+2 \min\{r+1/2,1\}}}, & \text{when } s < 1, \end{cases} \quad (\text{C.1})$$

for any $T \geq t_0 + 1$, where the constant $c_{1,1} = c_1 \bar{\lambda}^{\min\{2r+1,2\}} + c_2 \bar{\eta}^{-(2r+1)} (t_0+1)^{2\bar{\eta}\bar{\lambda}} + c_3 + c_4 (\sqrt{c}M + \sigma^2)$ is independent of T . Let $c_{1,1}$ be sufficiently large such that (C.1) holds true for $1 \leq T < t_0 + 1$.

We then finish the proof. $\square$

Proof of Theorem 2.4. Let $\theta_1 + \theta_2 = 1$ and $\alpha = 0$. If $T \geq t_0 + 1$, from Proposition A.2, Proposition B.1, Proposition B.4, Proposition B.6, Proposition B.9 and Proposition B.12 with $\alpha = 0$ and $0 < s \leq 1$, there holds

$$\begin{aligned}\mathbb{E}_{z^T}[\|H_{T+1} - H^\dagger\|_{\text{HS}}^2] &\leq c_1 (\bar{\lambda}(T+t_0)^{-\theta_2})^{2 \min\{r,1\}} \\ &\quad + c_2 \bar{\eta}^{-2r} (t_0+1)^{2\bar{\eta}\bar{\lambda}} (T+t_0)^{-2r(1-\theta_1)-2\bar{\eta}\bar{\lambda}} \\ &\quad + c_3 (T+t_0)^{-2\theta_2 \min\{r,1\}} \\ &\quad + c_4 (\sqrt{c}M + \sigma^2) (T+t_0)^{-(1+s)\theta_1 + s}.\end{aligned}$$

We choose $\theta_1 = \frac{s+2 \min\{r,1\}}{1+s+2 \min\{r,1\}}$ and $\theta_2 = \frac{1}{1+s+2 \min\{r,1\}}$, then

$$\mathbb{E}_{z^T}[\|H_{T+1} - H^\dagger\|_{\text{HS}}^2] \leq c_{1,2} (T+t_0)^{-\frac{2 \min\{r,1\}}{1+s+2 \min\{r,1\}}}, \quad (\text{C.2})$$

for any $T \geq t_0 + 1$, where $c_{1,2} = c_1 \bar{\lambda}^{2 \min\{r,1\}} + c_2 \bar{\eta}^{-2r} (t_0+1)^{2\bar{\eta}\bar{\lambda}} + c_3 + c_4 (\sqrt{c}M + \sigma^2)$ is a constant independent of T . Let $c_{1,2}$ be sufficiently large such that (C.2) holds true for $1 \leq T < t_0 + 1$.

The proof is complete. $\square$

Proof of Theorem 2.5. If the conditions $\eta_1(\kappa^2 + \lambda_1) \leq 1$ and

$$\eta_1 < \frac{1}{6c\kappa^2 \left(1 + \frac{1}{2e\theta_3}\right)}$$

hold, then, by Proposition A.2, Proposition B.1, Proposition B.4, Proposition B.11, and Proposition B.13 with $\alpha = 1/2$ and $0 < s \leq 1$, we obtain that

$$\begin{aligned}\mathbb{E}_{z^T}[\mathcal{E}(H_{T+1}) - \mathcal{E}(H^\dagger)] &\leq c_1 (\lambda_1 T^{-\theta_4})^{\min\{2r+1,2\}} \\ &\quad + c_2 \eta_1^{-(2r+1)} T^{-(2r+1)(1-\theta_3)} \exp\{-\tau \eta_1 \lambda_1 T^{1-\theta_4-\theta_3}\} \\ &\quad + \tilde{c}_4 \begin{cases} T^{-\theta_3} \log T, & \text{when } s = 1, \\ T^{-\theta_3}, & \text{when } s < 1. \end{cases}\end{aligned}$$

We choose $\theta_3 = \frac{2r+1}{2r+2}$ and $\theta_4 \geq \frac{2r+1}{(2r+2) \min\{2r+1,2\}}$, then

$$\mathbb{E}_{z^T}[\mathcal{E}(H_{T+1}) - \mathcal{E}(H^\dagger)] \leq c_{1,3} \begin{cases} T^{-\frac{2r+1}{2r+2}}, & \text{when } s < 1, \\ T^{-\frac{2r+1}{2r+2}} \log T, & \text{when } s = 1, \end{cases}$$

where $c_{1,3} = c_1 \lambda_1^{\min\{2r+1,2\}} + c_2 \eta_1^{-(2r+1)} + \tilde{c}_4$ is a constant independent of T .

The proof is complete. $\square$

Proof of Theorem 2.6. If the conditions $\eta_1(\kappa^2 + \lambda_1) \leq 1$ and

$$\eta_1 < \frac{1}{6c\kappa^2 \left(1 + \frac{1}{2e\theta_3}\right)}$$

hold, using Proposition A.2, Proposition B.1, Proposition B.4, Proposition B.11, and Proposition B.13 with $\alpha = 0$ and $0 < s \leq 1$, there holds

$$\begin{aligned} \mathbb{E}_{z^T} \left[\|H_{T+1} - H^\dagger\|_{\text{HS}}^2 \right] &\leq c_1 (\lambda_1 T^{-\theta_4})^{\min\{2r,2\}} \\ &\quad + c_2 \eta_1^{-2r} T^{-2r(1-\theta_3)} \exp\{-\tau \eta_1 \lambda_1 T^{1-\theta_4-\theta_3}\} \\ &\quad + \tilde{c}_4 T^{-(1+s)\theta_3+s \min\{1,\theta_3+\theta_4\}}. \end{aligned} \quad (\text{C.3})$$

Choosing $\theta_3 = \frac{2r+s}{1+2r+s}$ and $\theta_4 \geq \frac{2r}{(1+2r+s)\min\{2r,2\}}$, then

$$\mathbb{E}_{z^T} \left[\|H_{T+1} - H^\dagger\|_{\text{HS}}^2 \right] \leq c_{1,4} T^{-\frac{2r}{1+2r+s}},$$

where $c_{1,4} = c_1 \lambda_1^{\min\{2r,2\}} + c_2 \eta_1^{-2r} + \tilde{c}_4$ is a constant independent of T .

The proof is complete. $\square$

Remark 4. If we choose $\theta_3 + \theta_4 < 1$ in the above two proofs under the constant step size, then $\exp\{-\tau T^{1-\theta_4-\theta_3}\} = o(T^{-k})$ for any $k > 0$. Although the second term on the right-hand side of (C.3) decays faster than any polynomial, the overall learning rate would be slower than that achieved by our results.

D Convergence Analysis in High Probability

In this section, we derive the high-probability error bounds presented in Subsection 2.3. Our proofs are mainly based on the following proposition. This proposition is from (Tarres and Yao, 2014, Proposition A.3), and is an extension of (Pinelis, 1994, Theorem 3.4).

Proposition D.1. Let $(\mathcal{F}_i)_{i \geq 0}$ be a filtration, and denote by

$$\mathbb{E}_{i-1}[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_{i-1}]$$

the conditional expectation with respect to $\mathcal{F}_{i-1}$.

Let $(\xi_i)_{i \geq 1}$ be a martingale difference sequence in a Hilbert space with respect to $(\mathcal{F}_i)$, i.e., each ξ_i is $\mathcal{F}_i$ -measurable and satisfies $\mathbb{E}_{i-1}[\xi_i] = 0$. Suppose that $\|\xi_i\| \leq M_\xi$ and $\sum_{i=1}^t \mathbb{E}_{i-1} \|\xi_i\|^2 \leq \tau^2$ almost surely for some constant $M_\xi > 0$ and $\tau > 0$. Then, for any $\delta \in (0, 2/e)$, with probability at least $1 - \delta$, the following inequality holds:

$$\sup_{1 \leq k \leq t} \left\| \sum_{i=1}^k \xi_i \right\| \leq 2 \left(\frac{M_\xi}{3} + \tau \right) \log \frac{2}{\delta}.$$

Additionally,

$$\sup_{1 \leq k \leq t} \left\| \sum_{i=1}^k \xi_i \right\|^2 \leq 8 (M_\xi^2 + \tau^2) \log^2 \frac{2}{\delta}.$$

By Proposition A.3, since $\mathcal{T}_1$, $\mathcal{T}_2$ and $\mathcal{T}_3$ have already been bounded in Section B, our goal is to bound the remaining term $6 \left\| \sum_{t=1}^T \chi_t \right\|_{\text{HS}}^2$ with high probability. According to Proposition D.1, this requires the uniform bound on $\|\chi_t\|_{\text{HS}}$ for $1 \leq t \leq T$. Using (A.14), it is sufficient to bound $\|H_t\|_{\text{HS}}$. However, $\|H_t\|_{\text{HS}}$ may grow rapidly with increasing t . Therefore, we first establish a high-probability bound on $\|H_t\|_{\text{HS}}$, which motivates the decomposition of $H_t - H^\dagger$ into $L_t + R_t$ as follows.

Let us denote $\phi(x_t) \otimes \phi(x_t)$ by C_t . We define two random processes $(L_t)_{t \geq 1}$ and $(R_t)_{t \geq 1}$ recursively by

$$L_1 = -H^\dagger, \quad R_1 = \mathbf{0},$$

and for any $t \geq 1$,

$$\begin{aligned} L_{t+1} &:= L_t (I - \eta_t (C + \lambda_t I)) - \eta_t \lambda_t H^\dagger, \\ R_{t+1} &:= R_t (I - \eta_t (C_t + \lambda_t I)) + \eta_t (y_t - H^\dagger \phi(x_t)) \otimes \phi(x_t) + \eta_t L_t (C - C_t). \end{aligned} \quad (\text{D.1})$$

Note that for any $t \geq 1$, L_t is deterministic, while R_t depends on z^{t-1} and is independent of z_t . Moreover, by induction, one can verify that

$$L_t + R_t = H_t - H^\dagger \quad (\text{D.2})$$

for all $t \geq 1$.

We then provide a bound on L_t in Lemma D.2.

Lemma D.2. *Suppose that $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$ holds. Then, for any $t \geq 1$,*

$$\|L_t\|_{\text{HS}} \leq \|H^\dagger\|_{\text{HS}}.$$

Proof. We prove it by induction. For $t = 1$, $\|L_1\|_{\text{HS}} = \|H^\dagger\|_{\text{HS}}$. Since $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$, there holds $1 - \eta_t(\kappa^2 + \lambda_t) \geq 0$ for any $t \geq 1$. Thus,

$$\|L_{t+1}\|_{\text{HS}} \leq \|L_t\|_{\text{HS}}(1 - \eta_t \lambda_t) + \eta_t \lambda_t \|H^\dagger\|_{\text{HS}},$$

which implies that $\|L_t\|_{\text{HS}} \leq \|H^\dagger\|_{\text{HS}}$ for any $t \geq 1$. We then finish the proof. $\square$

By the following proposition, we can, with high probability, control the increasing rate of R_t in the online setting.

Proposition D.3. *Under Assumption 5, suppose that $\theta_1 + \theta_2 = 1$, $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$, and $\bar{\eta}\bar{\lambda} \geq \theta_1$. Then, with probability at least $1 - \delta$, there holds*

$$\|R_t\|_{\text{HS}} \leq d_2(t + t_0)^{\frac{1}{2} - \theta_1} \log(t + t_0) \log \frac{2}{\delta}, \quad 1 \leq t \leq T,$$

where $d_2 = d_2(\bar{\eta}, \bar{\lambda}, \theta_1, t_0, M_\rho, \|S^\dagger\|_{\text{HS}})$ is a constant independent of t , T , and δ .

Proof. Denote $(y_t - H^\dagger \phi(x_t)) \otimes \phi(x_t) + L_t(C - C_t)$ by K_t . Then, by applying induction to (D.1), R_{t+1} can be expressed as

$$R_{t+1} = \sum_{i=1}^t \eta_i K_i \prod_{j=i+1}^t (I - \eta_j (C_j + \lambda_j I)). \quad (\text{D.3})$$

Since $\mathbb{E}_{z_i}[K_i] = 0$, for each fixed t , consider the reversed sequence

$$\left(\eta_i K_i \prod_{j=i+1}^t (I - \eta_j (C_j + \lambda_j I)) \right)_{i=t, t-1, \dots, 1}.$$

Define the filtration $(\mathcal{F}_{t,\ell})_{\ell=0}^t$ by

$$\mathcal{F}_{t,0} := \{\emptyset, \emptyset^c\}, \quad \mathcal{F}_{t,\ell} := \sigma(z_j : j = t - \ell + 1, \dots, t), \quad \ell = 1, \dots, t.$$

Then the above reversed sequence forms a martingale difference sequence with respect to $(\mathcal{F}_{t,\ell})_{\ell=0}^t$. Indeed, for each $1 \leq i \leq t$, the product $\prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I))$ is $\sigma(z_{i+1}, \dots, z_t)$ -measurable, while K_i depends only on z_i . Hence, by the independence of the samples and the fact that $\mathbb{E}_{z_i}[K_i] = 0$,

$$\mathbb{E} \left[\eta_i K_i \prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I)) \middle| \sigma(z_{i+1}, \dots, z_t) \right] = 0.$$

Moreover, the conditional quadratic variation required in Proposition D.1 is exactly controlled by

$$\sum_{i=1}^t \mathbb{E}_{z_i} \left[\left\| \eta_i K_i \prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I)) \right\|_{\text{HS}}^2 \right],$$

which will be estimated below. We apply Proposition D.1 to bound R_{t+1} for each fixed t .

Using Lemma D.2 and Assumption 5, we have

$$\|K_t\|_{\text{HS}} \leq \kappa M_\rho + 3\kappa^2 \|H^\dagger\|_{\text{HS}}. \quad (\text{D.4})$$

By Lemma B.3,

$$\begin{aligned} \left\| \prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I)) \right\| &\leq \exp \left\{ - \sum_{j=i+1}^t \eta_j \lambda_j \right\} \\ &\leq \left(\frac{t+t_0+1}{i+t_0+1} \right)^{-\bar{\eta}\bar{\lambda}} \leq 2^{\bar{\eta}\bar{\lambda}} \left(\frac{t+t_0}{i+t_0} \right)^{-\bar{\eta}\bar{\lambda}}. \end{aligned} \quad (\text{D.5})$$

Combining (D.3), (D.4) and (D.5), and using the assumption that $\bar{\eta}\bar{\lambda} \geq \theta_1$, we obtain

$$\left\| \eta_i K_i \prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I)) \right\|_{\text{HS}} \leq (\kappa M_\rho + 3\kappa^2 \|H^\dagger\|_{\text{HS}}) 2^{\bar{\eta}\bar{\lambda}} \bar{\eta} (t+t_0)^{-\theta_1}, \quad 1 \leq i \leq t.$$

Moreover,

$$\begin{aligned} &\sum_{i=1}^t \mathbb{E}_{z_i} \left[\left\| \eta_i K_i \prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I)) \right\|_{\text{HS}}^2 \right] \\ &\leq (\kappa M_\rho + 3\kappa^2 \|H^\dagger\|_{\text{HS}})^2 2^{2\bar{\eta}\bar{\lambda}} \bar{\eta}^2 \sum_{i=1}^t (i+t_0)^{-2\theta_1} \left(\frac{t+t_0}{i+t_0} \right)^{-2\bar{\eta}\bar{\lambda}} \\ &\leq (\kappa M_\rho + 3\kappa^2 \|H^\dagger\|_{\text{HS}})^2 2^{4\bar{\eta}\bar{\lambda}} \frac{\bar{\eta}^2}{2\bar{\eta}\bar{\lambda} - 2\theta_1 + 1} (t+t_0+1)^{1-2\theta_1}. \end{aligned}$$

Then, by Proposition D.1, with probability at least $1 - \delta_{t+1}$,

$$\|R_{t+1}\|_{\text{HS}} \leq d_1 (t+t_0+1)^{\frac{1}{2}-\theta_1} \log \frac{2}{\delta_{t+1}},$$

for some constant d_1 that is independent of t , δ_{t+1} and T .

Now, for any $\delta \in (0, 2/e)$, choose $\delta_t = \delta(t+t_0)^{-2} t_0$ for any $1 \leq t \leq T$. Then $\sum_{t=1}^T \delta_t \leq \delta$ and

$$\|R_t\|_{\text{HS}} \leq d_2 (t+t_0)^{\frac{1}{2}-\theta_1} \log(t+t_0) \log \frac{2}{\delta}, \quad 1 \leq t \leq T,$$

where d_2 is a constant independent of t , T , and δ .

The proof is complete. $\square$

Recall that $\chi_t = \eta_t \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j (C + \lambda_j I))$ in Proposition A.3. Define

$$\tilde{\chi}_t := \chi_t \mathbb{1}_{A_t}, \quad (\text{D.6})$$

where

$$A_t := \left\{ \|R_t\|_{\text{HS}} \leq d_2(t+t_0)^{\frac{1}{2}-\theta_1} \log(t+t_0) \log \frac{2}{\delta} \right\}. \quad (\text{D.7})$$

Then, A_t is independent of z_t , and $\tilde{\chi}_t$ depends on $z^t = \{z_1, z_2, \dots, z_t\}$. Moreover, for any $t \geq 1$, we have $\mathbb{E}_{z_t} [\tilde{\chi}_t] = \mathbb{1}_{A_t} \mathbb{E}_{z_t} [\chi_t] = 0$. By Proposition D.3,

$$\mathbb{P}(\tilde{\chi}_t = \chi_t \text{ for any } 1 \leq t \leq T) \geq 1 - \delta.$$

In the next proposition, we provide bounds for $\sup_{1 \leq t \leq T} \|\tilde{\chi}_t\|_{\text{HS}}^2$ and $\sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2$ in preparation for applying Proposition D.1.

Proposition D.4. *Under Assumptions 3 and 5, suppose that $\theta_1 + \theta_2 = 1$, $\alpha \in [0, \frac{1}{2}]$, $(t_0 + 1)^{\theta_1} \geq \bar{\eta}(\kappa^2 + \bar{\lambda})$, $\bar{\eta}\bar{\lambda} \geq \theta_1$ and $\bar{\eta}\bar{\lambda} \geq 2\theta_1 - \frac{1}{2}$. Then,*

(1) *the Hilbert-Schmidt norm of $\tilde{\chi}_t$ is uniformly bounded as follows:*

$$\sup_{1 \leq t \leq T} \|\tilde{\chi}_t\|_{\text{HS}}^2 \leq M_1^2,$$

where

$$M_1^2 := 2d_3^2(T+t_0)^{-2\theta_1} + 2d_3^2(T+t_0)^{1-4\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta};$$

(2) *the total squared Hilbert-Schmidt norm in expectation is bounded by*

$$\sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 \leq \tau_1^2,$$

where τ_1 is defined as

$$\tau_1^2 := d_5 \begin{cases} (T+t_0)^{-\theta_1} + (T+t_0)^{1-3\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta}, \\ \text{if } 1 + 2\alpha - s > 1, \\ (T+t_0)^{-\theta_1} \log(T+t_0) + (T+t_0)^{1-3\theta_1} \log^3(T+t_0) \log^2 \frac{2}{\delta}, \\ \text{if } 1 + 2\alpha - s = 1, \\ (T+t_0)^{s-2\alpha-\theta_1(1+s-2\alpha)} + (T+t_0)^{1+s-2\alpha-\theta_1(3+s-2\alpha)} \log^2(T+t_0) \log^2 \frac{2}{\delta}, \\ \text{if } 0 \leq 1 + 2\alpha - s < 1. \end{cases}$$

Here, $d_3 = d_3(\bar{\eta}, \bar{\lambda}, \theta_1, t_0, M_\rho, \|S^\dagger\|_{\text{HS}}, \alpha)$ and $d_5 = d_5(\bar{\eta}, \bar{\lambda}, \theta_1, t_0, M_\rho, \|S^\dagger\|_{\text{HS}}, \alpha, s)$ are constants independent of T and δ .

Proof. (1) From Proposition A.3 and (D.2), using Lemma D.2 yields that

$$\begin{aligned} \|\tilde{\chi}_t\|_{\text{HS}} &\leq 2\eta_t \kappa \left(M_\rho + \kappa \|L_t + R_t + H^\dagger\|_{L_{\text{HS}}^\infty} \right) \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j (C + \lambda_j I)) \right\| \mathbb{1}_{A_t} \\ &\leq 2\eta_t \kappa \left(M_\rho + 2\kappa \|H^\dagger\|_{\text{HS}} + \kappa d_2(t+t_0)^{\frac{1}{2}-\theta_1} \log(t+t_0) \log \frac{2}{\delta} \right) \\ &\quad \times \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j (C + \lambda_j I)) \right\|. \end{aligned} \quad (\text{D.8})$$

If $\alpha > 0$, by Lemma B.2 (1) and Lemma B.3, we obtain

$$\begin{aligned} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\| &\leq \exp \left\{ - \sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{2(\kappa^{2\alpha} + (\alpha/e)^\alpha)}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^\alpha} \\ &\leq 2(\kappa^{2\alpha} + (\alpha/e)^\alpha) \left(\frac{T + t_0 + 1}{t + t_0 + 1} \right)^{-\bar{\eta}\bar{\lambda}}. \end{aligned} \quad (\text{D.9})$$

If $\alpha = 0$, then

$$\left\| \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\| \leq \exp \left\{ - \sum_{j=t+1}^T \eta_j \lambda_j \right\} \leq \left(\frac{T + t_0 + 1}{t + t_0 + 1} \right)^{-\bar{\eta}\bar{\lambda}}. \quad (\text{D.10})$$

Substituting (D.9) or (D.10) into (D.8) yields that

$$\|\tilde{\chi}_t\|_{\text{HS}} \leq d_3(T + t_0)^{-\bar{\eta}\bar{\lambda}}(t + t_0)^{\bar{\eta}\bar{\lambda} - \theta_1} \left(1 + (t + t_0)^{\frac{1}{2} - \theta_1} \log(t + t_0) \log \frac{2}{\delta} \right),$$

where d_3 is a constant independent of δ , T , and t . If $\bar{\eta}\bar{\lambda} \geq \theta_1$ and $\bar{\eta}\bar{\lambda} \geq 2\theta_1 - \frac{1}{2}$, the right-hand side of the above inequality achieves its maximum within $1 \leq t \leq T$ at $t = T$. Therefore, we obtain

$$\sup_{1 \leq t \leq T} \|\tilde{\chi}_t\|_{\text{HS}} \leq d_3(T + t_0)^{-\theta_1} + d_3(T + t_0)^{\frac{1}{2} - 2\theta_1} \log(T + t_0) \log \frac{2}{\delta}.$$

Thus,

$$\sup_{1 \leq t \leq T} \|\tilde{\chi}_t\|_{\text{HS}}^2 \leq 2d_3^2(T + t_0)^{-2\theta_1} + 2d_3^2(T + t_0)^{1 - 4\theta_1} \log^2(T + t_0) \log^2 \frac{2}{\delta}.$$

(2) By (D.6), we see that

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 &= \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z_t} \left[\left\| \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \mathbb{1}_{A_t} \right] \\ &\leq \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z_t} \left[\left\| (y_t - H_t \phi(x_t)) \otimes \phi(x_t) C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \mathbb{1}_{A_t} \right] \\ &= \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z_t} \left[\|y_t - H_t \phi(x_t)\|_{\mathcal{Y}}^2 \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \mathbb{1}_{A_t} \right], \end{aligned}$$

where the last inequality uses the definition of the Hilbert-Schmidt norm. By (D.2), Lemma D.2, and (D.7), it follows that

$$\|y_t - H_t \phi(x_t)\|_{\mathcal{Y}}^2 \mathbb{1}_{A_t} \leq 2M_\rho^2 + 2\kappa^2 \left(8\|H^\dagger\|_{\text{HS}}^2 + 2d_2^2(t + t_0)^{1 - 2\theta_1} \log^2(t + t_0) \log^2 \frac{2}{\delta} \right).$$

Then,

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 &\leq \sum_{t=1}^T \eta_t^2 \left(2M_\rho^2 + 2\kappa^2 \left(8\|H^\dagger\|_{\text{HS}}^2 + 2d_2^2(t + t_0)^{1 - 2\theta_1} \log^2(t + t_0) \log^2 \frac{2}{\delta} \right) \right) \\ &\quad \times \mathbb{E}_{z_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2. \end{aligned} \quad (\text{D.11})$$

By the definition of the trace of operators and using Assumption 3 and (3) in Lemma B.2, it follows that if $2\alpha + 1 - s > 0$,

$$\begin{aligned}
& \mathbb{E}_{z_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \\
&= \text{Tr} \left(C^{2\alpha+1} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right) \\
&\leq \text{Tr}(C^s) \left\| C^{2\alpha+1-s} \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I))^2 \right\| \\
&\leq \text{Tr}(C^s) \frac{2(\kappa^{2(2\alpha+1-s)}) + ((2\alpha + 1 - s)/(2e))^{2\alpha+1-s}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^{2\alpha+1-s}} \\
&\quad \times \exp \left\{ -2 \sum_{j=t+1}^T \eta_j \lambda_j \right\},
\end{aligned} \tag{D.12}$$

else if $2\alpha + 1 - s = 0$,

$$\mathbb{E}_{z_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \leq \text{Tr}(C^s) \exp \left\{ -2 \sum_{j=t+1}^T \eta_j \lambda_j \right\}. \tag{D.13}$$

Substituting (D.12) (or (D.13)) into (D.11), we deduce that there exists a constant d_4 independent of δ , t , and T , such that

$$\begin{aligned}
\sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 &\leq d_4 \sum_{t=1}^T (t + t_0)^{-2\theta_1} \left(1 + (t + t_0)^{1-2\theta_1} \log^2(T + t_0) \log^2 \frac{2}{\delta} \right) \\
&\quad \times \frac{\exp \left\{ -2 \sum_{j=t+1}^T \eta_j \lambda_j \right\}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^{2\alpha+1-s}}.
\end{aligned} \tag{D.14}$$

Now, we apply Proposition B.7 and Remark 3 with $v = 2\alpha + 1 - s$ to derive the bound. Let d_5 denote a constant independent of T and δ .

Case 1: If $2\alpha + 1 - s > 1$, then

$$\sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 \leq d_5 (T + t_0)^{-\theta_1} + d_5 (T + t_0)^{1-3\theta_1} \log^2(T + t_0) \log^2 \frac{2}{\delta}.$$

Case 2: If $2\alpha + 1 - s = 1$, then

$$\sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 \leq d_5 (T + t_0)^{-\theta_1} \log(T + t_0) + d_5 (T + t_0)^{1-3\theta_1} \log^3(T + t_0) \log^2 \frac{2}{\delta}.$$

Case 3: If $0 < 2\alpha + 1 - s < 1$, then

$$\sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 \leq d_5 (T + t_0)^{s-2\alpha-\theta_1(1+s-2\alpha)} + d_5 (T + t_0)^{1+s-2\alpha-\theta_1(3+s-2\alpha)} \log^2(T + t_0) \log^2 \frac{2}{\delta}.$$

Case 4: If $2\alpha + 1 - s = 0$, since

$$\begin{aligned} \exp \left\{ -2 \sum_{j=t+1}^T \eta_j \lambda_j \right\} &= \exp \left\{ -2\bar{\eta}\bar{\lambda} \sum_{j=t+1}^T (j+t_0)^{-1} \right\} \\ &\leq \left(\frac{T+t_0+1}{t+t_0+1} \right)^{-2\bar{\eta}\bar{\lambda}}, \end{aligned}$$

substituting this into (D.14) gives

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_{z_t} \|\tilde{\chi}_t\|_{\text{HS}}^2 &\leq d_5 (T+t_0)^{-2\bar{\eta}\bar{\lambda}} \left((T+t_0)^{2\bar{\eta}\bar{\lambda}-2\theta_1+1} + (T+t_0)^{2\bar{\eta}\bar{\lambda}+2-4\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta} \right) \\ &\leq d_5 \left((T+t_0)^{1-2\theta_1} + (T+t_0)^{2-4\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta} \right), \end{aligned}$$

which is consistent with the bound in Case 3.

The proof is complete. $\square$

Since $\tilde{\chi}_t$ is $\sigma(z_1, z_2, \dots, z_t)$ measurable and $\mathbb{E}_{z_t} [\tilde{\chi}_t] = 0$, $(\tilde{\chi}_t)_{1 \leq t \leq T}$ is a martingale difference sequence. Based on Proposition D.1, we derive the high-probability error bounds.

Proof of Theorem 2.7. By Proposition A.3 with $\alpha = \frac{1}{2}$,

$$\left\| (H_{T+1} - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + 6 \left\| \sum_{t=1}^T \chi_t \right\|_{\text{HS}}^2. \quad (\text{D.15})$$

Using Proposition D.3, there holds

$$\mathbb{P}(\tilde{\chi}_t = \chi_t \text{ for any } 1 \leq t \leq T) \geq 1 - \delta. \quad (\text{D.16})$$

Choose $\theta_1 = \frac{\min\{2r+1, 2\}}{1+\min\{2r+1, 2\}}$ and $\theta_2 = \frac{1}{1+\min\{2r+1, 2\}}$. Applying Proposition B.1, Proposition B.4 and Proposition B.8 with $\alpha = \frac{1}{2}$ and $T \geq t_0 + 1$, we have

$$\mathcal{T}_1 \leq c_1 \lambda_T^{\min\{2r+1, 2\}} = c_1 \bar{\lambda}^{\min\{2r+1, 2\}} (T+t_0)^{-\frac{\min\{2r+1, 2\}}{1+\min\{2r+1, 2\}}}, \quad (\text{D.17})$$

$$\begin{aligned} \mathcal{T}_2 &\leq c_2 \bar{\eta}^{-(2r+1)} (t_0+1)^{2\bar{\eta}\bar{\lambda}} (T+t_0)^{-(2r+1)(1-\theta_1)-2\bar{\eta}\bar{\lambda}} \\ &\leq c_2 \bar{\eta}^{-(2r+1)} (t_0+1)^{2\bar{\eta}\bar{\lambda}} (T+t_0)^{-\frac{\min\{2r+1, 2\}}{1+\min\{2r+1, 2\}}}, \end{aligned} \quad (\text{D.18})$$

and

$$\mathcal{T}_3 \leq c_3 (T+t_0)^{-2\theta_2 \min\{r, 1\} + \theta_1 - 1} \leq c_3 (T+t_0)^{-\frac{\min\{2r+1, 2\}}{1+\min\{2r+1, 2\}}}. \quad (\text{D.19})$$

Using Proposition D.1 and Proposition D.4, we deduce that with probability at least $1 - \delta$,

$$\left\| \sum_{i=1}^T \tilde{\chi}_t \right\|_{\text{HS}}^2 \leq 8 (M_1^2 + \tau_1^2) \log^2 \frac{2}{\delta}$$

with

$$\begin{aligned} M_1^2 &= 2d_3^2 (T+t_0)^{-2\theta_1} + 2d_3^2 (T+t_0)^{1-4\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta} \\ &\leq 2d_3^2 (T+t_0)^{-\theta_1} + 2d_3^2 (T+t_0)^{1-3\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta}, \end{aligned}$$

and

$$\tau_1^2 = d_5 \begin{cases} (T+t_0)^{-\theta_1} + (T+t_0)^{1-3\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta}, & \text{when } s < 1, \\ (T+t_0)^{-\theta_1} \log(T+t_0) + (T+t_0)^{1-3\theta_1} \log^3(T+t_0) \log^2 \frac{2}{\delta}, & \text{when } s = 1. \end{cases}$$

Therefore,

$$\begin{aligned} \left\| \sum_{i=1}^T \tilde{\chi}_t \right\|_{\text{HS}}^2 &\leq 8(2d_3^2 + d_5) \left((T+t_0)^{-\theta_1} + (T+t_0)^{1-3\theta_1} \log^2(T+t_0) \log^2 \frac{2}{\delta} \right) \\ &\quad \times \log^2 \frac{2}{\delta} \begin{cases} 1, & \text{when } s < 1, \\ \log(T+t_0), & \text{when } s = 1. \end{cases} \end{aligned} \quad (\text{D.20})$$

If $T \geq t_0 + 1$, combining (D.15), (D.16), (D.17), (D.18), (D.19), and (D.20), we obtain that there exists some constant $c_{2,1}$ independent of T and δ , such that

$$\begin{aligned} &\left\| (H_{T+1} - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \\ &\leq c_{2,1} \begin{cases} (T+t_0)^{-\theta_1} \log^2 \frac{2}{\delta} + (T+t_0)^{1-3\theta_1} \log^2(T+t_0) \log^4 \frac{2}{\delta}, & s < 1, \\ (T+t_0)^{-\theta_1} \log(T+t_0) \log^2 \frac{2}{\delta} + (T+t_0)^{1-3\theta_1} \log^3(T+t_0) \log^4 \frac{2}{\delta}, & s = 1. \end{cases} \end{aligned}$$

holds with probability at least $1 - 2\delta$.

Since it is easy to verify that $\left\| (H_{T+1} - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2$ for $1 \leq T < t_0 + 1$ can be bounded uniformly by a constant, we can choose $c_{2,1}$ to be sufficiently large such that the bound also holds true for $1 \leq T < t_0 + 1$.

The proof is complete. $\square$

Proof of Corollary 2.8. For any $t \geq 1$ and any $\delta \in (0, 2/e)$, apply Theorem 2.7 with $\delta_t = (t+t_0)^{-2}t_0\delta$, so that $\sum_{t \geq 1} \delta_t \leq \delta$. When $s < 1$, we have

$$\mathcal{E}(h_{t+1}) - \mathcal{E}(h^\dagger) \lesssim (t+t_0)^{-\theta_1} \log^4 \frac{2}{\delta_t} \lesssim (t+t_0)^{-\theta_1} \log^4(t+t_0) \log^4 \frac{2}{\delta}.$$

When $s = 1$, similarly, we have

$$\mathcal{E}(h_{t+1}) - \mathcal{E}(h^\dagger) \lesssim (t+t_0)^{-\theta_1} \log^5(t+t_0) \log^4 \frac{2}{\delta}.$$

The proof is complete. $\square$

Proof of Theorem 2.9. By Proposition A.3 with $\alpha = 0$,

$$\|H_{T+1} - H^\dagger\|_{\text{HS}}^2 \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + 6 \left\| \sum_{t=1}^T \chi_t \right\|_{\text{HS}}^2. \quad (\text{D.21})$$

According to Proposition D.3, we have

$$\mathbb{P}(\tilde{\chi}_t = \chi_t \text{ for any } 1 \leq t \leq T) \geq 1 - \delta. \quad (\text{D.22})$$

Applying Proposition B.1, Proposition B.4, and Proposition B.6 with $\alpha = 0$ and $T \geq t_0 + 1$, we obtain

$$\mathcal{T}_1 \leq c_1 \lambda_T^{\min\{2r, 2\}} = c_1 \bar{\lambda}^{\min\{2r, 2\}} (T+t_0)^{-\min\{2r, 2\}\theta_2}, \quad (\text{D.23})$$

$$\mathcal{T}_2 \leq c_2 \bar{\eta}^{-2r} (t_0 + 1)^{2\bar{\eta}\bar{\lambda}} (T+t_0)^{-2r(1-\theta_1)-2\bar{\eta}\bar{\lambda}}, \quad (\text{D.24})$$

and

$$\mathcal{T}_3 \leq c_3(T + t_0)^{-2\theta_2 \min\{r, 1\}}. \quad (\text{D.25})$$

Using Proposition D.1 and Proposition D.4, we deduce that with probability at least $1 - \delta$,

$$\left\| \sum_{i=1}^T \tilde{\chi}_t \right\|_{\text{HS}}^2 \leq 8(M_1^2 + \tau_1^2) \log^2 \frac{2}{\delta}.$$

Additionally, we have

$$M_1^2 = 2d_3^2(T + t_0)^{-2\theta_1} + 2d_3^2(T + t_0)^{1-4\theta_1} \log^2(T + t_0) \log^2 \frac{2}{\delta},$$

and

$$\tau_1^2 = d_5(T + t_0)^{s-\theta_1(1+s)} + d_5(T + t_0)^{1+s-\theta_1(3+s)} \log^2(T + t_0) \log^2 \frac{2}{\delta}.$$

Thus,

$$\begin{aligned} \left\| \sum_{i=1}^T \tilde{\chi}_t \right\|_{\text{HS}}^2 &\leq 8(2d_3^2 + d_5) \left((T + t_0)^{s-\theta_1(1+s)} + (T + t_0)^{1+s-\theta_1(3+s)} \right. \\ &\quad \left. \times \log^2(T + t_0) \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta}. \end{aligned} \quad (\text{D.26})$$

Combining (D.21), (D.22), (D.23), (D.24), (D.25), and (D.26), let $c_{2,2}$ denote a constant independent of T and δ ,

(1) if $2r + s < 1$, choose $\theta_1 = \frac{1+2\min\{r, 1\}+s}{3+2\min\{r, 1\}+s} = \frac{1+2r+s}{3+2r+s}$ and

$$\theta_2 = \frac{2}{3 + 2\min\{r, 1\} + s} = \frac{2}{3 + 2r + s},$$

then,

$$\begin{aligned} \|H_{T+1} - H^\dagger\|_{\text{HS}}^2 &\leq c_{2,2}(T + t_0)^{-\frac{4\min\{r, 1\}}{3+2\min\{r, 1\}+s}} \log^2(T + t_0) \log^4 \frac{2}{\delta} \\ &= c_{2,2}(T + t_0)^{-\frac{4r}{3+2r+s}} \log^2(T + t_0) \log^4 \frac{2}{\delta} \end{aligned}$$

holds with probability at least $1 - 2\delta$;

(2) if $2\min\{r, 1\} + s \geq 1$, i.e., $2r + s \geq 1$, choose $\theta_1 = \frac{2\min\{r, 1\}+s}{1+2\min\{r, 1\}+s}$ and $\theta_2 = \frac{1}{1+2\min\{r, 1\}+s}$, then

$$\begin{aligned} \|H_{T+1} - H^\dagger\|_{\text{HS}}^2 &\leq c_{2,2} \left((T + t_0)^{-\frac{2\min\{r, 1\}}{1+2\min\{r, 1\}+s}} + (T + t_0)^{-\frac{4\min\{r, 1\}+s-1}{1+2\min\{r, 1\}+s}} \log^2(T + t_0) \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \\ &\lesssim (T + t_0)^{-\frac{2\min\{r, 1\}}{1+2\min\{r, 1\}+s}} \log^4 \frac{2}{\delta} \end{aligned}$$

holds with probability at least $1 - 2\delta$.

Since $\|H_{T+1} - H^\dagger\|_{\text{HS}}^2$ for $1 \leq T < t_0 + 1$ can be uniformly bounded by a constant, we can choose $c_{2,2}$ to be sufficiently large so that the bounds also hold in this case.

The proof is then complete. $\square$

Proof of Corollary 2.10. For any $\delta \in (0, 2/e)$, using Theorem 2.9 with $\delta_t = (t + t_0)^{-2} t_0 \delta$, we derive the desired bounds. $\square$

Next, we focus on the finite-horizon setting and derive the corresponding high-probability error bounds.

Proposition D.5. Under Assumption 5, let $T \geq 2$, $t_0 = \theta_1 = \theta_2 = 0$, $\eta_t = \bar{\eta} = \eta_1 T^{-\theta_3}$ and $\lambda_t = \bar{\lambda} = \lambda_1 T^{-\theta_4}$. Suppose that $\eta_1(\kappa^2 + \lambda_1) \leq 1$. Then, for any $\delta \in (0, 2/e)$, with probability at least $1 - \delta$, there holds

$$\|R_t\|_{\text{HS}} \leq \frac{2d_6}{\log 2} T^{\frac{1}{2}-\theta_3} \log T \log \frac{2}{\delta}, \quad 1 \leq t \leq T,$$

where $d_6 = d_6(\eta_1, \theta_3, M_\rho, \|S^\dagger\|_{\text{HS}})$ is a constant independent of t , T , and δ .

Proof. The proof follows the same strategy as Proposition D.3. Define K_t as in the proof of Proposition D.3, and represent R_{t+1} in the same summation form. By (D.4), there holds

$$\left\| \eta_i K_i \prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I)) \right\|_{\text{HS}} \leq \eta_1 (\kappa M_\rho + 3\kappa^2 \|H^\dagger\|_{\text{HS}}) T^{-\theta_3},$$

and

$$\sum_{i=1}^t \mathbb{E}_{z_i} \left[\left\| \eta_i K_i \prod_{j=i+1}^t (I - \eta_j(C_j + \lambda_j I)) \right\|_{\text{HS}}^2 \right] \leq \eta_1^2 (\kappa M_\rho + 3\kappa^2 \|H^\dagger\|_{\text{HS}})^2 T^{1-2\theta_3}.$$

Then, by Proposition D.1, for some constant d_6 independent of t , T , and δ_{t+1} , with probability at least $1 - \delta_{t+1}$, it holds that

$$\|R_{t+1}\|_{\text{HS}} \leq d_6 T^{\frac{1}{2}-\theta_3} \log \frac{2}{\delta_{t+1}}.$$

For $\delta \in (0, 2/e)$, choose $\delta_t = \frac{\delta}{T}$ for any $1 \leq t \leq T$. Then we obtain

$$\|R_t\|_{\text{HS}} \leq \frac{2d_6}{\log 2} T^{\frac{1}{2}-\theta_3} \log T \log \frac{2}{\delta}, \quad 1 \leq t \leq T.$$

The proof is complete. $\square$

We now define

$$\bar{\chi}_t := \chi_t \mathbb{1}_{\bar{A}_t}, \quad (\text{D.27})$$

where

$$\bar{A}_t := \left\{ \|R_t\|_{\text{HS}} \leq \frac{2d_6}{\log 2} T^{\frac{1}{2}-\theta_3} \log T \log \frac{2}{\delta} \right\}. \quad (\text{D.28})$$

Note that $\bar{A}_t$ is independent of z_t , and for any $t \in \mathbb{N}_T$, we have $\mathbb{E}_{z_t}[\bar{\chi}_t] = 0$. Moreover, by Proposition D.5,

$$\mathbb{P}(\bar{\chi}_t = \chi_t \text{ for any } 1 \leq t \leq T) \geq 1 - \delta.$$

Proposition D.6. Suppose Assumptions 3 and 5 hold. Let $\alpha \in [0, \frac{1}{2}]$, $t_0 = \theta_1 = \theta_2 = 0$, $\eta_t \equiv \bar{\eta} = \eta_1 T^{-\theta_3}$ with $0 < \theta_3 < 1$, and $\lambda_t \equiv \bar{\lambda} = \lambda_1 T^{-\theta_4}$ with $\theta_4 > 0$. Assume $T \geq 2$ and $\eta_1(\kappa^2 + \lambda_1) \leq 1$. Then,

(1) The Hilbert-Schmidt norm of $\bar{\chi}_t$ is uniformly bounded as follows:

$$\sup_{1 \leq t \leq T} \|\bar{\chi}_t\|_{\text{HS}}^2 \leq M_2^2,$$

where

$$M_2^2 := d_7 T^{-2\theta_3} \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right).$$

(2) The total squared Hilbert-Schmidt norm in expectation is bounded by

$$\sum_{t=1}^T \mathbb{E}_{z_t} \|\bar{\chi}_t\|_{\text{HS}}^2 \leq \tau_2^2,$$

where τ_2 is defined as

$$\tau_2^2 := d_8 \delta_3 \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \begin{cases} T^{-(1-2\alpha+s)\theta_3+(s-2\alpha)\min\{1,\theta_3+\theta_4\}}, & \text{when } 2\alpha < s \leq 1+2\alpha, \\ T^{-\theta_3} \log T, & \text{when } 2\alpha = s, \\ T^{-\theta_3}, & \text{when } 2\alpha > s. \end{cases}$$

Here d_7 depends only on $\eta_1, \theta_3, M_\rho, \|S^\dagger\|_{\text{HS}}$, and α , while d_8 depends only on $\eta_1, \lambda_1, \theta_3, \theta_4, M_\rho, \|S^\dagger\|_{\text{HS}}, \alpha$, and s ; both constants are independent of T and δ .

Proof. (1) By Proposition A.3, (D.2), and Lemma D.2, for any $1 \leq t \leq T$, we have

$$\begin{aligned} \|\bar{\chi}_t\|_{\text{HS}} &\leq 2\eta_t \kappa \left(M_\rho + \kappa \|L_t + R_t + H^\dagger\|_{L_{\text{HS}}^\infty} \right) \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\mathbb{A}_t} \\ &\leq 2\eta_1 \kappa^{1+2\alpha} \left(M_\rho + 2\kappa \|H^\dagger\|_{\text{HS}} + \kappa \frac{2d_6}{\log 2} T^{\frac{1}{2}-\theta_3} \log T \log \frac{2}{\delta} \right) T^{-\theta_3}, \end{aligned} \quad (\text{D.29})$$

where the last inequality uses $\left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\| \leq \kappa^{2\alpha}$. Hence, there exists a constant d_7 independent of t, T , and δ , such that

$$\sup_{1 \leq t \leq T} \|\bar{\chi}_t\|_{\text{HS}}^2 \leq d_7 T^{-2\theta_3} \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right).$$

(2) The argument is parallel to that for $\tilde{\chi}_t$. We therefore only indicate the changes arising from the constant step size and regularization parameter. By (D.27),

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_{z_t} \|\bar{\chi}_t\|_{\text{HS}}^2 &= \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z_t} \left[\left\| \mathcal{B}_t C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \right\|_{\text{HS}}^2 \mathbb{1}_{\bar{\mathcal{A}}_t} \right] \\ &\leq \sum_{t=1}^T \eta_t^2 \mathbb{E}_{z_t} \left[\|y_t - H_t \phi(x_t)\|_{\mathcal{Y}}^2 \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_\kappa}^2 \mathbb{1}_{\bar{\mathcal{A}}_t} \right]. \end{aligned}$$

By Assumption 5, (D.2), Lemma D.2, and (D.28), we obtain

$$\|y_t - H_t \phi(x_t)\|_{\mathcal{Y}}^2 \mathbb{1}_{\bar{\mathcal{A}}_t} \leq 2M_\rho^2 + 2\kappa^2 \left(8\|H^\dagger\|_{\text{HS}}^2 + 2 \left(\frac{2d_6}{\log 2} \right)^2 T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right).$$

Substituting into the earlier bound yields

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_{z_t} \|\bar{\chi}_t\|_{\text{HS}}^2 &\leq \sum_{t=1}^T \eta_t^2 \left(2M_\rho^2 + 2\kappa^2 \left(8\|H^\dagger\|_{\text{HS}}^2 + 2 \left(\frac{2d_6}{\log 2} \right)^2 T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \right) \\ &\quad \times \mathbb{E}_{z_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_\kappa}^2. \end{aligned} \quad (\text{D.30})$$

Then, if $2\alpha + 1 - s > 0$,

$$\begin{aligned}
& \mathbb{E}_{z_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \\
&= \text{Tr} \left(C^{2\alpha+1} (I - \bar{\eta}(C + \bar{\lambda}I))^{2(T-t)} \right) \\
&\leq \text{Tr}(C^s) \left\| C^{2\alpha+1-s} (I - \bar{\eta}(C + \bar{\lambda}I))^{2(T-t)} \right\| \\
&\leq \text{Tr}(C^s) \frac{2(\kappa^{2(2\alpha+1-s)}) + ((2\alpha+1-s)/(2e))^{2\alpha+1-s}}{1 + ((T-t)\bar{\eta})^{2\alpha+1-s}} \\
&\quad \times \exp \{ -2(T-t)\bar{\eta}\bar{\lambda} \},
\end{aligned} \tag{D.31}$$

and if $2\alpha + 1 - s = 0$, then

$$\mathbb{E}_{z_t} \left\| C^\alpha \prod_{j=t+1}^T (I - \eta_j(C + \lambda_j I)) \phi(x_t) \right\|_{\mathcal{H}_K}^2 \leq \text{Tr}(C^s) \exp \{ -2(T-t)\bar{\eta}\bar{\lambda} \}. \tag{D.32}$$

Substituting (D.31) or (D.32) into (D.30) yields

$$\sum_{t=1}^T \mathbb{E}_{z_t} \|\bar{\chi}_t\|_{\text{HS}}^2 \leq d_8 T^{-2\theta_3} \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \sum_{t=0}^{T-1} \frac{\exp \{ -2\eta_1 \lambda_1 t T^{-\theta_4 - \theta_3} \}}{1 + (t\bar{\eta})^{2\alpha+1-s}},$$

where d_8 is a constant independent of t , T , and δ . Using Proposition B.10 with $v = 1 + 2\alpha - s$, we obtain

$$\begin{aligned}
\sum_{t=1}^T \mathbb{E}_{z_t} \|\bar{\chi}_t\|_{\text{HS}}^2 &\leq d_8 \delta_3 T^{-2\theta_3} \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \\
&\quad \times \begin{cases} T^{(1+2\alpha-s)\theta_3 + (s-2\alpha)\min\{1, \theta_3 + \theta_4\}}, & \text{when } 2\alpha < s \leq 1 + 2\alpha, \\ T^{\theta_3} \log T, & \text{when } 2\alpha = s, \\ T^{\theta_3}, & \text{when } 2\alpha > s, \end{cases} \\
&= d_8 \delta_3 \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \\
&\quad \times \begin{cases} T^{-(1-2\alpha+s)\theta_3 + (s-2\alpha)\min\{1, \theta_3 + \theta_4\}}, & \text{when } 2\alpha < s \leq 1 + 2\alpha, \\ T^{-\theta_3} \log T, & \text{when } 2\alpha = s, \\ T^{-\theta_3}, & \text{when } 2\alpha > s. \end{cases}
\end{aligned}$$

The proof is complete. $\square$

Next, we prove the high-probability bounds for prediction and estimation errors in the finite-horizon setting.

Proof of Theorem 2.11. By Proposition A.3 with $\alpha = \frac{1}{2}$, we have

$$\left\| (H_{T+1} - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + 6 \left\| \sum_{t=1}^T \chi_t \right\|_{\text{HS}}^2, \tag{D.33}$$

where $\mathcal{T}_3 = 0$. Using Proposition D.5, it holds that

$$\mathbb{P}(\bar{\chi}_t = \chi_t \text{ for any } 1 \leq t \leq T) \geq 1 - \delta. \tag{D.34}$$

Let $\theta_3 = \frac{2r+1}{2r+2}$ and choose $\theta_4 \geq \frac{2r+1}{(2r+2)\min\{2r+1,2\}}$. Applying Proposition B.1 and Proposition B.4 with $\alpha = \frac{1}{2}$, $t_0 = \theta_1 = \theta_2 = 0$, $\bar{\eta} = \eta_1 T^{-\theta_3}$ and $\bar{\lambda} = \lambda_1 T^{-\theta_4}$, we obtain

$$\mathcal{T}_1 \leq c_1 \lambda_T^{\min\{2r+1,2\}} = c_1 (\lambda_1 T^{-\theta_4})^{\min\{2r+1,2\}} \leq c_1 \lambda_1^{\min\{2r+1,2\}} T^{-\frac{2r+1}{2r+2}}, \quad (\text{D.35})$$

and

$$\begin{aligned} \mathcal{T}_2 &\leq c_2 \bar{\eta}^{-(2r+1)} T^{-(2r+1)} \exp\{-\tau \bar{\eta} \bar{\lambda} T\} \\ &\leq c_2 \eta_1^{-(2r+1)} T^{-\frac{2r+1}{2r+2}}. \end{aligned} \quad (\text{D.36})$$

Using Proposition D.1 and Proposition D.6, we conclude that with probability at least $1 - \delta$,

$$\left\| \sum_{i=1}^T \bar{\chi}_t \right\|_{\text{HS}}^2 \leq 8 (M_2^2 + \tau_2^2) \log^2 \frac{2}{\delta}.$$

Additionally, for $\alpha = \frac{1}{2}$, we have

$$\begin{aligned} M_2^2 &= d_7 T^{-2\theta_3} \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \\ &\leq d_7 \left(T^{-\theta_3} + T^{1-3\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right), \end{aligned}$$

and

$$\tau_2^2 = d_8 \delta_3 \left(T^{-\theta_3} + T^{1-3\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \begin{cases} \log T, & \text{when } s = 1, \\ 1, & \text{when } s < 1. \end{cases}$$

Therefore,

$$\left\| \sum_{i=1}^T \bar{\chi}_t \right\|_{\text{HS}}^2 \leq 8(d_7 + d_8 \delta_3) \left(T^{-\theta_3} + T^{1-3\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \begin{cases} \log T, & \text{when } s = 1, \\ 1, & \text{when } s < 1. \end{cases} \quad (\text{D.37})$$

Combining (D.33), (D.34), (D.35), (D.36), and (D.37), we conclude that there exists a constant $c_{2,3}$ independent of T , such that

$$\left\| (H_{T+1} - H^\dagger) C^{\frac{1}{2}} \right\|_{\text{HS}}^2 \leq c_{2,3} \begin{cases} T^{-\theta_3} \log^2 \frac{2}{\delta} + T^{1-3\theta_3} \log^2 T \log^4 \frac{2}{\delta}, & \text{when } s < 1, \\ T^{-\theta_3} \log T \log^2 \frac{2}{\delta} + T^{1-3\theta_3} \log^3 T \log^4 \frac{2}{\delta}, & \text{when } s = 1, \end{cases}$$

holds with probability at least $1 - 2\delta$.

The proof is then complete. $\square$

Proof of Theorem 2.12. By Proposition A.3 with $\alpha = 0$, we have

$$\|H_{T+1} - H^\dagger\|_{\text{HS}}^2 \leq \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + 6 \left\| \sum_{t=1}^T \chi_t \right\|_{\text{HS}}^2, \quad (\text{D.38})$$

where $\mathcal{T}_3 = 0$. According to Proposition D.5, it holds that

$$\mathbb{P}(\bar{\chi}_t = \chi_t \text{ for any } 1 \leq t \leq T) \geq 1 - \delta. \quad (\text{D.39})$$

Applying Proposition B.1 and Proposition B.4 with $\alpha = 0$, $t_0 = \theta_1 = \theta_2 = 0$, $\bar{\eta} = \eta_1 T^{-\theta_3}$ and $\bar{\lambda} = \lambda_1 T^{-\theta_4}$, we obtain

$$\mathcal{T}_1 \leq c_1 \lambda_T^{\min\{2r,2\}} = c_1 (\lambda_1 T^{-\theta_4})^{\min\{2r,2\}}, \quad (\text{D.40})$$

and

$$\begin{aligned} \mathcal{T}_2 &\leq c_2 \bar{\eta}^{-2r} T^{-2r} \exp\{-\tau \bar{\eta} \bar{\lambda} T\} \\ &\leq c_2 \eta_1^{-2r} T^{-2r(1-\theta_3)}. \end{aligned} \quad (\text{D.41})$$

Using Proposition D.1 and Proposition D.6, we conclude that, with probability at least $1 - \delta$,

$$\left\| \sum_{i=1}^T \bar{\chi}_t \right\|_{\text{HS}}^2 \leq 8 (M_2^2 + \tau_2^2) \log^2 \frac{2}{\delta}. \quad (\text{D.42})$$

Additionally, setting $\alpha = 0$, we have

$$M_2^2 = d_7 T^{-2\theta_3} \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right),$$

and

$$\begin{aligned} \tau_2^2 &= d_8 \delta_3 \left(1 + T^{1-2\theta_3} \log^2 T \log^2 \frac{2}{\delta} \right) T^{-(1+s)\theta_3 + s \min\{1, \theta_3 + \theta_4\}} \\ &= d_8 \delta_3 \left(T^{-(1+s)\theta_3 + s} + T^{-(3+s)\theta_3 + 1+s} \log^2 T \log^2 \frac{2}{\delta} \right) \end{aligned}$$

when $\theta_3 + \theta_4 \geq 1$. Therefore, if $\theta_3 + \theta_4 \geq 1$, then

$$\left\| \sum_{i=1}^T \bar{\chi}_t \right\|_{\text{HS}}^2 \leq 8(d_7 + d_8 \delta_3) \left(T^{-(1+s)\theta_3 + s} + T^{-(3+s)\theta_3 + 1+s} \log^2 T \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta}. \quad (\text{D.43})$$

Let $c_{2,4}$ be a constant independent of T and δ . Combining (D.38), (D.39), (D.40), (D.41), (D.42), and (D.43), we obtain the following estimates:

(1) If $2r + s \geq 1$, choose $\theta_3 = \frac{2r+s}{1+2r+s}$ and $\theta_4 \geq \frac{r}{(1+2r+s) \min\{r, 1\}}$, then

$$\begin{aligned} \|H_{T+1} - H^\dagger\|_{\text{HS}}^2 &\leq c_{2,4} \left(T^{-\frac{2r}{1+2r+s}} + T^{-\frac{4r+s-1}{1+2r+s}} \log^2 T \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \\ &\lesssim T^{-\frac{2r}{1+2r+s}} \log^4 \frac{2}{\delta} \end{aligned}$$

holds with probability at least $1 - 2\delta$.

(2) If $2r + s < 1$, choose $\theta_3 = \frac{1+2r+s}{3+2r+s}$ and $\theta_4 \geq \frac{2r}{(3+2r+s) \min\{r, 1\}} = \frac{2r}{(3+2r+s)r}$, then

$$\begin{aligned} \|H_{T+1} - H^\dagger\|_{\text{HS}}^2 &\leq c_{2,4} \left(T^{-\frac{1+2r-s}{3+2r+s}} + T^{-\frac{4r}{3+2r+s}} \log^2 T \log^2 \frac{2}{\delta} \right) \log^2 \frac{2}{\delta} \\ &\lesssim T^{-\frac{4r}{3+2r+s}} \log^2 T \log^4 \frac{2}{\delta} \end{aligned}$$

holds with probability at least $1 - 2\delta$.

The proof is then complete. $\square$

A Proof of Proposition 2.1

Proof. Since W is self-adjoint and positive, $W^{1/2}$ is also self-adjoint and positive. Let $\mathcal{H}_0 = \text{span}\{K(x, \cdot)y = \mathcal{K}(x, \cdot)Wy : x \in \mathcal{X}, y \in \mathcal{Y}\}$ and $\mathcal{B}_0 = \text{span}\{W^{1/2}y \otimes \phi(x) : x \in \mathcal{X}, y \in \mathcal{Y}\}$. Then, it is clear that $\mathcal{H}_K = \overline{\mathcal{H}_0}$ and $\mathcal{B}_{\text{HS}}(\mathcal{H}_K, W^{1/2}\mathcal{Y}) = \overline{\mathcal{B}_0}$.

We define the mapping $W_0 : \mathcal{H}_0 \rightarrow \mathcal{B}_0$ by $\sum_{i=1}^n \alpha_i K(x_i, \cdot)y_i \mapsto \sum_{i=1}^n \alpha_i W^{1/2}y_i \otimes \phi(x_i)$ for any $n \in \mathbb{N}$, $x_1, \dots, x_n \in \mathcal{X}$ and $y_1, \dots, y_n \in \mathcal{Y}$. We see that W_0 is well-defined and linear. Moreover, for any

$\sum_{i=1}^n \alpha_i K(x_i, \cdot) y_i = \sum_{i=1}^n \alpha_i \mathcal{K}(x_i, \cdot) W y_i \in \mathcal{H}_0$, we have

$$\begin{aligned}
\left\| W_0 \left(\sum_{i=1}^n \alpha_i K(x_i, \cdot) y_i \right) \right\|_{\text{HS}}^2 &= \sum_{i,j=1}^n \alpha_i \alpha_j \langle W^{1/2} y_i \otimes \phi(x_i), W^{1/2} y_j \otimes \phi(x_j) \rangle_{\text{HS}} \\
&= \sum_{i,j=1}^n \alpha_i \alpha_j \mathcal{K}(x_i, x_j) \langle W y_i, y_j \rangle_{\mathcal{Y}} \\
&= \sum_{i,j=1}^n \alpha_i \alpha_j \langle K(x_i, x_j) y_i, y_j \rangle_{\mathcal{Y}} \\
&= \left\| \sum_{i=1}^n \alpha_i K(x_i, \cdot) y_i \right\|_{\mathcal{H}}^2.
\end{aligned}$$

By extending W_0 to $W : \mathcal{H} \rightarrow \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \overline{W^{1/2} \mathcal{Y}})$, we conclude that $\mathcal{H}$ is isometric to

$$\mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \overline{W^{1/2} \mathcal{Y}}) \subset \mathcal{B}_{\text{HS}}(\mathcal{H}_{\mathcal{K}}, \mathcal{Y}).$$

Next, we show that $h(x) = (Wh)(\phi(x))$ for all $h \in \mathcal{H}$. For any $y \in \mathcal{Y}$,

$$\begin{aligned}
\langle y, h(x) \rangle_{\mathcal{Y}} &= \langle K(x, \cdot) y, h \rangle_{\mathcal{H}} = \langle W(K(x, \cdot) y), Wh \rangle_{\text{HS}} = \langle W^{1/2} y \otimes \phi(x), Wh \rangle_{\text{HS}} \\
&= \text{Tr} \left(\left(W^{1/2} y \otimes \phi(x) \right)^* (Wh) \right) = \left\langle y, W^{1/2} (Wh) \phi(x) \right\rangle_{\mathcal{Y}},
\end{aligned}$$

where the property $\langle y, h(x) \rangle_{\mathcal{Y}} = \langle K(x, \cdot) y, h \rangle_{\mathcal{H}}$ is used in the first equality. Thus, we conclude that $h(x) = W^{1/2} (Wh) \phi(x)$. The uniqueness of this representation is obvious. This completes the proof. $\square$

B Proof of Proposition 2.2

Lemma B.1. *Suppose that $\bar{\phi} = \mathbb{E}[\phi(x)]$. The moment condition (2.3) holds if*

$$\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^4 \right] \leq c \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \right] \right)^2 \quad (\text{B.1})$$

holds for some constant $c > 0$.

Proof. First, using that $\mathbb{E}[\phi(x) - \bar{\phi}] = 0$, we see that

$$\begin{aligned}
\left(\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \right] \right)^2 &= \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi} + \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \right] \right)^2 \\
&= \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \right] \right)^2 + \langle \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^4 + 2 \langle \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \right].
\end{aligned} \quad (\text{B.2})$$

Using $\mathbb{E}[\phi(x) - \bar{\phi}] = 0$ again, it holds that

$$\begin{aligned}
\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_{\mathcal{K}}}^4 \right] &= \mathbb{E} \left[\left(\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 + \langle \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 + 2 \langle \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}} \langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}} \right)^2 \right] \\
&= \mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^4 \right] + \langle \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^4 + 6 \langle \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2 \right] \\
&\quad + 4 \langle \bar{\phi}, f \rangle_{\mathcal{H}_{\mathcal{K}}} \mathbb{E} \left[\langle \phi(x) - \bar{\phi} \rangle_{\mathcal{H}_{\mathcal{K}}}^3 \right].
\end{aligned} \quad (\text{B.3})$$

By Hölder's inequality and (2.3), we obtain that

$$\begin{aligned}
& 4 \langle \bar{\phi}, f \rangle_{\mathcal{H}_K} \mathbb{E} \left[\langle \phi(x) - \bar{\phi} \rangle_{\mathcal{H}_K}^3 \right] \\
& \leq 4 \langle \bar{\phi}, f \rangle_{\mathcal{H}_K} \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi} \rangle_{\mathcal{H}_K}^4 \right] \right)^{3/4} \\
& \leq 4c^{3/4} \langle \bar{\phi}, f \rangle_{\mathcal{H}_K} \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi} \rangle_{\mathcal{H}_K}^2 \right] \right)^{3/2} \\
& \leq 2c^{3/4} \langle \bar{\phi}, f \rangle_{\mathcal{H}_K}^2 \mathbb{E} \left[\langle \phi(x) - \bar{\phi} \rangle_{\mathcal{H}_K}^2 \right] + 2c^{3/4} \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi} \rangle_{\mathcal{H}_K}^2 \right] \right)^2.
\end{aligned} \tag{B.4}$$

Then, taking (B.4) back into (B.3) yields that

$$\begin{aligned}
\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_K}^4 \right] & \leq \left(c + 2c^{3/4} \right) \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi} \rangle_{\mathcal{H}_K}^2 \right] \right)^2 + \langle \bar{\phi}, f \rangle_{\mathcal{H}_K}^4 \\
& \quad + \left(6 + 2c^{3/4} \right) \langle \bar{\phi}, f \rangle_{\mathcal{H}_K}^2 \mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_K}^2 \right].
\end{aligned} \tag{B.5}$$

Combining (B.5) with (B.2), we conclude that

$$\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_K}^4 \right] \leq \max \left\{ c + 2c^{3/4}, 3 + c^{3/4} \right\} \left(\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_K}^2 \right] \right)^2,$$

which completes this proof. $\square$

Proof of Proposition 2.2. Recall that $\Sigma := \mathbb{E} \left[(\phi(x) - \bar{\phi}) \otimes (\phi(x) - \bar{\phi}) \right]$. Since Σ is compact and self-adjoint, it admits the spectral decomposition:

$$\Sigma = \sum_{k \geq 1} \lambda_k \phi_k \otimes \phi_k.$$

We claim that $\phi(x) - \bar{\phi} \in \overline{\text{ran}(\Sigma)}$ almost surely. To prove this, for any $f \in \ker(\Sigma)$, it holds that

$$\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_K}^2 \right] = \langle \Sigma f, f \rangle_{\mathcal{H}_K} = 0.$$

This implies that $\phi(x) - \bar{\phi} \in \ker(\Sigma)^\perp = \overline{\text{ran}(\Sigma)}$ almost surely. Therefore, we have

$$\phi(x) - \bar{\phi} = \sum_{k \geq 1} \sqrt{\lambda_k} \xi_k \phi_k,$$

where $\xi_k := \frac{\langle \phi(x) - \bar{\phi}, \phi_k \rangle}{\sqrt{\lambda_k}}$, $\mathbb{E}[\xi_k] = 0$ and $\mathbb{E}[\|\xi_k\|_{\mathcal{H}_K}^2] = 1$ for all $k \geq 1$.

Now, Assume that $\{\xi_k\}_{k \geq 1}$ consists of independent random variables. We will show that (2.3) holds if $\{\mathbb{E}[\xi_k^4]\}_{k \geq 1}$ is uniformly bounded, i.e., there exists a constant $C > 0$ such that $\mathbb{E}[\xi_k^4] \leq C$ for all $k \geq 1$. Since the ξ_k are mean-zero and independent, it follows that

$$\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_K}^4 \right] = \sum_{k \geq 1} \lambda_k^2 \langle f, \phi_k \rangle_{\mathcal{H}_K}^4 \mathbb{E}[\xi_k^4] + 6 \sum_{i \neq j} \lambda_i \lambda_j \langle f, \phi_i \rangle_{\mathcal{H}_K}^2 \langle f, \phi_j \rangle_{\mathcal{H}_K}^2,$$

and

$$\left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_K}^2 \right] \right)^2 = \sum_{k \geq 1} \lambda_k^2 \langle f, \phi_k \rangle_{\mathcal{H}_K}^4 + 2 \sum_{i \neq j} \lambda_i \lambda_j \langle f, \phi_i \rangle_{\mathcal{H}_K}^2 \langle f, \phi_j \rangle_{\mathcal{H}_K}^2.$$

Using $\mathbb{E}[\xi_k^4] \leq C$, we obtain

$$\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_K}^4 \right] \leq \max \left\{ C, \frac{1}{3} \right\} \left(\mathbb{E} \left[\langle \phi(x) - \bar{\phi}, f \rangle_{\mathcal{H}_K}^2 \right] \right)^2.$$

By Lemma B.1, there exists a constant $c > 0$, such that

$$\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_K}^4 \right] \leq c \left(\mathbb{E} \left[\langle \phi(x), f \rangle_{\mathcal{H}_K}^2 \right] \right)^2.$$

The proof is then complete. $\square$

C An Auxiliary Equivalence Related to Assumption 4

The following proposition provides an equivalent formulation related to Assumption 4. Its proof has essentially been established in our previous work (Shi and Yang, 2024). We include it here for completeness and to keep the present paper reasonably self-contained.

Proposition C.1. *The following three statements are equivalent.*

(1) *There exists a constant $c > 0$ such that for any $f \in \mathcal{H}_K$,*

$$\mathbb{E} \left[\langle v, f \rangle_{\mathcal{H}_K}^4 \right] \leq c \left(\mathbb{E} \left[\langle v, f \rangle_{\mathcal{H}_K}^2 \right] \right)^2.$$

(2) *For any separable Hilbert space $\mathcal{Y}$, there exists a constant $c > 0$ such that for any compact linear operator $A : \mathcal{H}_K \rightarrow \mathcal{Y}$, there holds*

$$\mathbb{E} \left[\|Av\|_{\mathcal{Y}}^4 \right] \leq c \left(\mathbb{E} \left[\|Av\|_{\mathcal{Y}}^2 \right] \right)^2.$$

(3) *There exists a separable Hilbert space $\mathcal{Y}$ and a constant $c > 0$ such that for any compact linear operator $A : \mathcal{H}_K \rightarrow \mathcal{Y}$, there holds*

$$\mathbb{E} \left[\|Av\|_{\mathcal{Y}}^4 \right] \leq c \left(\mathbb{E} \left[\|Av\|_{\mathcal{Y}}^2 \right] \right)^2.$$

Proof. (1) implies (2): Let $A : \mathcal{H}_K \rightarrow \mathcal{Y}$ be any compact linear operator. Then A^*A is a compact, self-adjoint, positive operator on $\mathcal{H}_K$. Hence, by the spectral theorem,

$$A^*A = \sum_{j \geq 1} \lambda_j e_j \otimes e_j,$$

where $\{\lambda_j\}_{j \geq 1} \subset [0, \infty)$ are the eigenvalues of A^*A , and $\{e_j\}_{j \geq 1}$ is an orthonormal system in $\mathcal{H}_K$. Therefore,

$$\|Av\|_{\mathcal{Y}}^2 = \langle v, A^*Av \rangle_{\mathcal{H}_K} = \sum_{j \geq 1} \lambda_j \langle v, e_j \rangle_{\mathcal{H}_K}^2,$$

and thus

$$\begin{aligned} \mathbb{E} \left[\|Av\|_{\mathcal{Y}}^4 \right] &= \mathbb{E} \left[\left(\sum_{j \geq 1} \lambda_j \langle v, e_j \rangle_{\mathcal{H}_K}^2 \right)^2 \right] \\ &= \sum_{i, j \geq 1} \lambda_i \lambda_j \mathbb{E} \left[\langle v, e_i \rangle_{\mathcal{H}_K}^2 \langle v, e_j \rangle_{\mathcal{H}_K}^2 \right] \\ &\leq \sum_{i, j \geq 1} \lambda_i \lambda_j \sqrt{\mathbb{E} \left[\langle v, e_i \rangle_{\mathcal{H}_K}^4 \right]} \sqrt{\mathbb{E} \left[\langle v, e_j \rangle_{\mathcal{H}_K}^4 \right]}, \end{aligned}$$

where we used the Cauchy–Schwarz inequality. By (1), it follows that

$$\mathbb{E} \left[\|Av\|_{\mathcal{Y}}^4 \right] \leq c \left(\sum_{j \geq 1} \lambda_j \mathbb{E} \left[\langle v, e_j \rangle_{\mathcal{H}_K}^2 \right] \right)^2.$$

Since

$$\sum_{j \geq 1} \lambda_j \mathbb{E} \left[\langle v, e_j \rangle_{\mathcal{H}_K}^2 \right] = \mathbb{E} \left[\sum_{j \geq 1} \lambda_j \langle v, e_j \rangle_{\mathcal{H}_K}^2 \right] = \mathbb{E} \left[\|Av\|_{\mathcal{Y}}^2 \right],$$

we conclude that

$$\mathbb{E} [\|Av\|_{\mathcal{Y}}^4] \leq c (\mathbb{E} [\|Av\|_{\mathcal{Y}}^2])^2.$$

(2) implies (3): This is immediate.

(3) implies (1): Let $\mathcal{Y}$ and $c > 0$ be as in (3). Choose any $g \in \mathcal{Y}$ with $\|g\|_{\mathcal{Y}} = 1$. For a fixed $f \in \mathcal{H}_{\mathcal{K}}$, define the rank-one operator $A : \mathcal{H}_{\mathcal{K}} \rightarrow \mathcal{Y}$ by

$$Ah := \langle h, f \rangle_{\mathcal{H}_{\mathcal{K}}} g, \quad \forall h \in \mathcal{H}_{\mathcal{K}}.$$

Then A is compact. Moreover,

$$\|Av\|_{\mathcal{Y}}^2 = |\langle v, f \rangle_{\mathcal{H}_{\mathcal{K}}}|^2 \|g\|_{\mathcal{Y}}^2 = |\langle v, f \rangle_{\mathcal{H}_{\mathcal{K}}}|^2.$$

Applying (3) to this operator A , we obtain

$$\mathbb{E} [\langle v, f \rangle_{\mathcal{H}_{\mathcal{K}}}^4] = \mathbb{E} [\|Av\|_{\mathcal{Y}}^4] \leq c (\mathbb{E} [\|Av\|_{\mathcal{Y}}^2])^2 = c (\mathbb{E} [\langle v, f \rangle_{\mathcal{H}_{\mathcal{K}}}^2])^2.$$

This proves (1). $\square$

D Proof of Proposition B.7

In this subsection, our goal is to bound

$$\sum_{t=1}^T \exp \left\{ - \sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v}.$$

Lemma D.1. *Let $v > 0$, $p \in \mathbb{R}$, $T \geq t_0 + 1$ and $t_0 \geq 1$. The step size η_t is set as (A.4). Then, there holds*

$$\sum_{t=1}^{T/2} \frac{(t+t_0)^p}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \leq \delta' \begin{cases} (T+t_0)^{-(1-\theta_1)v+p+1}, & \text{when } p > -1, \\ (T+t_0)^{-(1-\theta_1)v} \log(T+t_0), & \text{when } p = -1, \\ (T+t_0)^{-(1-\theta_1)v}, & \text{when } p < -1, \end{cases}$$

where δ' is a constant independent of T and t_0 .

Proof. Since $\sum_{j=t+1}^T \eta_j \geq \bar{\eta} [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]$, it follows that

$$\sum_{t=1}^{T/2} \frac{(t+t_0)^p}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \leq \frac{1}{\min\{1, \bar{\eta}^v\}} \sum_{t=1}^{T/2} \frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v}.$$

As $t+t_0+1 \leq \frac{3}{4}(T+t_0+1)$ when $t \leq T/2$ and $T \geq t_0+1$, we obtain

$$\begin{aligned} \sum_{t=1}^{T/2} \frac{(t+t_0)^p}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} &\leq \frac{(1 - (3/4)^{1-\theta_1})^{-v}}{\min\{1, \bar{\eta}^v\}} \sum_{t=1}^{T/2} \frac{(t+t_0)^p}{(T+t_0)^{(1-\theta_1)v}} \\ &\leq \frac{(1 - (3/4)^{1-\theta_1})^{-v}}{\min\{1, \bar{\eta}^v\}} (T+t_0)^{-(1-\theta_1)v} \int_0^{\frac{T}{2}+1} (x+t_0)^p dx \\ &\leq \frac{(1 - (3/4)^{1-\theta_1})^{-v}}{\min\{1, \bar{\eta}^v\}} (T+t_0)^{-(1-\theta_1)v} \begin{cases} \frac{1}{p+1} (T+t_0)^{p+1}, & \text{when } p > -1, \\ \log(T+t_0), & \text{when } p = -1, \\ \frac{t_0^{p+1}}{-1-p} \leq \frac{1}{-1-p}, & \text{when } p < -1, \end{cases} \\ &\leq \delta' \begin{cases} (T+t_0)^{-(1-\theta_1)v+p+1}, & \text{when } p > -1, \\ (T+t_0)^{-(1-\theta_1)v} \log(T+t_0), & \text{when } p = -1, \\ (T+t_0)^{-(1-\theta_1)v}, & \text{when } p < -1, \end{cases} \end{aligned}$$

where

$$\delta' = \frac{(1 - (3/4)^{1-\theta_1})^{-v}}{\min\{1, \bar{\eta}^v\}} \begin{cases} \frac{1}{p+1}, & \text{when } p > -1, \\ 1, & \text{when } p = -1, \\ \frac{1}{-1-p}, & \text{when } p < -1, \end{cases}$$

which is independent of T and t_0 .

The proof is then finished. $\square$

Lemma D.2. *Let $v > 0$, $p \in \mathbb{R}$, $T \geq t_0 + 1$ and $t_0 \geq 1$. The step size η_t is set as (A.4). Then, there holds*

$$\sum_{t=T/2}^T \frac{(t+t_0)^p}{1 + \left(\sum_{j=t+1}^T \eta_j\right)^v} \leq \delta'' \begin{cases} (T+t_0)^{p+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{p+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{p+1-v(1-\theta_1)}, & \text{when } v < 1, \end{cases}$$

where δ'' is a constant independent of T and t_0 .

Proof. It is obvious that

$$\begin{aligned} & \sum_{t=T/2}^T \frac{(t+t_0)^p}{1 + \left(\sum_{j=t+1}^T \eta_j\right)^v} \\ & \leq \frac{1}{\min\{1, \bar{\eta}^v\}} \sum_{t=T/2}^{T-1} \frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v} + (T+t_0)^p. \end{aligned} \quad (\text{D.1})$$

Next, we bound $\sum_{t=T/2}^{T-1} \frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v}$. If $p > 0$, it holds that

$$\frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v} \leq \int_{t+1}^{t+2} \frac{(u+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (u+t_0)^{1-\theta_1}]^v} du.$$

If $p \leq 0$, there holds $(t+t_0)^p \leq 3^{-p}(t+t_0+2)^p$. Thus we have

$$\frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v} \leq 3^{-p} \int_{t+1}^{t+2} \frac{(u+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (u+t_0)^{1-\theta_1}]^v} du.$$

Therefore,

$$\begin{aligned} & \sum_{t=T/2}^{T-1} \frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v} \\ & \leq \max\{3^{-p}, 1\} \int_{T/2+1}^{T+1} \frac{(u+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (u+t_0)^{1-\theta_1}]^v} du. \end{aligned}$$

Let $\xi = (T+t_0+1)^{1-\theta_1} - (u+t_0)^{1-\theta_1}$, then $d\xi = -(1-\theta_1)(u+t_0)^{-\theta_1} du$, then

$$\begin{aligned} & \sum_{t=T/2}^{T-1} \frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v} \\ & \leq \frac{\max\{3^{-p}, 1\}}{1-\theta_1} \int_0^{(T+t_0+1)^{1-\theta_1} - (T/2+t_0+1)^{1-\theta_1}} \frac{(u+t_0)^{p+\theta_1}}{1 + \xi^v} d\xi \\ & \leq \frac{\max\{3^{-p}, 1\}}{1-\theta_1} \int_0^{(T+t_0+1)^{1-\theta_1} (1-2^{\theta_1-1})} \frac{(u+t_0)^{p+\theta_1}}{1 + \xi^v} d\xi. \end{aligned} \quad (\text{D.2})$$

Since $u+t_0 \in [T/2+t_0+1, T+t_0+1]$, whenever $p+\theta_1 > 0$ or not, we have

$$(u+t_0)^{p+\theta_1} \leq 2^{|p+\theta_1|} (T+t_0)^{p+\theta_1}. \quad (\text{D.3})$$

Hence, substituting (D.3) to (D.2) yields that

$$\begin{aligned}
& \sum_{t=T/2}^{T-1} \frac{(t+t_0)^p}{1 + [(T+t_0+1)^{1-\theta_1} - (t+t_0+1)^{1-\theta_1}]^v} \\
& \leq \frac{\max\{3^{-p}, 1\}}{1-\theta_1} 2^{|p+\theta_1|} (T+t_0)^{p+\theta_1} \left(1 + \int_1^{(T+t_0+1)^{1-\theta_1}(1-2^{\theta_1-1})} \frac{1}{\xi^v} d\xi \right) \\
& \leq \frac{\max\{3^{-p}, 1\}}{1-\theta_1} 2^{|p+\theta_1|} (T+t_0)^{p+\theta_1} \begin{cases} \frac{v}{v-1}, & \text{when } v > 1, \\ (2-\theta_1) \log(T+t_0+1), & \text{when } v = 1, \\ \frac{(1-2^{\theta_1-1})^{1-v}}{1-v} (T+t_0+1)^{(1-\theta_1)(1-v)}, & \text{when } v < 1, \end{cases} \quad (\text{D.4}) \\
& = \tilde{\delta}'' \begin{cases} (T+t_0)^{p+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{p+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{p+1-v(1-\theta_1)}, & \text{when } v < 1, \end{cases}
\end{aligned}$$

with some constant $\tilde{\delta}''$ independent of T and t_0 . Combining (D.4) with (D.1) yields that there exists a constant δ'' independent of T and t_0 , such that

$$\sum_{t=T/2}^T \frac{(t+t_0)^p}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \leq \delta'' \begin{cases} (T+t_0)^{p+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{p+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{p+1-v(1-\theta_1)}, & \text{when } v < 1, \end{cases} \quad (\text{D.5})$$

which completes the proof. $\square$

Proposition D.3. *Let $v > 0$, $\theta \in \mathbb{R}$, $t_0 \geq 1$ and $T \geq t_0 + 1$. The step size η_t is set as (A.4). Suppose that $\bar{\eta}\bar{\lambda} > \theta - 1$ and $\theta_1 + \theta_2 = 1$. Then, there holds*

$$\sum_{t=1}^T \exp \left\{ - \sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \leq \delta_1 \begin{cases} (T+t_0)^{-\theta+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{-\theta+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{-\theta+1-v(1-\theta_1)}, & \text{when } v < 1, \end{cases}$$

where δ_1 is a constant independent of T and t_0 .

Proof. From Lemma B.3 (2), we have

$$\begin{aligned}
& \sum_{t=1}^T \exp \left\{ - \sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \leq \sum_{t=1}^T \left(\frac{t+t_0+1}{T+t_0+1} \right)^{\bar{\eta}\bar{\lambda}} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \\
& \leq 2^{\bar{\eta}\bar{\lambda}} (T+t_0)^{-\bar{\eta}\bar{\lambda}} \sum_{t=1}^T \frac{(t+t_0)^{\bar{\eta}\bar{\lambda}-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v}.
\end{aligned}$$

Using Lemma D.1 and Lemma D.2, we obtain

$$\begin{aligned}
& \sum_{t=1}^T \exp \left\{ - \sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \\
& \leq 2^{\bar{\eta}\bar{\lambda}} (T+t_0)^{-\bar{\eta}\bar{\lambda}} \left(\delta' (T+t_0)^{-(1-\theta_1)v+p+1} + \delta'' \begin{cases} (T+t_0)^{p+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{p+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{p+1-v(1-\theta_1)}, & \text{when } v < 1, \end{cases} \right)
\end{aligned}$$

where $p = \bar{\eta}\bar{\lambda} - \theta > -1$. Therefore,

$$\begin{aligned} & \sum_{t=1}^T \exp \left\{ - \sum_{j=t+1}^T \eta_j \lambda_j \right\} \frac{(t+t_0)^{-\theta}}{1 + \left(\sum_{j=t+1}^T \eta_j \right)^v} \\ & \leq 2^{\bar{\eta}\bar{\lambda}} (\delta' + \delta'') \begin{cases} (T+t_0)^{-\theta+\theta_1}, & \text{when } v > 1, \\ (T+t_0)^{-\theta+\theta_1} \log(T+t_0), & \text{when } v = 1, \\ (T+t_0)^{-\theta+1-v(1-\theta_1)}, & \text{when } v < 1. \end{cases} \end{aligned} \quad (\text{D.6})$$

We finish the proof by setting $\delta_1 = 2^{\bar{\eta}\bar{\lambda}} (\delta' + \delta'')$, which is independent of T and t_0 . $\square$

E Extension to General Kernel Setting

We now discuss an extension of the regularized SGD analysis beyond the kernel class in Assumption 1. This choice of kernel has been employed in functional regression with structured output learning, as noted in Section 1. We now turn to extending the class of operator-valued kernels, further showing the generality and applicability of our analysis to a broader range of nonlinear operator learning problems.

We now consider an alternative setting for vector-valued RKHS and briefly list the conditions below.

- (1) Let $\mathcal{H}$ be separable, which is true if the spaces $\mathcal{X}$ and $\mathcal{Y}$ are separable and K is a Mercer kernel (Carmeli et al., 2006, Corollary 5.2). A kernel K is Mercer if and only if the RKHS induced by K is a subspace of the space of continuous operators from $\mathcal{X}$ to $\mathcal{Y}$, which in turn holds if and only if K is locally bounded and $K(x, \cdot)$ is strongly continuous for any $x \in \mathcal{X}$ (Carmeli et al., 2006, Proposition 5.1).
- (2) We assume that the operator-valued kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{B}(\mathcal{Y})$ satisfies that $K(x, x)$ is compact for any $x \in \mathcal{X}$.
- (3) We further assume that K is strongly measurable. Under assumptions (1) and (2), this is equivalent to requiring that each element in $\mathcal{H}$ is a measurable function (Carmeli et al., 2006, Proposition 3.3).
- (4) By Corollary 4.6 and Proposition 4.8 in (Carmeli et al., 2006), if $K(x, x)$ is an operator of trace class (which implies compactness obviously) for almost all $x \in \mathcal{X}$ and

$$\mathbb{E} [\text{Tr} (K(x, x))] < \infty, \quad (\text{E.1})$$

then the inclusion $\iota : \mathcal{H} \rightarrow L^2(\mathcal{X}, \mathcal{Y})$ is well-defined and Hilbert-Schmidt. As a result, the operator $L_K := \iota^* \iota$ is trace-class, which plays a role similar to that of C . And thus, we assume Assumption 3 holds for L_K with $s \in (0, 1]$, i.e., $\text{Tr}(L_K^s) < +\infty$.

The conditions listed above are required when conducting error analysis for kernels beyond the special case considered in Assumption 1. We remark that when the kernel is chosen as Eq. (1.2), i.e., $K(x, x') = \mathcal{K}(x, x')W$ with W being a self-adjoint and positive operator, the above conditions are satisfied if W is trace-class and the scalar-valued kernel $\mathcal{K}$ is a Mercer kernel such that $\sup_{x \in \mathcal{X}} \mathcal{K}(x, x) < \infty$.

We now outline the framework for generalizing the conclusions of this paper to the general scenario discussed above, while leaving the detailed proof to future work. It is straightforward to observe that for any $h \in \mathcal{H}$, the operator L_K satisfies

$$L_K h = \mathbb{E} [K(x, \cdot) h(x)].$$

Moreover, for any $h \in \mathcal{H}$, since the noise ϵ is centered and independent of x , there holds

$$\begin{aligned}\mathcal{E}(h) - \mathcal{E}(h^\dagger) &= \mathbb{E} \|h(x) - h^\dagger(x)\|_{\mathcal{Y}}^2 \\ &= \mathbb{E} [\langle K(x, \cdot)(h(x) - h^\dagger(x)), h - h^\dagger \rangle_{\mathcal{H}}] \\ &= \|L_K^{1/2}(h - h^\dagger)\|_{\mathcal{H}}^2.\end{aligned}$$

Our goal is to bound the prediction error $\|L_K^{1/2}(h - h^\dagger)\|_{\mathcal{H}}^2$ and estimation error $\|h - h^\dagger\|_{\mathcal{H}}^2$ for the regularized SGD estimator $h = h_{T+1}$. Similar to the approach in the proofs of our main results, we can derive analogs of equations (A.5), (A.6), Proposition A.2, and Proposition A.3. Next, under assumptions similar to those in Section 2, we can carry out the error analysis, which we leave to future work.

We point out that the framework discussed in this subsection does not cover the case considered in Assumption 1, as the kernel $\mathcal{K}(x, x)I$ is not a compact operator and thus fails to satisfy the assumptions required in the current setting. Hence, the framework developed here and the one based on Assumption 1 are mutually exclusive. Nevertheless, combining the kernel choices from Assumption 1 and those introduced in this subsection can significantly broaden the applicability of our analysis developed in this paper.

F Connection with PCA Encoder-decoder Framework

In this appendix, we briefly discuss how the regularized SGD framework may be combined with a PCA-based encoder-decoder strategy. The purpose of this discussion is only to illustrate a possible implementation route in discretized settings; it is not part of the proved results of the paper. A full analysis would require additional estimates for empirical PCA and projection errors, which are beyond the scope of the present work.

Let $(\mathcal{X}, \langle \cdot \rangle_{\mathcal{X}}, \|\cdot\|_{\mathcal{X}})$ and $(\mathcal{Y}, \langle \cdot \rangle_{\mathcal{Y}}, \|\cdot\|_{\mathcal{Y}})$ be two real separable Hilbert spaces. Suppose that

$$h^\dagger : \mathcal{X} \rightarrow \mathcal{Y} \tag{F.1}$$

is a potentially nonlinear operator. Given i.i.d. samples $\{x_t, y_t\}_{t=1}^T \sim \rho$, where $y_t = h^\dagger(x_t) + \epsilon_t$, and ϵ_t denotes centered i.i.d. noise independent of x_t , our goal is to solve the prediction problem, i.e., to minimize the prediction error $\mathcal{E}(h) = \mathbb{E}_\rho [\|h(x) - y\|_{\mathcal{Y}}^2]$ for some estimator h . To this end, we apply the PCA technique to project the input and output samples onto finite-dimensional Euclidean spaces. We then approximate the mapping from the finite-dimensional input space to the finite-dimensional output space using kernel methods. Below, we briefly review the PCA technique.

The primary function of PCA is to extract the principal features of the data. High-dimensional data often suffers from the curse of dimensionality, and dimensionality reduction—achieved by identifying and retaining the most significant information—serves as an effective remedy. This constitutes the central role of PCA. In our setting, we employ PCA to reduce the samples from an infinite-dimensional space to a finite-dimensional one. The PCA algorithm applied to a random input x in $\mathcal{X}$ seeks to minimize the reconstruction error $\mathbb{E} [\|(I - P)x\|_{\mathcal{X}}^2]$ over $\Pi_{d_{\mathcal{X}}}$, the set of all orthogonal projections P with rank $d_{\mathcal{X}}$. Given the covariance operator of x defined as $\Sigma_x := \mathbb{E}[x \otimes x]$, there exist eigenvalue-eigenvector pairs $\{\lambda_i^{d_{\mathcal{X}}}, \phi_i^{d_{\mathcal{X}}}\}_{i \geq 1}$ satisfying $\langle \Sigma_x \phi_i^{d_{\mathcal{X}}}, \phi_j^{d_{\mathcal{X}}} \rangle_{\mathcal{X}} = \lambda_i^{d_{\mathcal{X}}} \delta_{ij}$ and $\lambda_1^{d_{\mathcal{X}}} \geq \lambda_2^{d_{\mathcal{X}}} \geq \dots \geq 0$, where $\delta_{ij} = 1$ if $i = j$, otherwise 0. It can be shown that the optimal PCA projection is given by

$$P_{d_{\mathcal{X}}}^{\mathcal{X}} = \arg \min_{P \in \Pi_{d_{\mathcal{X}}}} \mathbb{E} [\|(I - P)x\|_{\mathcal{X}}^2] = \mathcal{D}_{d_{\mathcal{X}}}^{\mathcal{X}} \circ \mathcal{E}_{d_{\mathcal{X}}}^{\mathcal{X}},$$

where the encoder $\mathcal{E}_{d_{\mathcal{X}}}^{\mathcal{X}} : \mathcal{X} \rightarrow \mathbb{R}^{d_{\mathcal{X}}}$ is defined as

$$\mathcal{E}_{d_{\mathcal{X}}}^{\mathcal{X}}(x) := (\langle x, \phi_i^{\mathcal{X}} \rangle_{\mathcal{X}})_{i=1}^{d_{\mathcal{X}}},$$

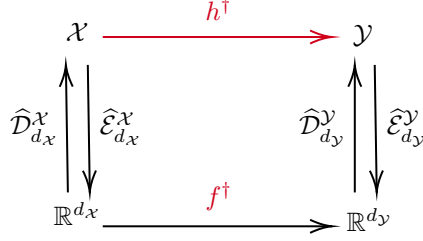


Figure 3: Commutative diagram of PCA encoder-decoder framework

and the decoder $\mathcal{D}_{d_X}^\mathcal{X} : \mathbb{R}^{d_X} \rightarrow \mathcal{X}$ is defined as

$$\mathcal{D}_{d_X}^\mathcal{X}(\eta) := \sum_{i=1}^{d_X} \eta_i \phi_i^\mathcal{X} = (\mathcal{E}_{d_X}^\mathcal{X})^*(\eta).$$

It then follows that $\mathbb{E} \left[\|(I - P_{d_X}^\mathcal{X})x\|_\mathcal{X}^2 \right] = \sum_{i > d_X} \lambda_i^{d_X}$, see (Lanthaler et al., 2022, Theorem 3.8).

In practice, it is usually difficult to obtain Σ_x directly, so we typically use the empirical covariance operator $\Sigma_x^T = \frac{1}{T} \sum_{i=1}^T x_i \otimes x_i$ as a substitute, thus deriving the empirical PCA. Following the same procedure, we naturally obtain the empirical encoder $\widehat{\mathcal{E}}_{d_X}^\mathcal{X}$ and empirical decoder $\widehat{\mathcal{D}}_{d_X}^\mathcal{X}$. Similarly, by replacing Σ_x^T , the input random variable x , and the rank d_X with $\Sigma_y^T := \frac{1}{T} \sum_{i=1}^T y_i \otimes y_i$, the output random variable y and d_Y , respectively, we apply empirical PCA to y in $\mathcal{Y}$ with rank d_Y , and obtain the empirical encoder $\widehat{\mathcal{E}}_{d_Y}^\mathcal{Y}$ and empirical decoder $\widehat{\mathcal{D}}_{d_Y}^\mathcal{Y}$.

We now formulate the estimator as $h := \widehat{\mathcal{D}}_{d_Y}^\mathcal{Y} \circ f \circ \widehat{\mathcal{E}}_{d_X}^\mathcal{X}$. It is then natural to choose $f^\dagger = \widehat{\mathcal{E}}_{d_Y}^\mathcal{Y} \circ h^\dagger \circ \widehat{\mathcal{D}}_{d_X}^\mathcal{X}$. This formulation naturally gives rise to a commutative diagram, as illustrated in Figure 3. The works (Bhattacharya et al., 2021; Lanthaler, 2023) represent f using neural networks. In contrast, we represent f in an RKHS induced by a matrix-valued kernel $k : \mathbb{R}^{d_X} \times \mathbb{R}^{d_X} \rightarrow \mathcal{B}(\mathbb{R}^{d_Y})$. Specifically, we consider kernels of the form $k(u, v) = \phi(\|u - v\|_{\mathbb{R}^{d_X}})$, where $\phi : [0, \infty) \rightarrow \mathbb{R}$ is a radial function, such that k is positive definite for $d_X > 0$. This property can be equivalently characterized by requiring ϕ to be completely monotone. Notable examples satisfying this condition include the inverse multiquadrics $\phi(x) = (c^2 + x^2)^{-\beta}$ and the Gaussian kernel $\phi(x) = e^{-\alpha|x|^2}$ for any $c > 0$, $\beta > 0$, and $\alpha > 0$; see (Wendland, 2004). Let $\mathcal{H}_k$ denote the RKHS induced by the matrix-valued kernel k . With a slight abuse of notation, we define the prediction error as

$$\mathcal{E}(f) = \mathcal{E}(h) := \mathbb{E} \left[\|h(x) - y\|_\mathcal{Y}^2 \right] = \mathbb{E} \left[\left\| \widehat{\mathcal{D}}_{d_Y}^\mathcal{Y} \circ f \circ \widehat{\mathcal{E}}_{d_X}^\mathcal{X}(x) - y \right\|_\mathcal{Y}^2 \right], \quad \forall f \in \mathcal{H}_K,$$

where $h = \widehat{\mathcal{D}}_{d_Y}^\mathcal{Y} \circ f \circ \widehat{\mathcal{E}}_{d_X}^\mathcal{X}$. Using the identity $(\widehat{\mathcal{D}}_{d_X}^\mathcal{X})^* = \widehat{\mathcal{E}}_{d_X}^\mathcal{X}$, we compute the Fréchet derivative of $\mathcal{E}(f)$ in $\mathcal{H}_k$, and obtain

$$\nabla \mathcal{E}(f) = 2\mathbb{E} \left[\phi \left(\left\| \widehat{\mathcal{E}}_{d_X}^\mathcal{X}(x) - \cdot \right\| \right) \widehat{\mathcal{E}}_{d_Y}^\mathcal{Y} \left(\widehat{\mathcal{D}}_{d_Y}^\mathcal{Y} \circ f \circ \widehat{\mathcal{E}}_{d_X}^\mathcal{X} x - y \right) \right].$$

Based on samples, we derive the regularized SGD iteration with $f_1 = 0$, and

$$f_{t+1} = f_t - \eta \left(\phi \left(\left\| \widehat{\mathcal{E}}_{d_X}^\mathcal{X}(x_t) - \cdot \right\| \right) \left(f_t \left(\widehat{\mathcal{E}}_{d_X}^\mathcal{X} x_t \right) - \widehat{\mathcal{E}}_{d_Y}^\mathcal{Y} y_t \right) + \lambda f_t \right),$$

where η is the step size. This can be interpreted as an SGD scheme based on the samples $\{\widehat{\mathcal{E}}_{d_X}^\mathcal{X} x_t, \widehat{\mathcal{E}}_{d_Y}^\mathcal{Y} y_t\}_{t=1}^T$ in $\mathcal{H}_k$. Accordingly, we define $h_1 = 0$ and

$$h_{t+1} = h_t - \eta \left(\phi \left(\left\| \widehat{\mathcal{E}}_{d_X}^\mathcal{X}(x_t - \cdot) \right\| \right) \widehat{P}_{d_Y}^\mathcal{Y} (h_t(x_t) - y_t) + \lambda h_t \right), \quad (\text{F.2})$$

where $\widehat{P}_{d_Y}^\mathcal{Y} := \widehat{\mathcal{D}}_{d_Y}^\mathcal{Y} \circ \widehat{\mathcal{E}}_{d_Y}^\mathcal{Y}$ is the empirical projection operator.

Under suitable assumptions, we can assert that, $\phi\left(\left\|\widehat{\mathcal{E}}_{d_{\mathcal{X}}}^{\mathcal{X}}(x - \cdot)\right\|\right)\widehat{P}_{d_{\mathcal{Y}}}^{\mathcal{Y}}$ converges to $\phi(\|x - \cdot\|_{\mathcal{X}})I_{\mathcal{Y}}$ as $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$ tend to ∞ . Hence, the SGD iteration (F.2) can be approached by the following scheme with $\widetilde{h}_1 = 0$, and

$$\widetilde{h}_{t+1} = \widetilde{h}_t - \eta \left(\phi(\|x_t - \cdot\|_{\mathcal{X}})(\widetilde{h}_t(x_t) - y_t) + \lambda \widetilde{h}_t \right),$$

which is the setting analyzed in this paper.

To summarize, using the PCA encoder-decoder as a concrete example, we see that our analysis can seamlessly align with the encoder-decoder framework. Rigorous proofs will be provided in our future work.

G Detailed Setup of the Numerical Experiments

We describe the matrix realization used in the numerical experiments. The input variable is represented through the first m basis functions

$$e_k(u) = \sqrt{2} \cos(k\pi u), \quad u \in [0, 1], \quad 1 \leq k \leq m,$$

and generated by

$$x = \sum_{k=1}^m \sqrt{\mu_k} \xi_k e_k, \quad \mu_k = k^{-\alpha},$$

where $\xi_k \sim \mathcal{N}(0, 1)$ are independent. We consider the scalar kernel

$$\mathcal{K}(x, x') = \sum_{k \geq 1} \nu_k \langle x, e_k \rangle_{L^2} \langle x', e_k \rangle_{L^2}, \quad \nu_k = k^{-\beta},$$

and the operator-valued kernel $K(x, x') = \mathcal{K}(x, x')I_{\mathcal{Y}}$ with $\mathcal{Y} = L^2[0, 1]$.

Let $\{\psi_k\}_{k \geq 1}$ denote the canonical orthonormal basis of $\mathcal{H}_{\mathcal{K}}$,

$$\psi_k = \sqrt{\nu_k} \langle \cdot, e_k \rangle_{L^2},$$

so that

$$\phi(x) = \mathcal{K}(x, \cdot) = \sum_{k=1}^m \sqrt{\mu_k \nu_k} \xi_k \psi_k.$$

The covariance operator $C : \mathcal{H}_{\mathcal{K}} \rightarrow \mathcal{H}_{\mathcal{K}}$ is diagonal in this basis:

$$C = \sum_{k=1}^m \mu_k \nu_k \psi_k \otimes \psi_k.$$

We choose the target operator as

$$H^\dagger = S^\dagger C^r, \quad S^\dagger \psi_k = e_k, \quad 1 \leq k \leq m.$$

Then

$$h^\dagger(x) = H^\dagger \phi(x), \quad \epsilon_t = \frac{\sigma}{\sqrt{m}} \sum_{k=1}^m \zeta_k^t e_k,$$

where $\zeta_k^t \sim \mathcal{N}(0, 1)$ are independent, and hence

$$x_t = \sum_{k=1}^m \sqrt{\mu_k} \xi_k^t e_k, \quad y_t = \sum_{k=1}^m \left((\mu_k \nu_k)^{r+1/2} \zeta_k^t + \frac{\sigma}{\sqrt{m}} \zeta_k^t \right) e_k.$$

Writing the estimator in matrix form as

$$H_t = \sum_{k_1, k_2=1}^m H_{k_1, k_2}^{(t)} e_{k_1} \otimes \psi_{k_2}, \quad H^{(t)} = (H_{k_1, k_2}^{(t)})_{1 \leq k_1, k_2 \leq m},$$

and introducing

$$A = \text{diag}(\sqrt{\mu_1 \nu_1}, \dots, \sqrt{\mu_m \nu_m}), \quad B = \text{diag}((\mu_1 \nu_1)^{r+1/2}, \dots, (\mu_m \nu_m)^{r+1/2}),$$

together with

$$z^t = A \xi^t, \quad y^t = B \xi^t + \frac{\sigma}{\sqrt{m}} \zeta^t,$$

where

$$\xi^t = (\xi_1^t, \dots, \xi_m^t)^\top, \quad \zeta^t = (\zeta_1^t, \dots, \zeta_m^t)^\top,$$

the regularized SGD iteration becomes

$$H^{(1)} = 0, \quad H^{(t+1)} = (1 - \eta_t \lambda_t) H^{(t)} - \eta_t (H^{(t)} z^t - y^t) (z^t)^\top.$$

Finally, letting

$$H_0 = \text{diag}(\mu_1^r \nu_1^r, \dots, \mu_m^r \nu_m^r),$$

the prediction and estimation errors are computed by

$$\mathcal{E}(h_{t+1}) - \mathcal{E}(h^\dagger) = \|(H^{(t+1)} - H_0)A\|_F^2, \quad \|h_{t+1} - h^\dagger\|_{\mathcal{H}}^2 = \|H^{(t+1)} - H_0\|_F^2,$$

where $\|\cdot\|_F$ denotes the Frobenius norm of a matrix.

This completes the numerical setup used for the experiments in the main text, for which we take

$$r = 1.0, \quad m = 200, \quad \alpha = 0.3, \quad \beta = 0.9, \quad \sigma = 0.1.$$